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(12) **United States Patent**
Hua et al.

(10) **Patent No.: US 12,410,234 B2**

(45) **Date of Patent: Sep. 9, 2025**

(54) **COMPOSITIONS AND METHODS FOR
RETRIEVING TUMOR-RELATED
ANTIBODIES AND ANTIGENS**

(58) **Field of Classification Search**

None

See application file for complete search history.

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(56) **References Cited**

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(73) Assignee: **The Trustees of the University of
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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1217 days.

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(21) Appl. No.: **17/050,299**

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§ 371 (c)(1),

(2) Date: **Oct. 23, 2020**

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PCT Pub. Date: **Oct. 31, 2019**

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Related U.S. Application Data

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26, 2018.

(51) **Int. Cl.**

A61K 47/68 (2017.01)

A61K 40/11 (2025.01)

A61K 40/31 (2025.01)

A61K 40/42 (2025.01)

C07K 14/705 (2006.01)

C07K 14/725 (2006.01)

C07K 16/28 (2006.01)

A61K 38/00 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 14/70578** (2013.01); **A61K 40/11**
(2025.01); **A61K 40/31** (2025.01); **A61K**
40/4201 (2025.01); **A61K 40/4224** (2025.01);
A61K 47/68033 (2023.08); **A61K 47/6849**
(2017.08); **C07K 14/7051** (2013.01); **C07K**
14/70517 (2013.01); **C07K 14/70521**
(2013.01); **C07K 16/2896** (2013.01); **A61K**
38/00 (2013.01); **C07K 2317/53** (2013.01);
C07K 2317/565 (2013.01); **C07K 2317/569**
(2013.01); **C07K 2317/622** (2013.01); **C07K**
2319/02 (2013.01); **C07K 2319/03** (2013.01);
C07K 2319/30 (2013.01); **C07K 2319/33**
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(57)

ABSTRACT

The present invention includes compositions and methods for retrieving tumor-related antibodies and antigens. In one aspect, the invention includes a method for Sequential Tumor-related Antibody and antigen Retrieving (STAR) which directly and efficiently identifies potent antibodies that can specifically bind to tumor-related antigens on the tumor cell surface. In another aspect, the invention includes a CAR comprising a nanobody, a transmembrane domain, and an intracellular domain, wherein the nanobody is retrieved by a STAR method. In another aspect, the invention includes a CAR T system that targets CD13 and treats acute myeloid leukemia. In another aspect, the invention includes a CAR T system and ADC that targets CDH17 and treats NETs and other types of tumors expressing this antigen, with tolerable toxicities.

17 Claims, 44 Drawing Sheets

Specification includes a Sequence Listing.

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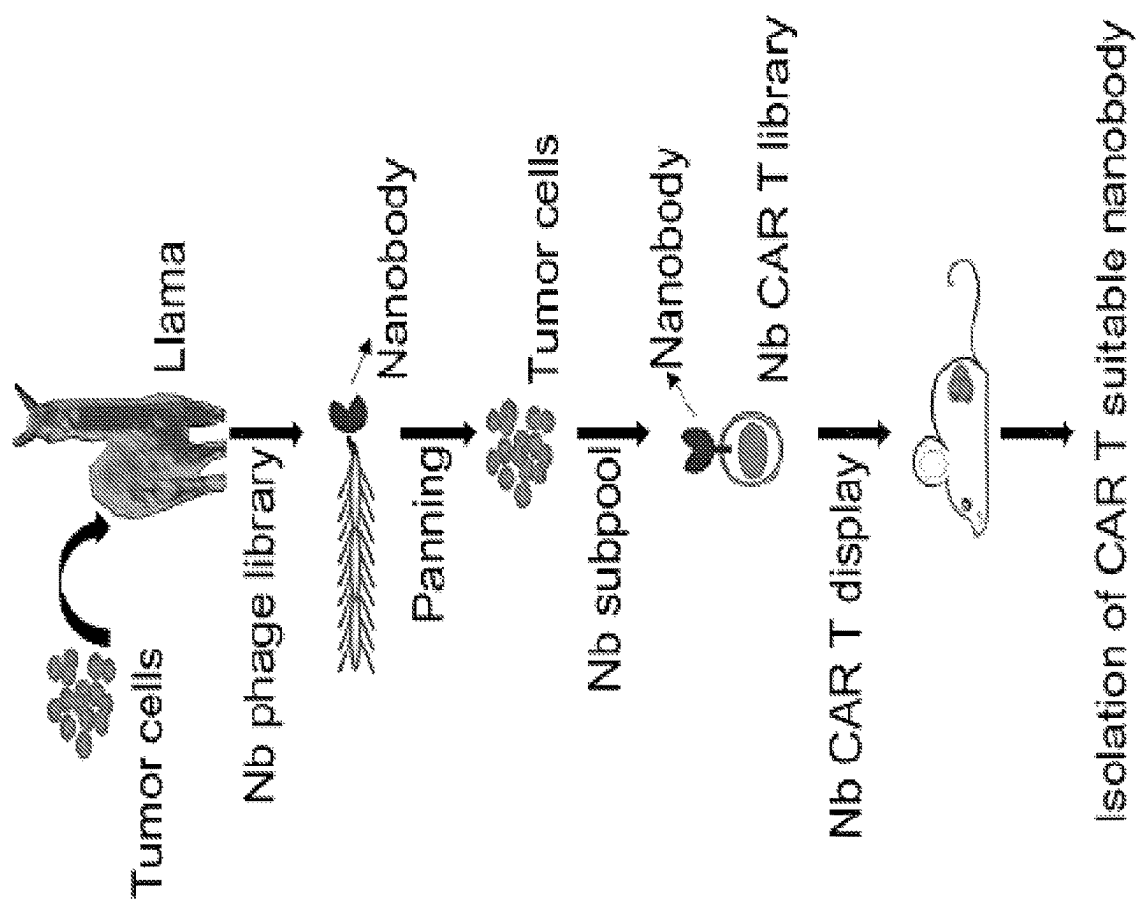


FIG. 1A

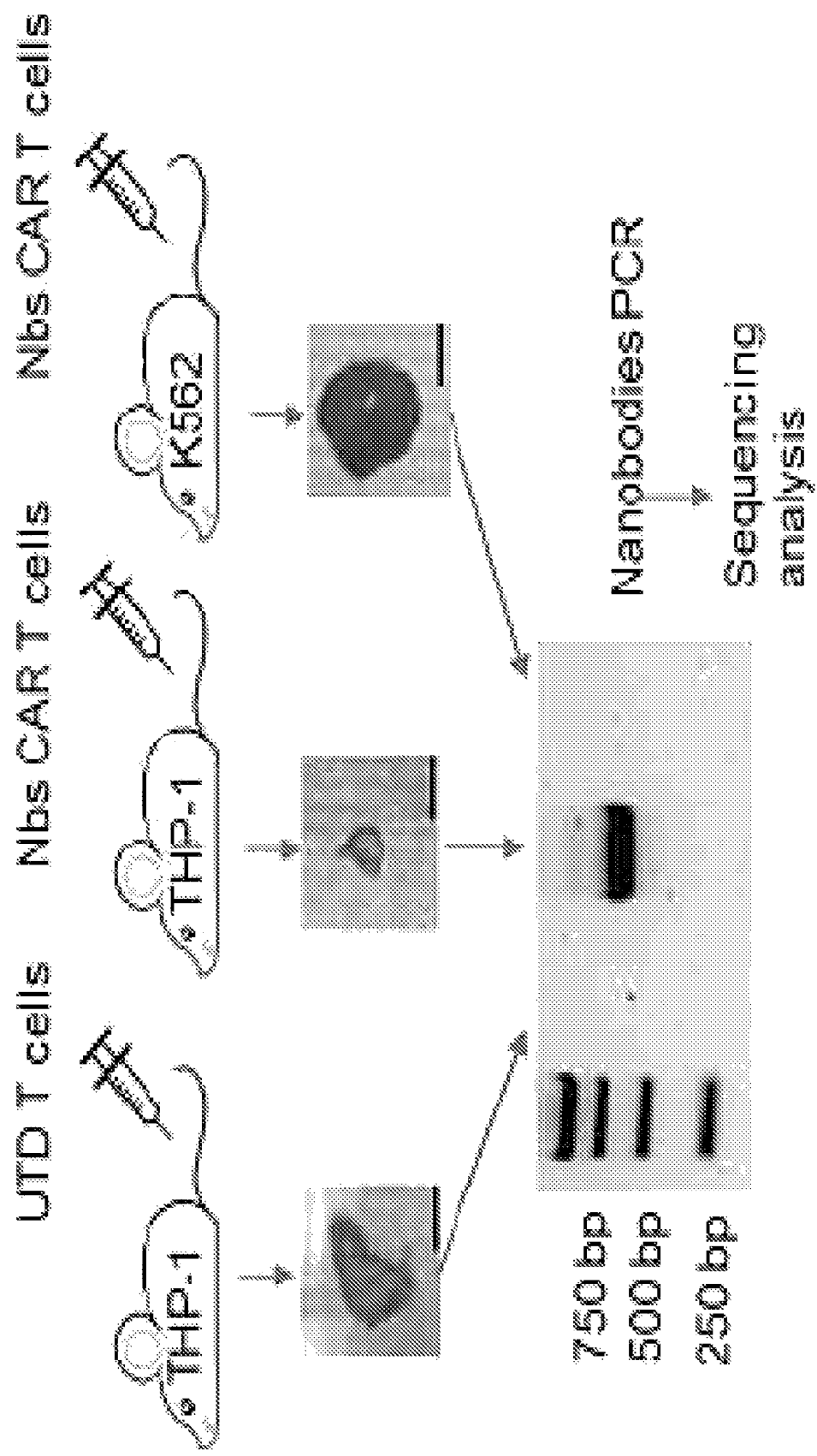


FIG. 1B

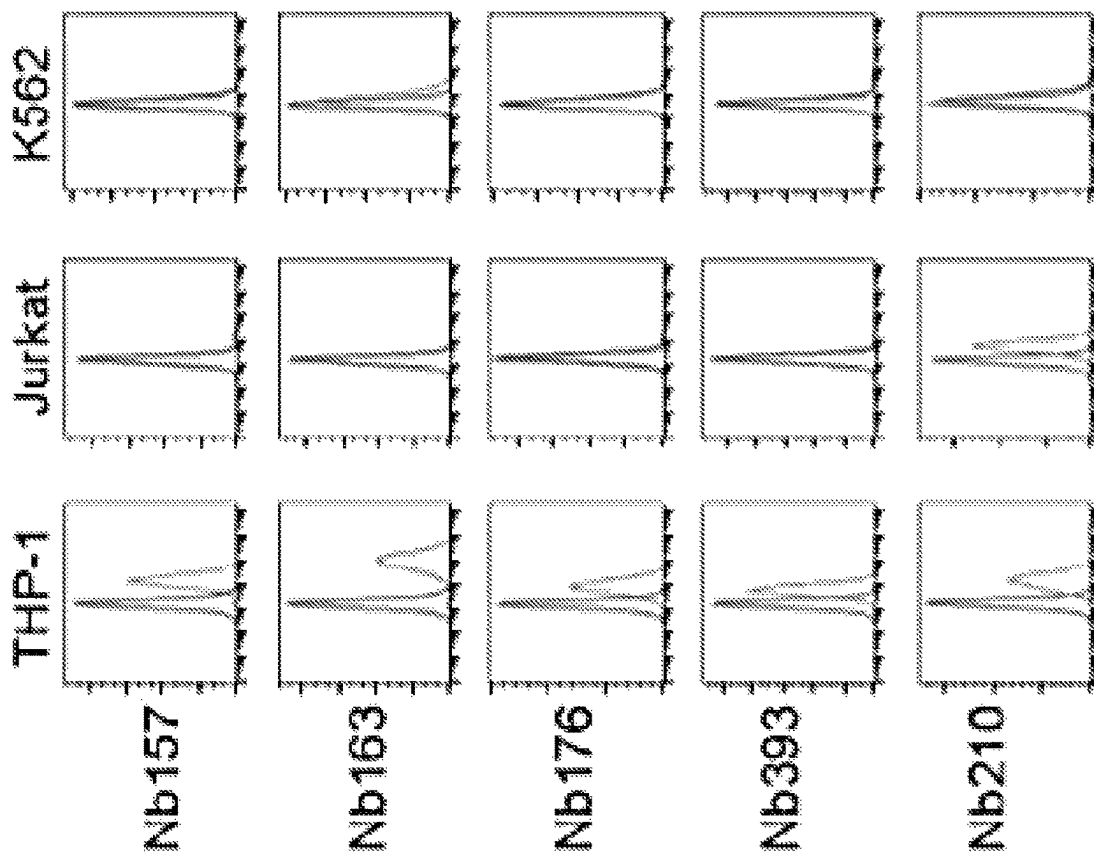


FIG. 1C

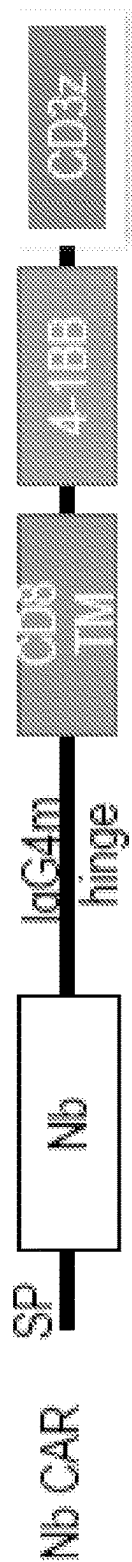


FIG. 2A

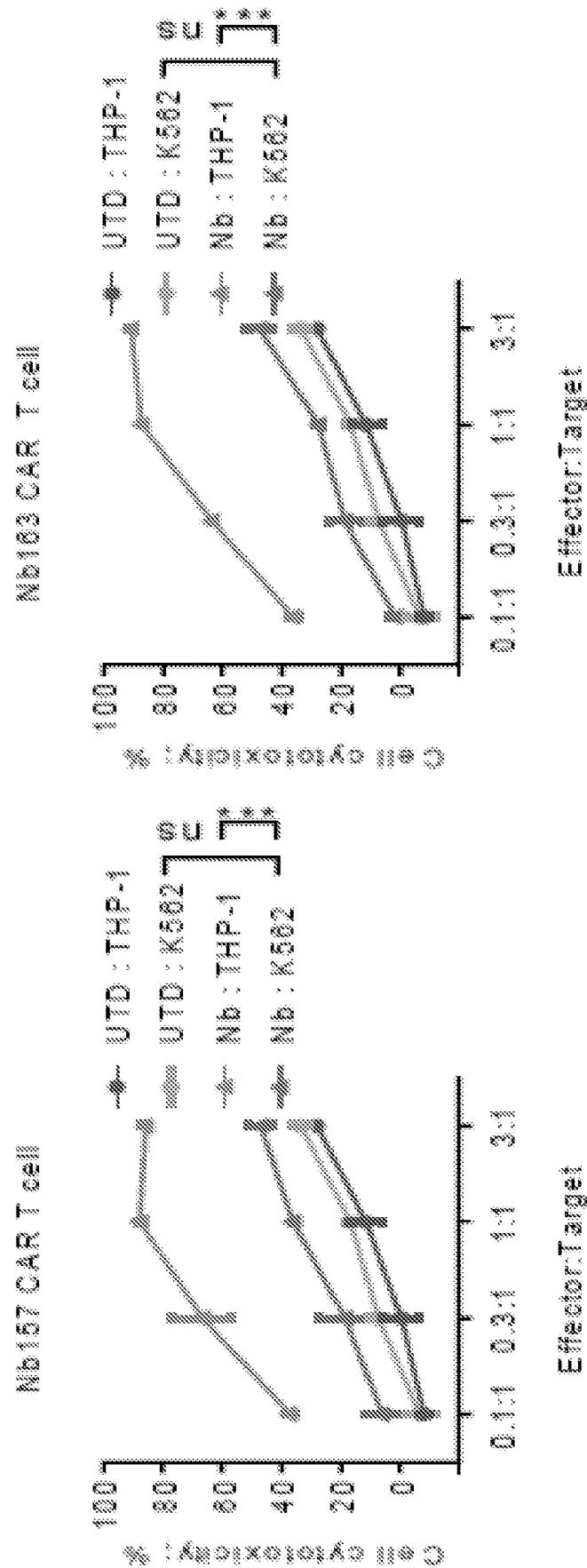


FIG. 2B

FIG. 2C

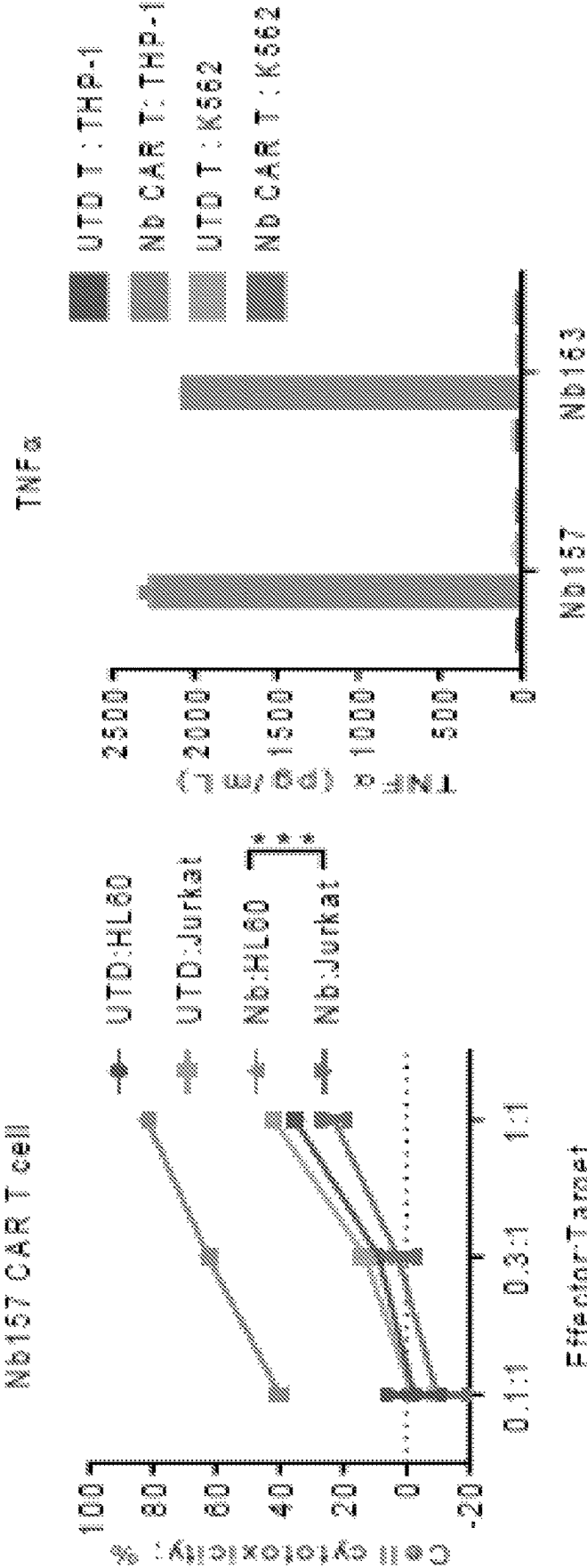


FIG. 2D

FIG. 2E

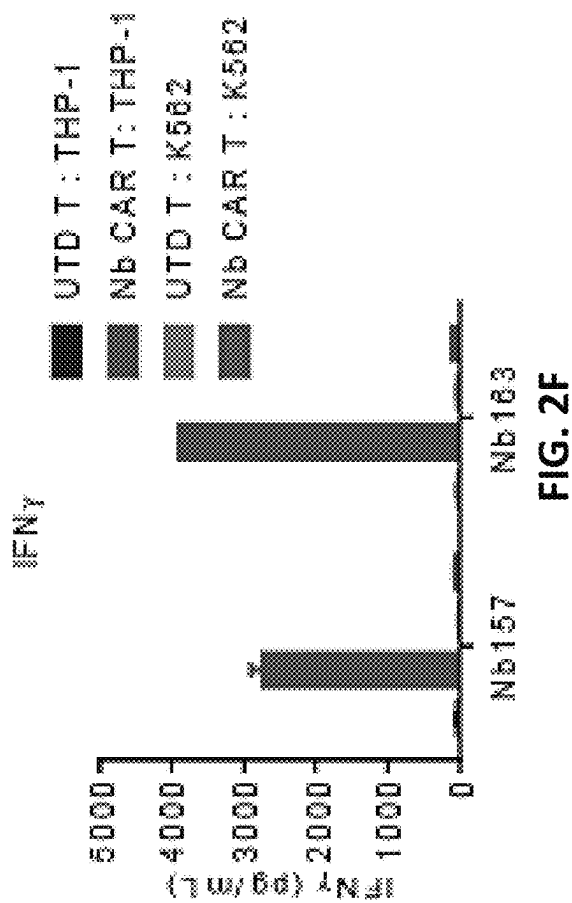


FIG. 2F

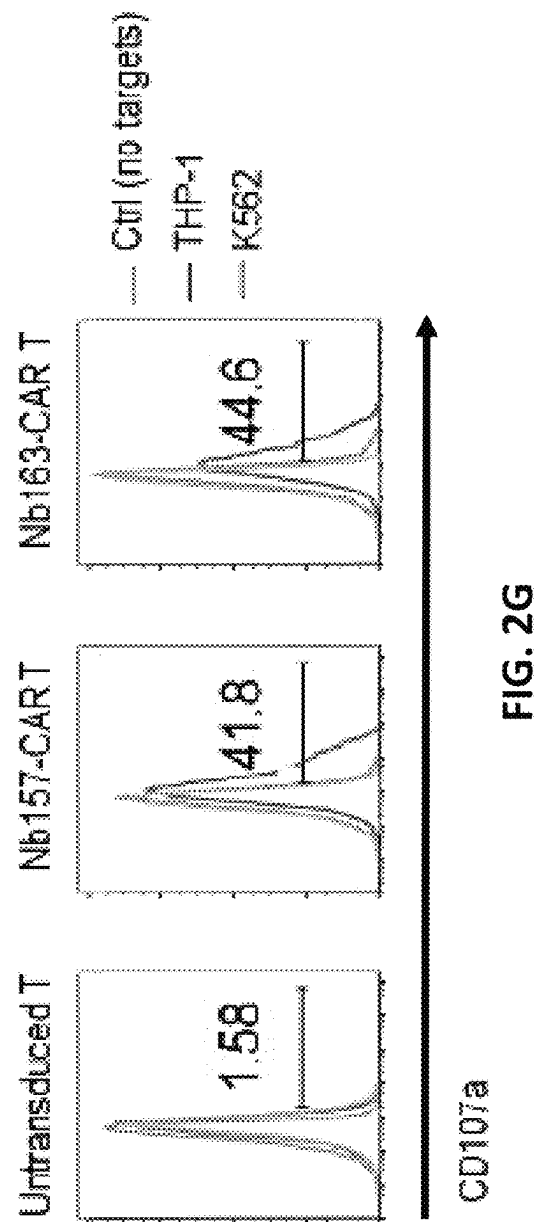


FIG. 2G

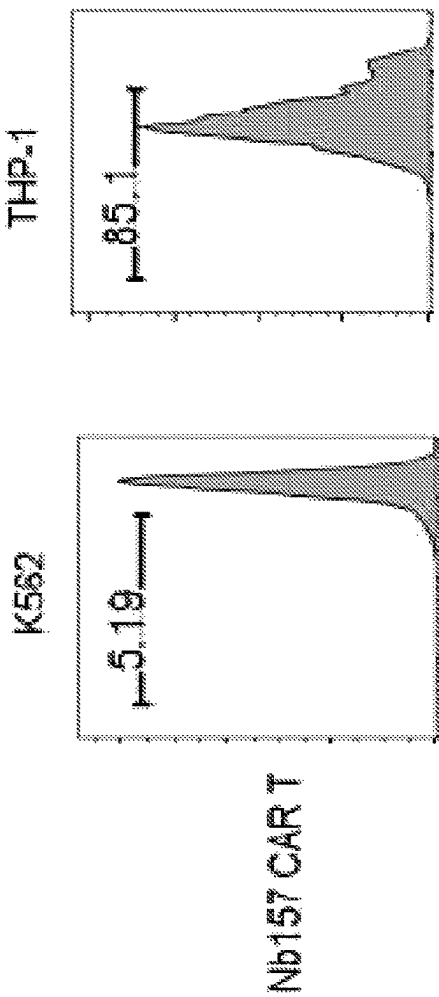
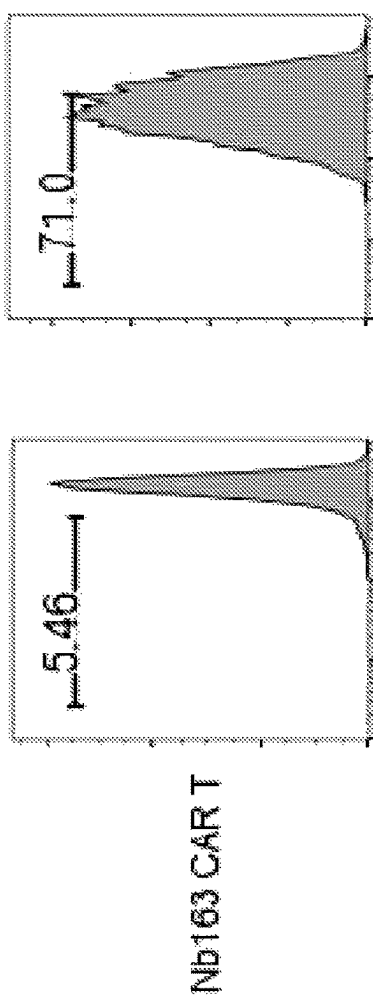


FIG. 2I

FIG. 2H



CellTrace

FIG. 2K

FIG. 2J

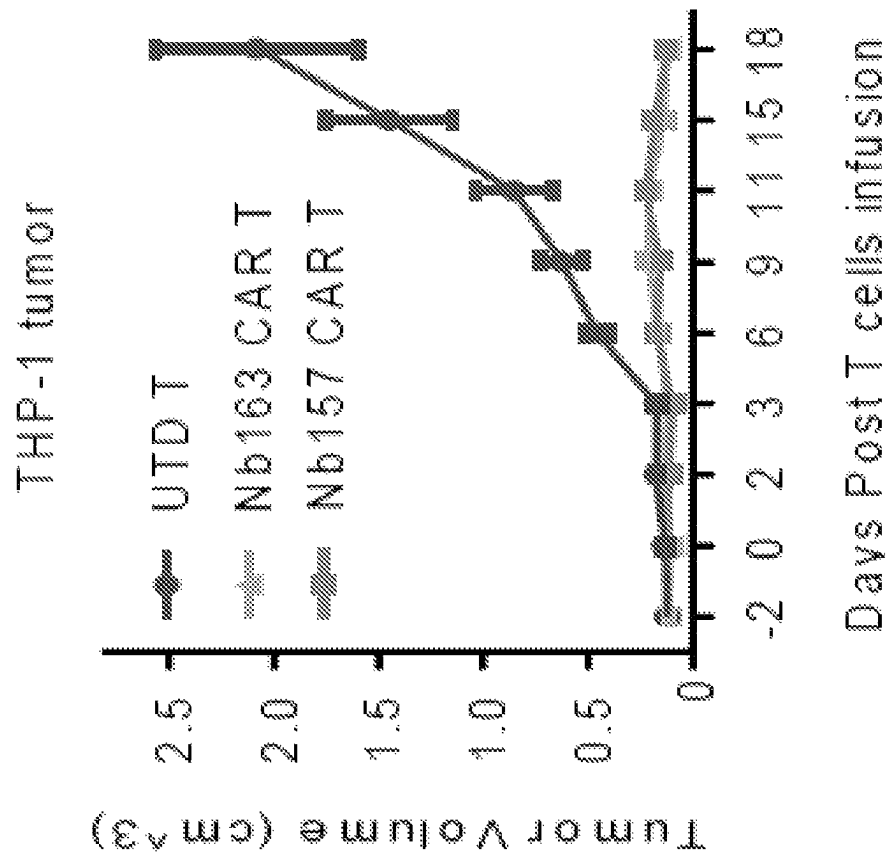


FIG. 3A

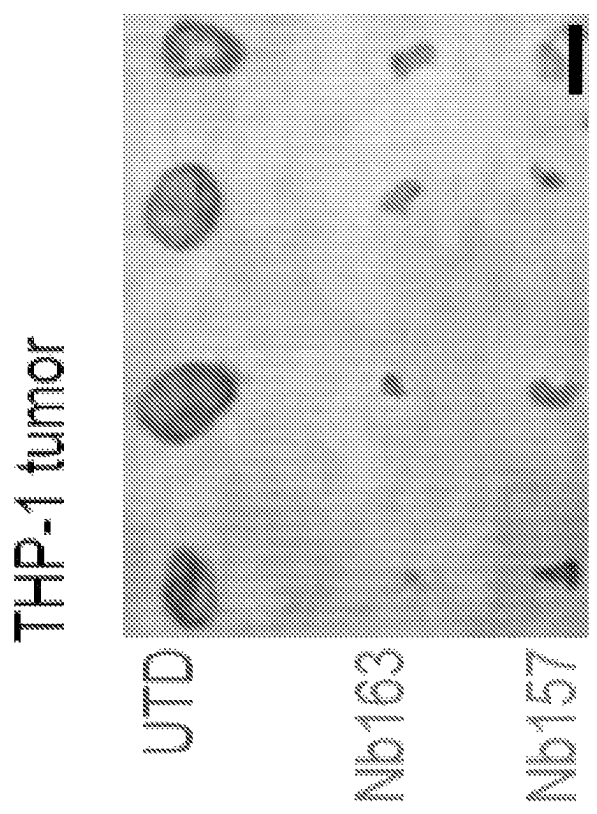


FIG. 3B

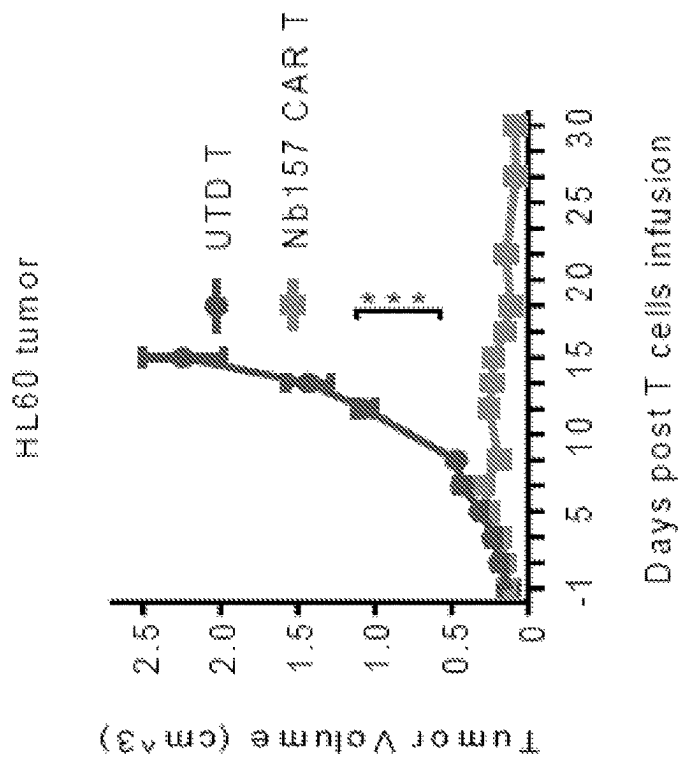


FIG. 3D

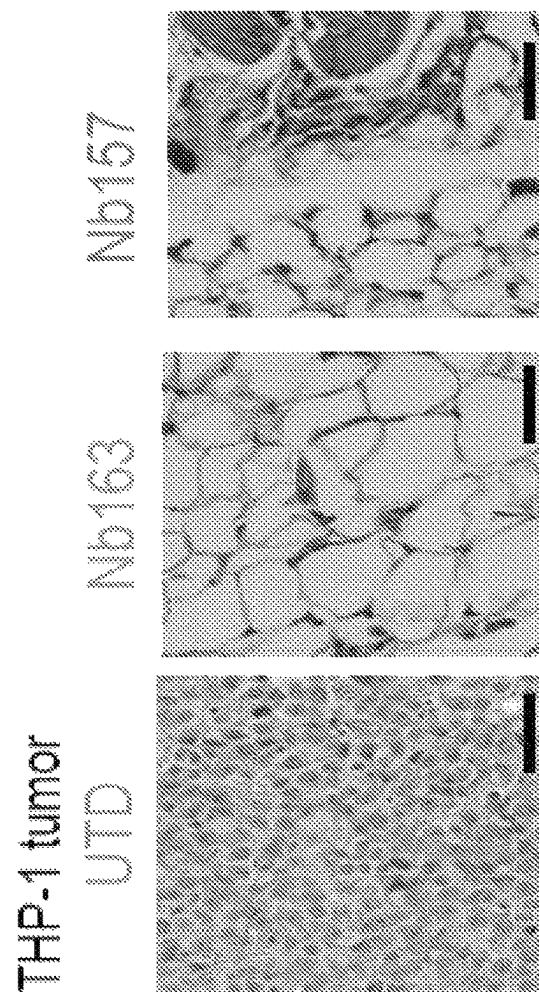


FIG. 3C

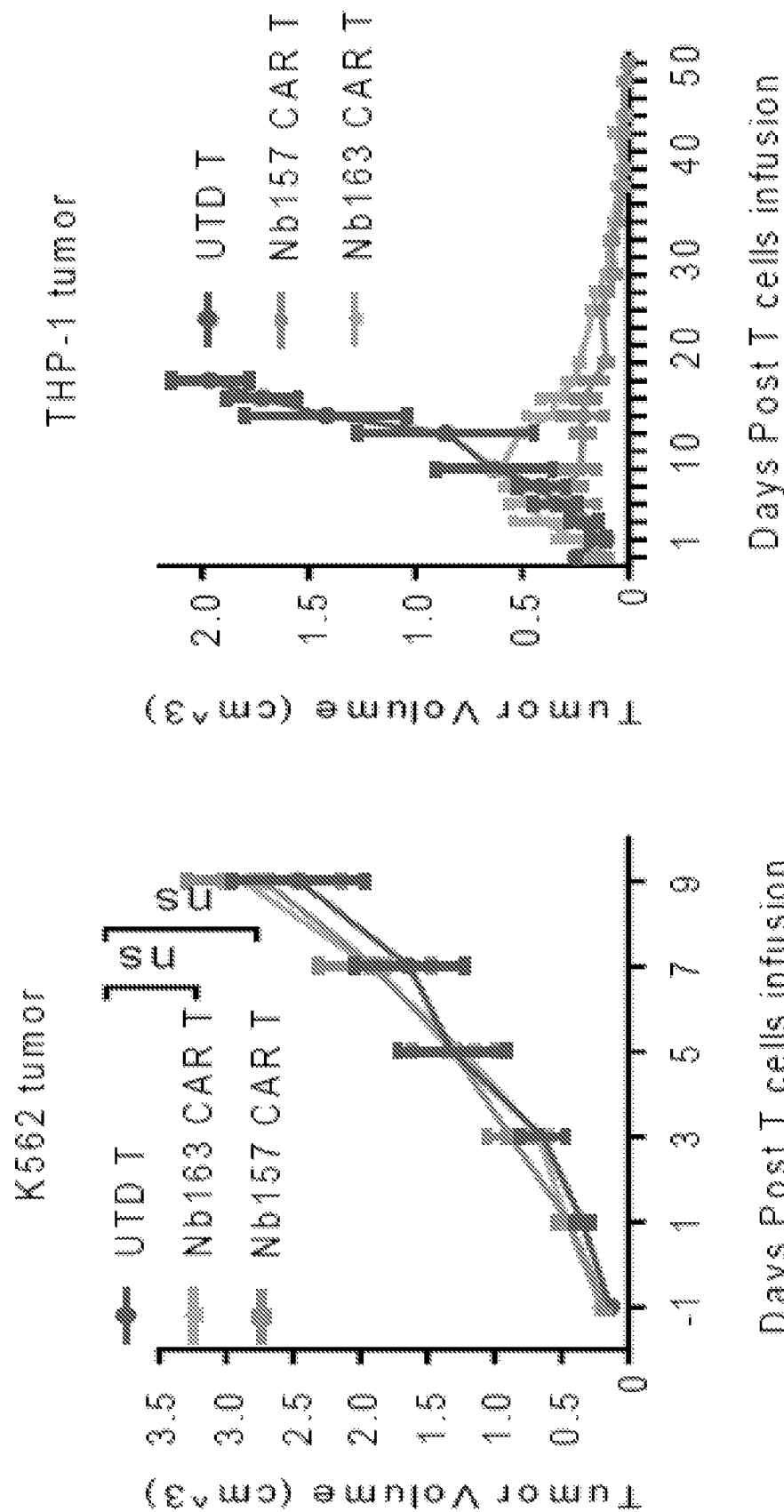


FIG. 3F

FIG. 3E

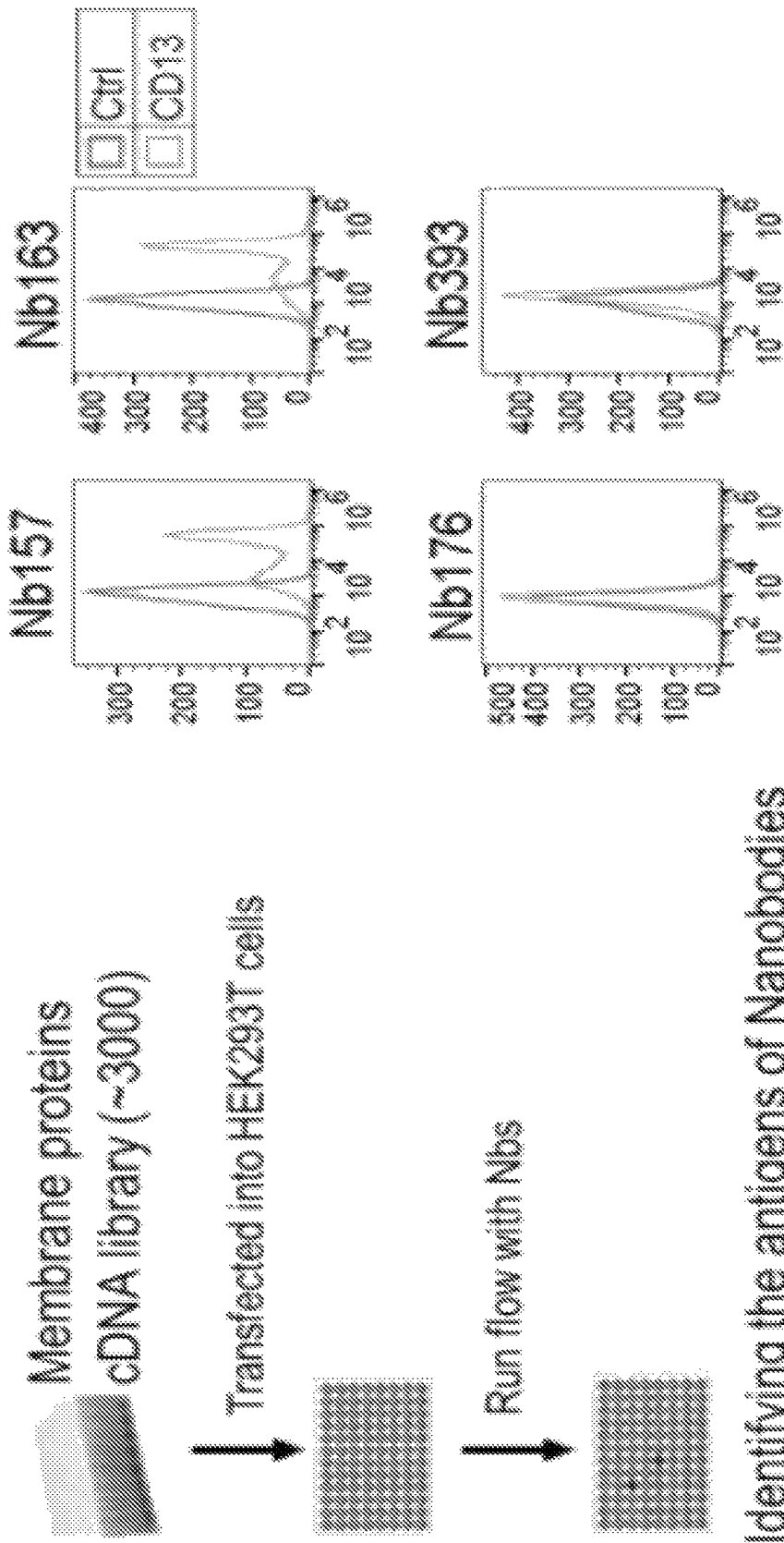


FIG. 4A

FIG. 4B

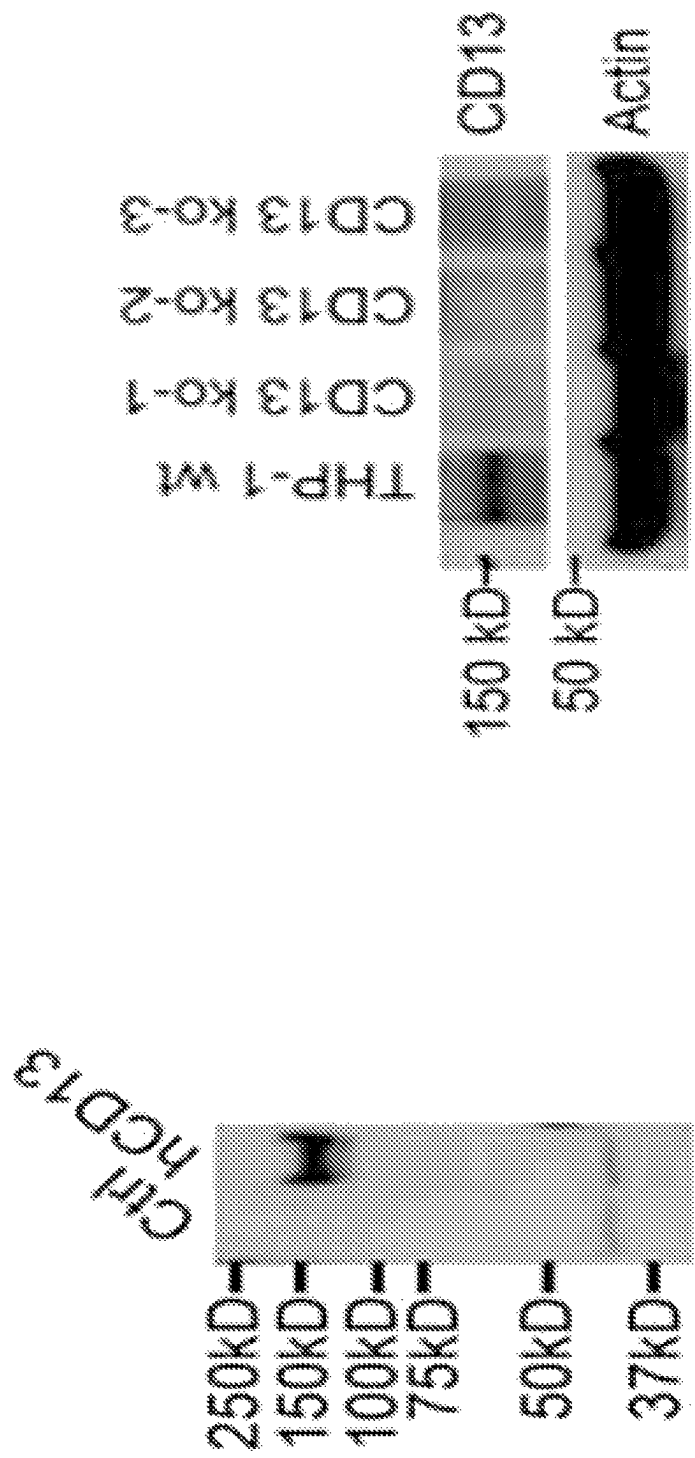


FIG. 4D

FIG. 4C

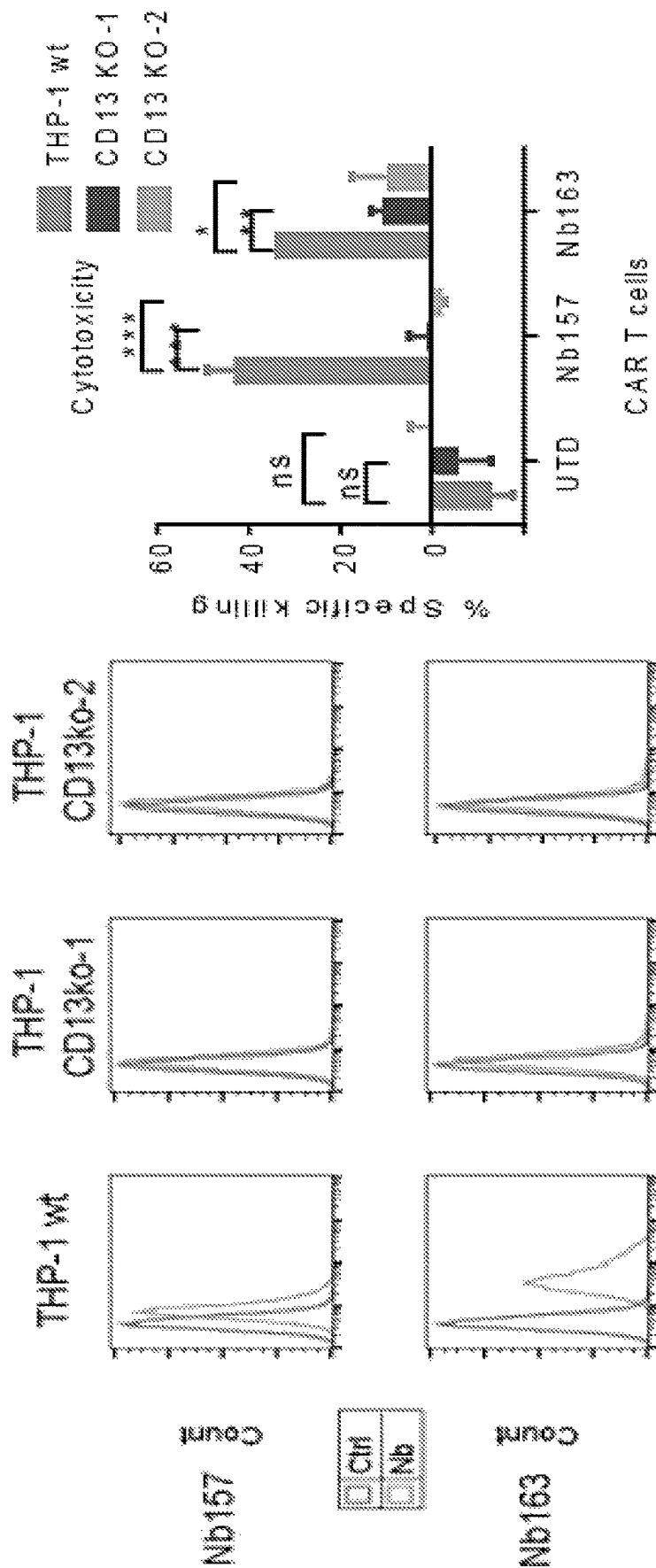


FIG. 4F

FIG. 4E

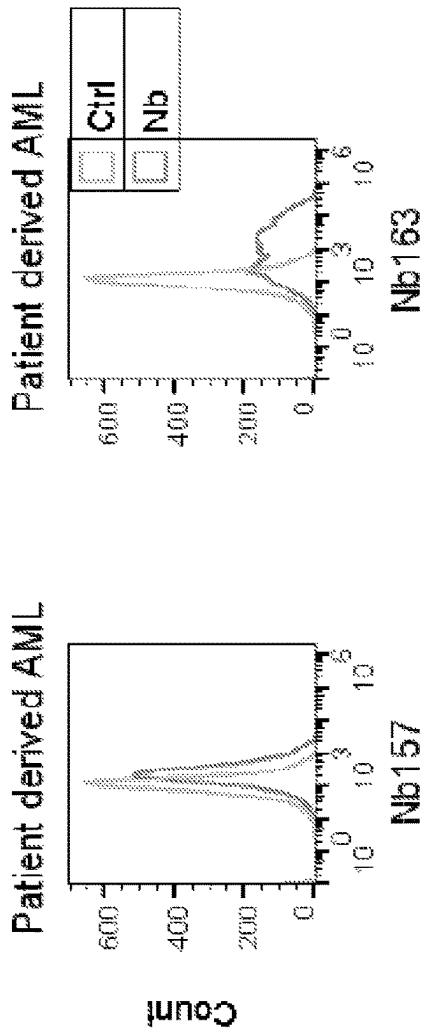


FIG. 5A

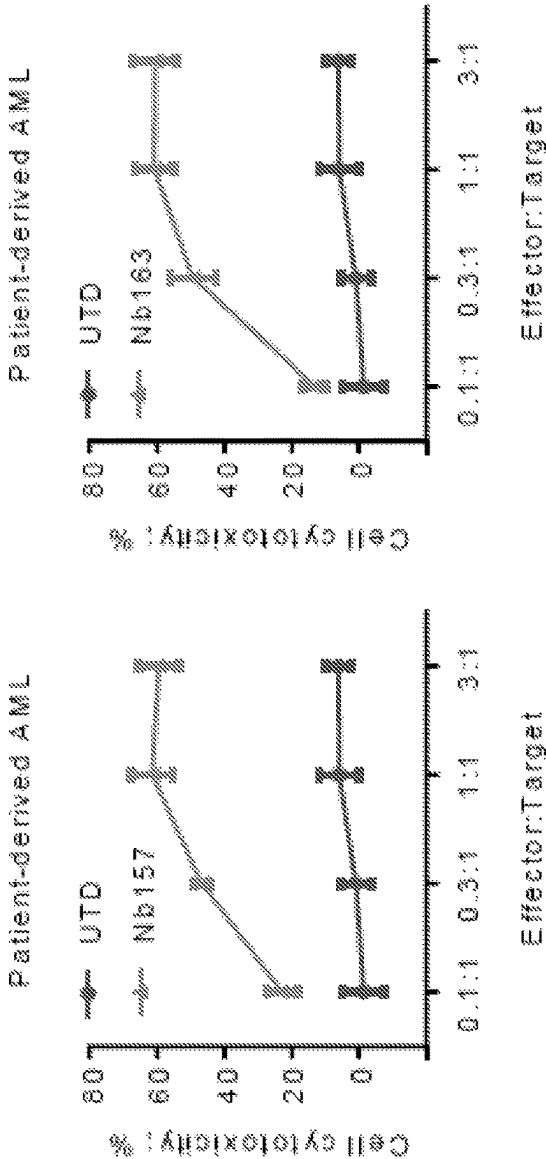


FIG. 5C

Patient derived AML

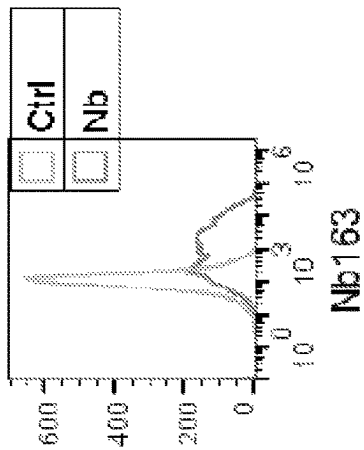


FIG. 5B

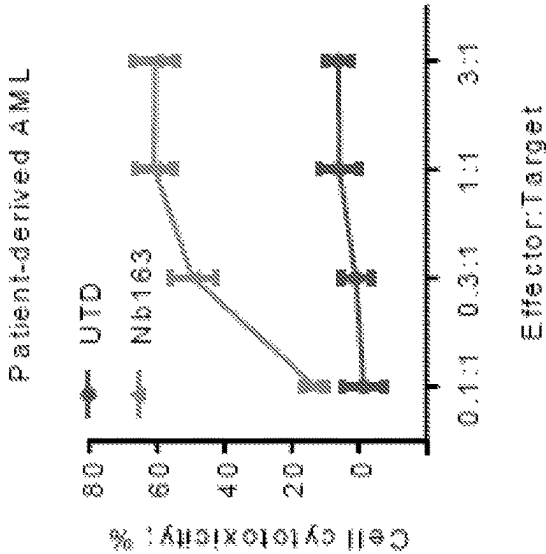


FIG. 5D

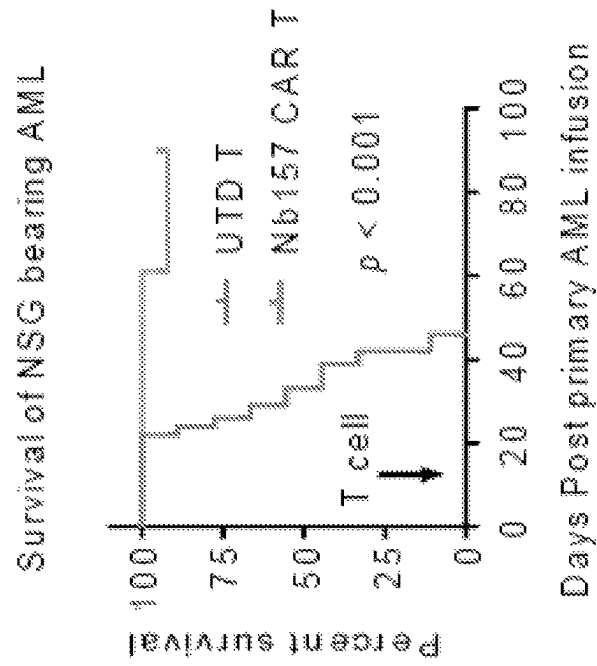


FIG. 5E

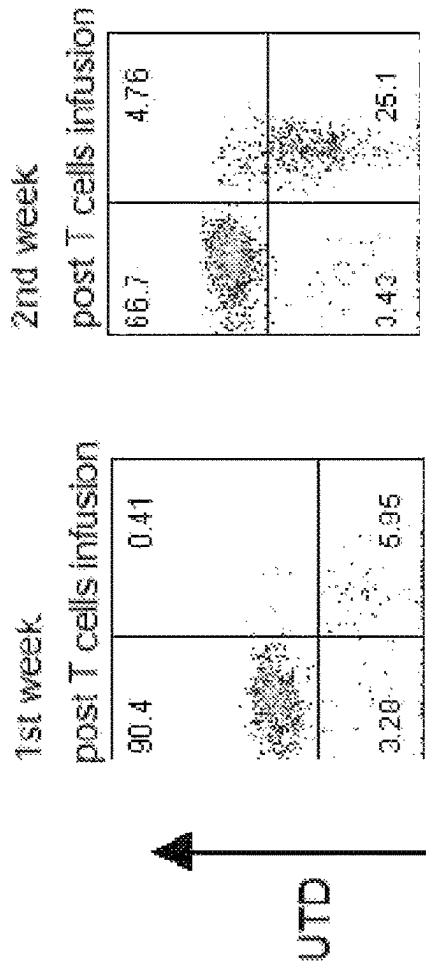


FIG. 5G

FIG. 5H



FIG. 5G

FIG. 5I

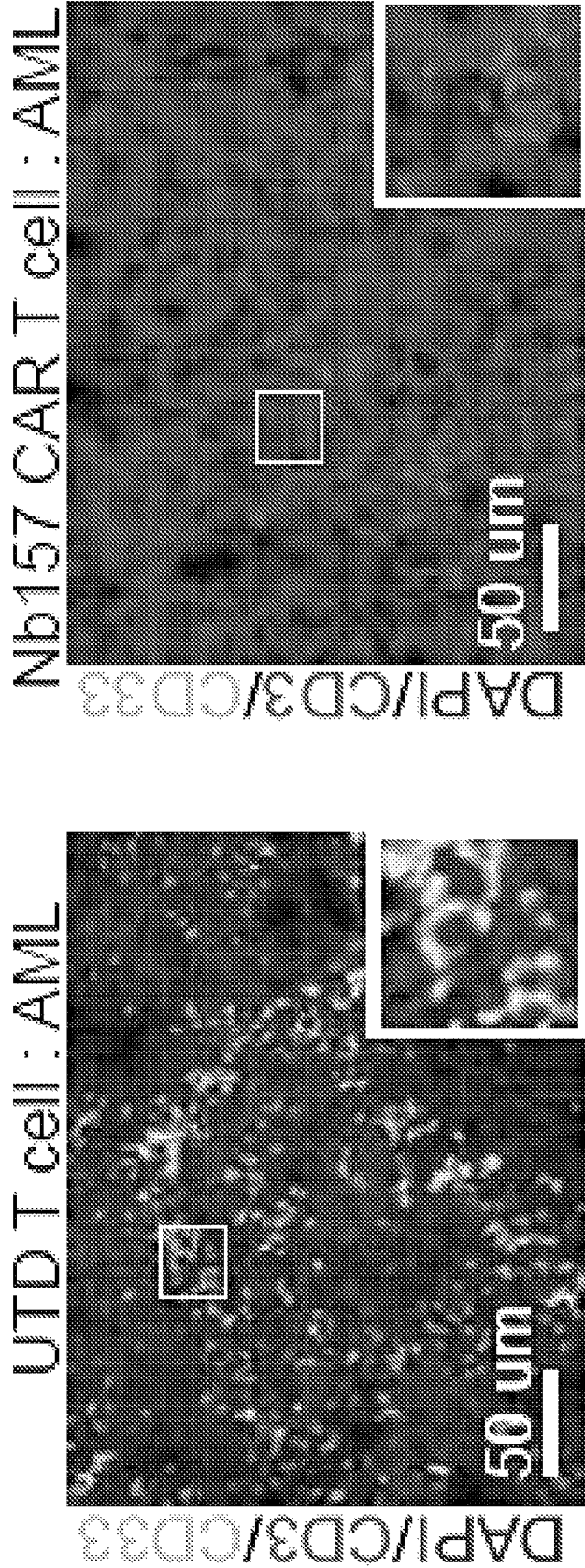


FIG. 5J

FIG. 5K

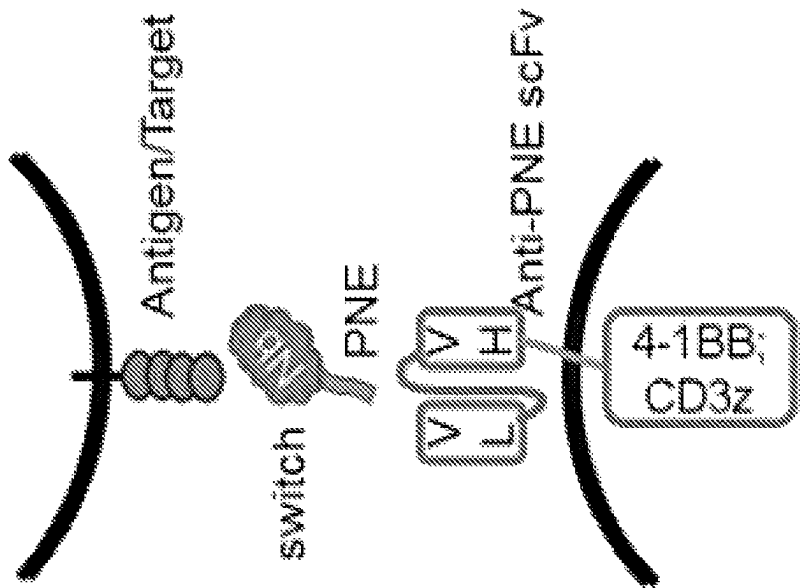


FIG. 6B

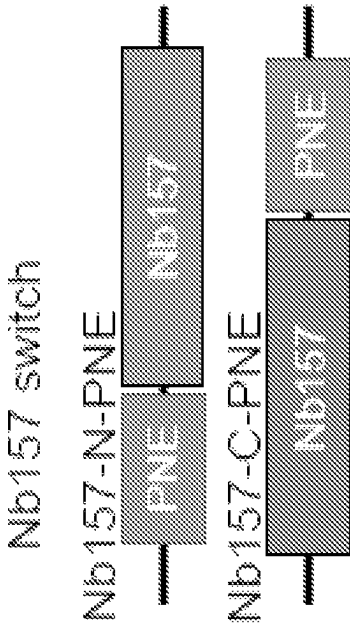


FIG. 6A

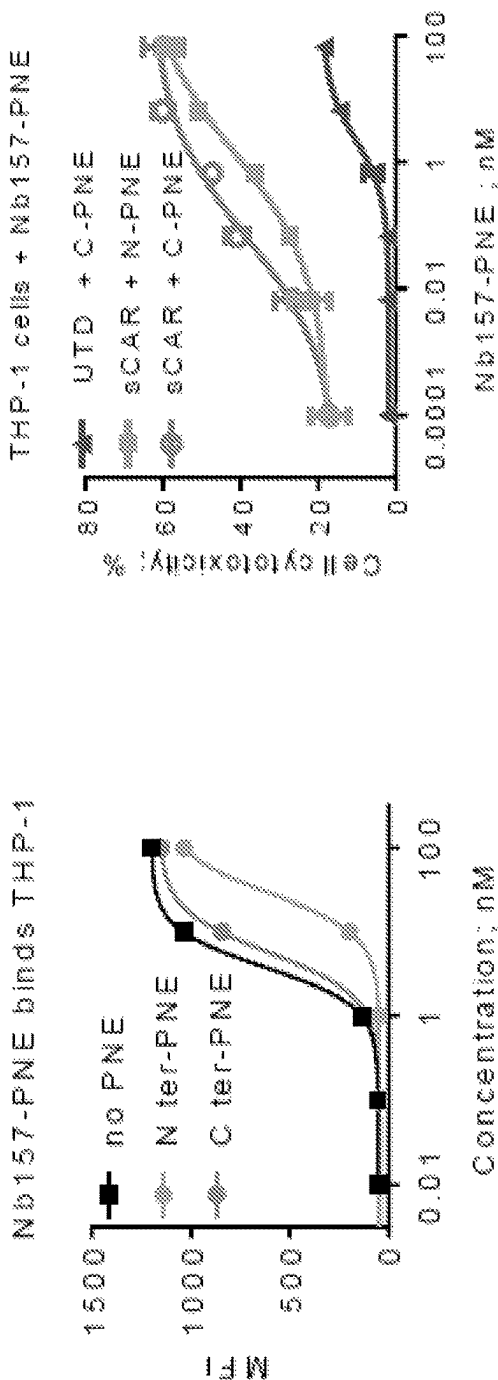


FIG. 6D

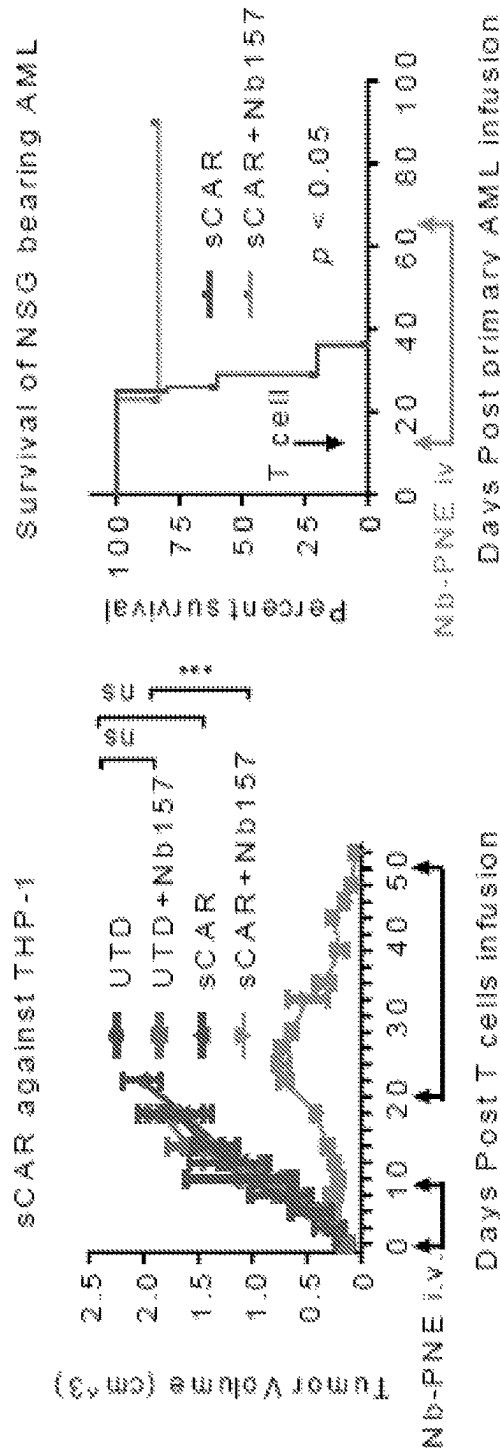


FIG. 6E

FIG. 6F

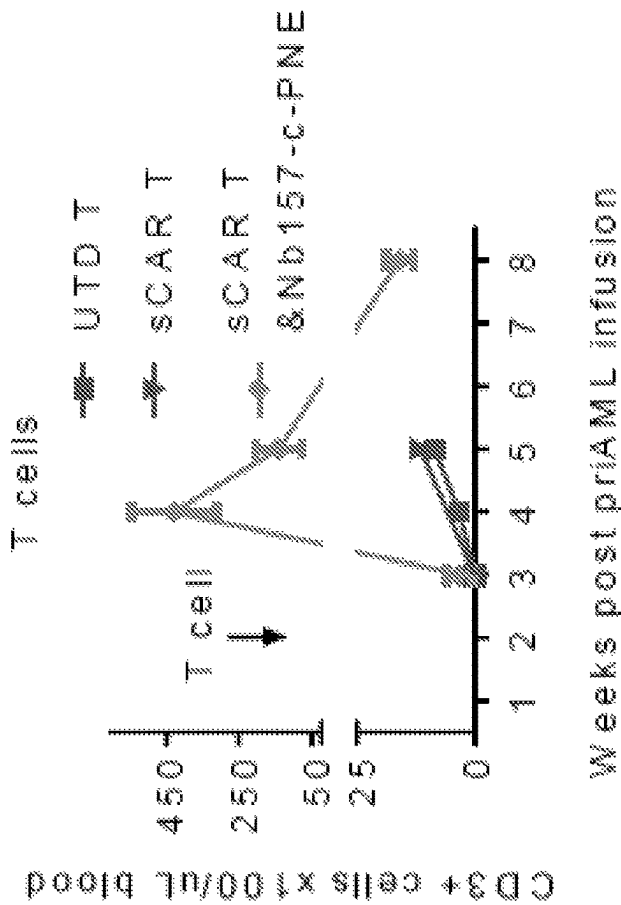


FIG. 6H

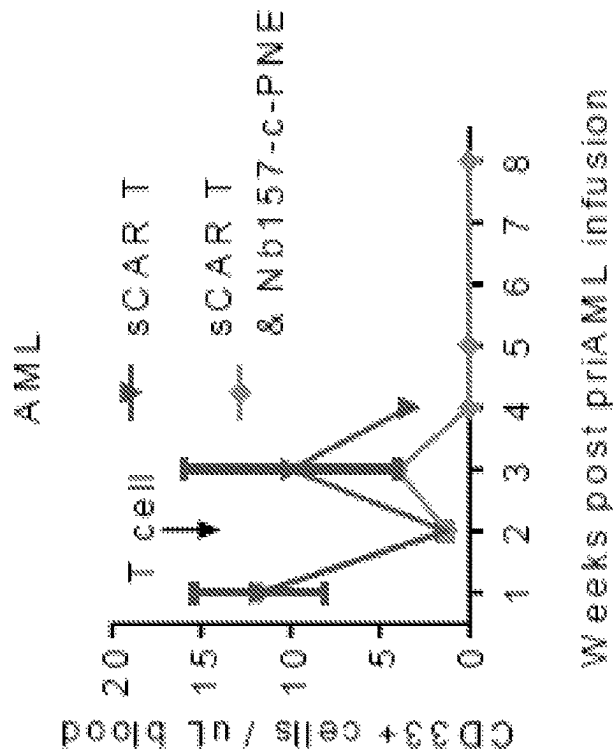


FIG. 6G

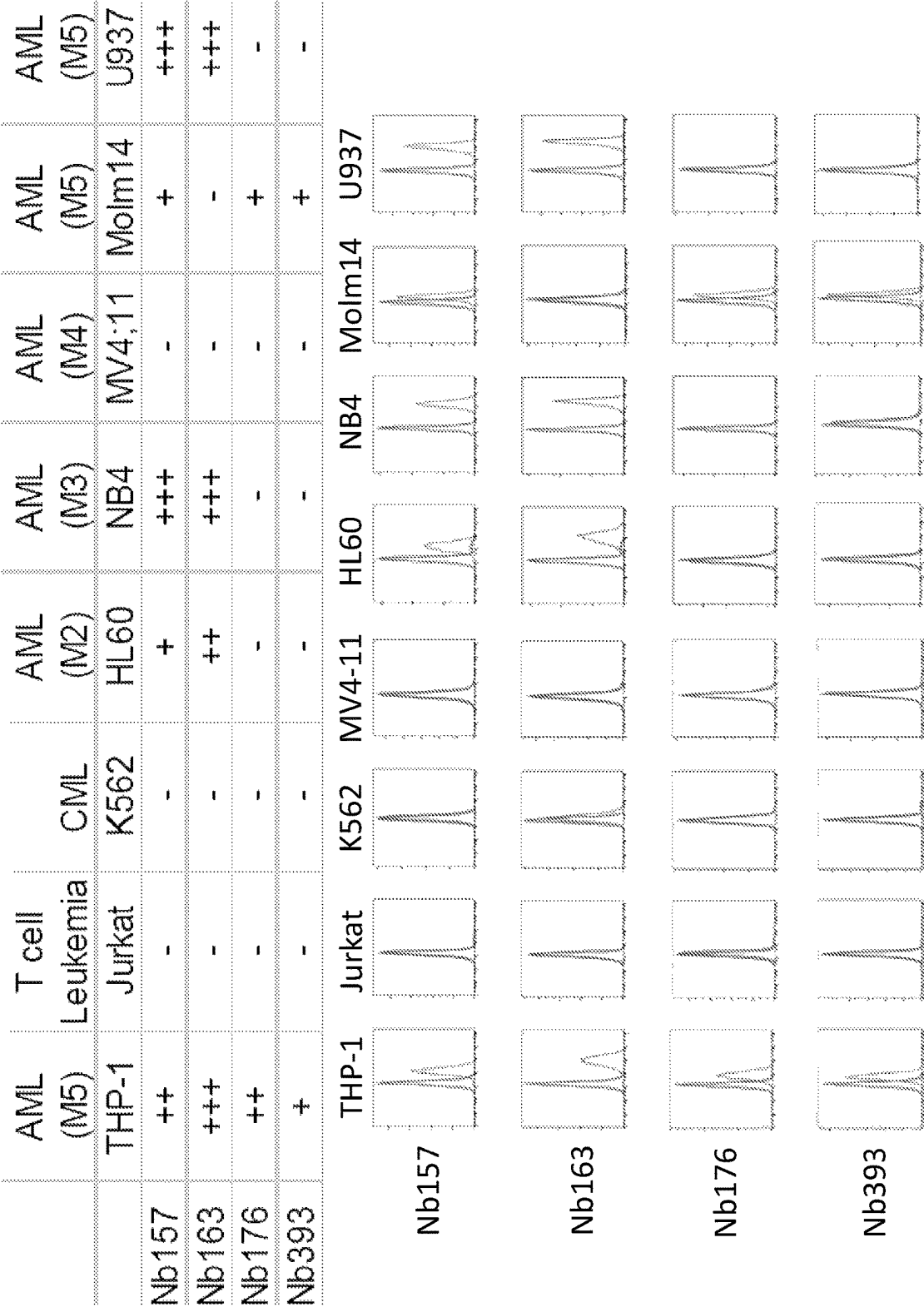


FIG. 7

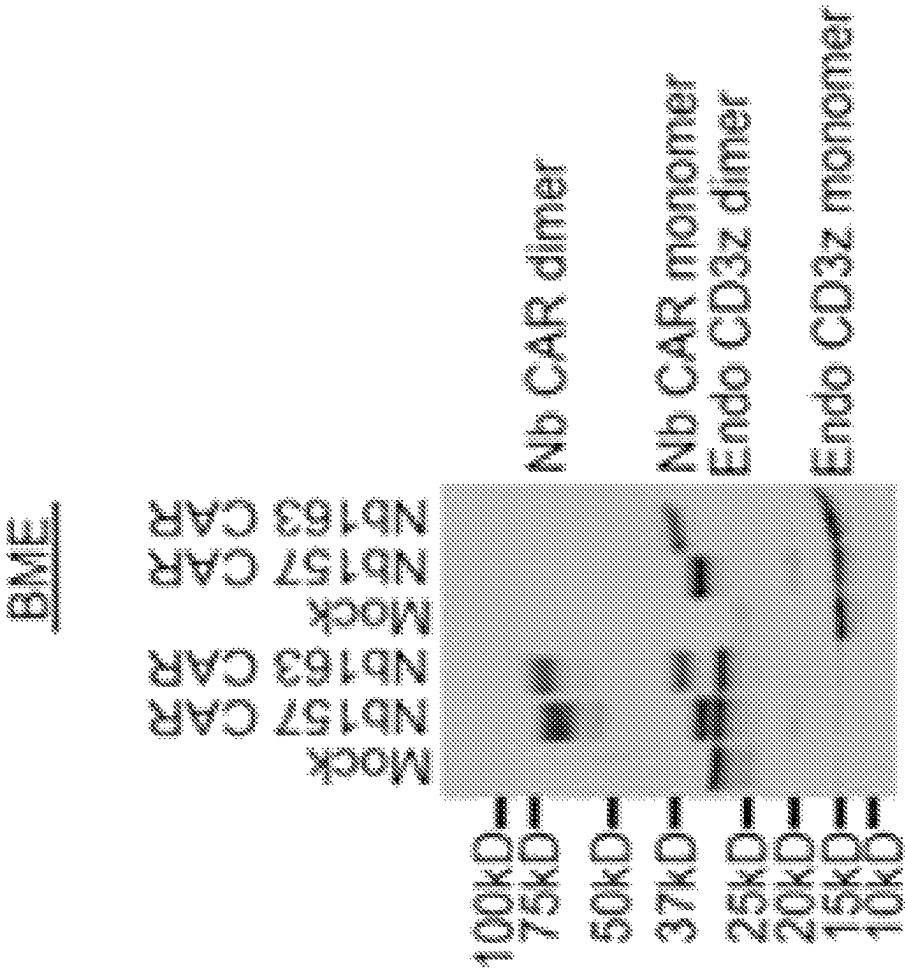


FIG. 8A

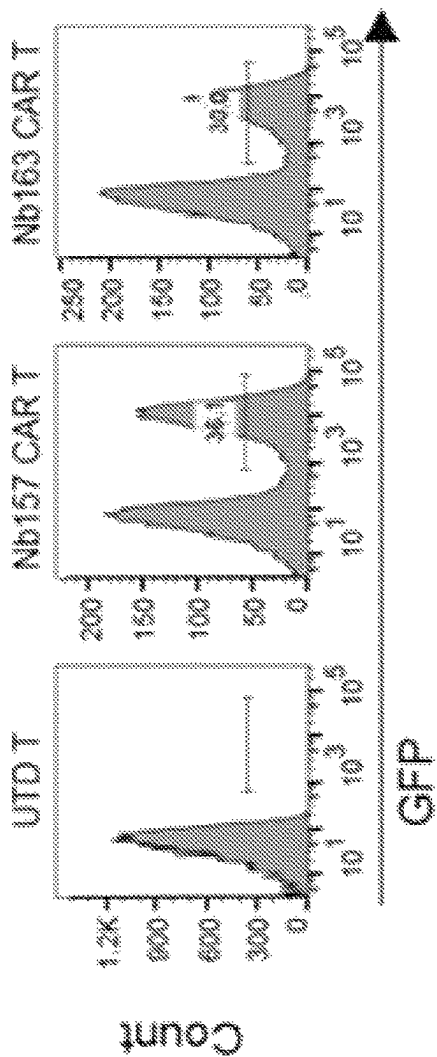


FIG. 8B

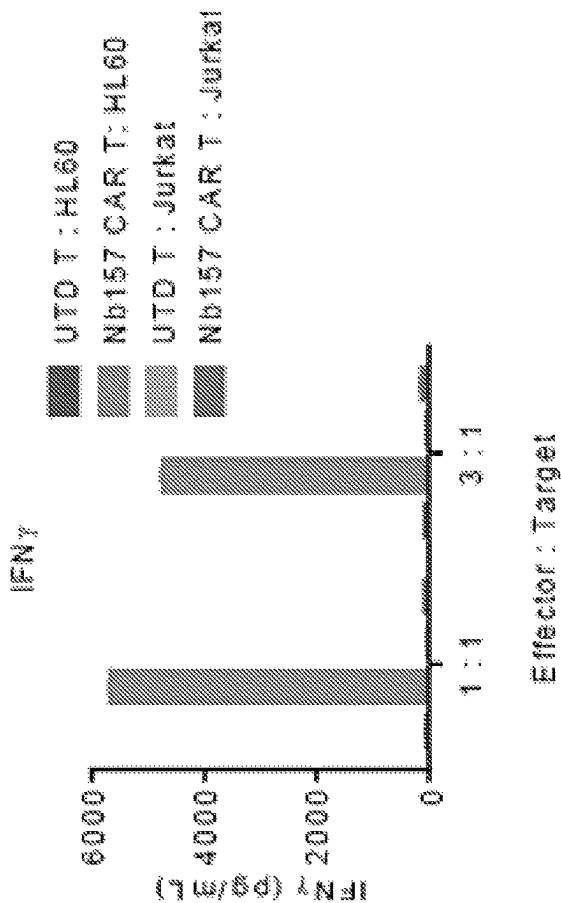


FIG. 8C

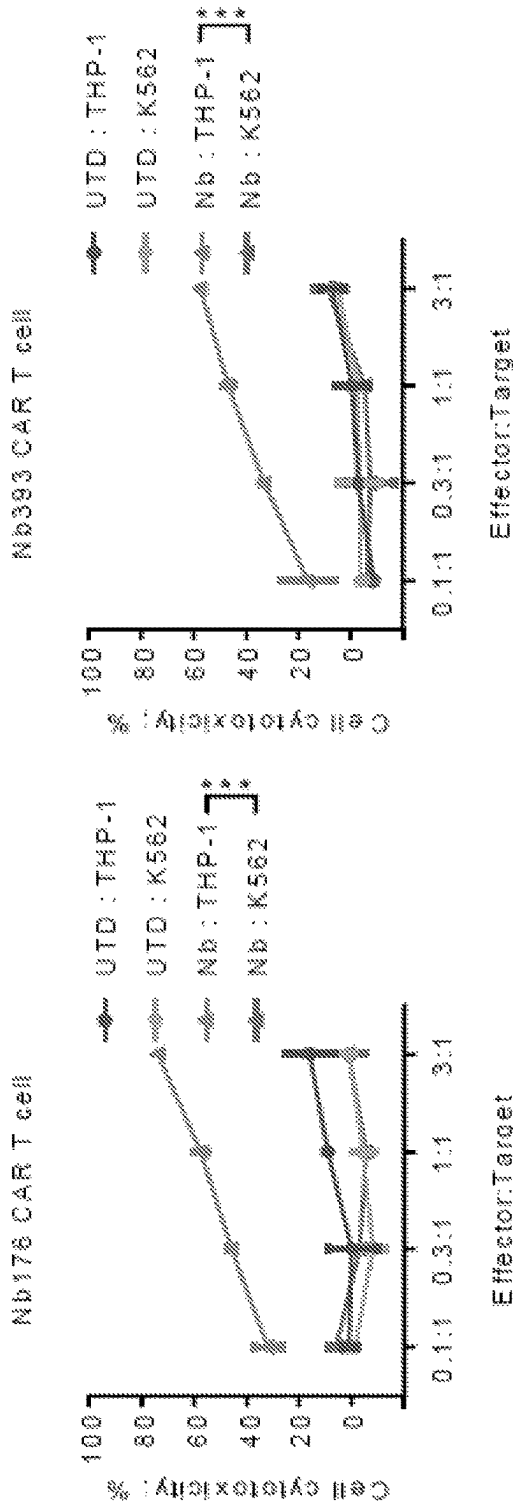


FIG. 9A

FIG. 9B

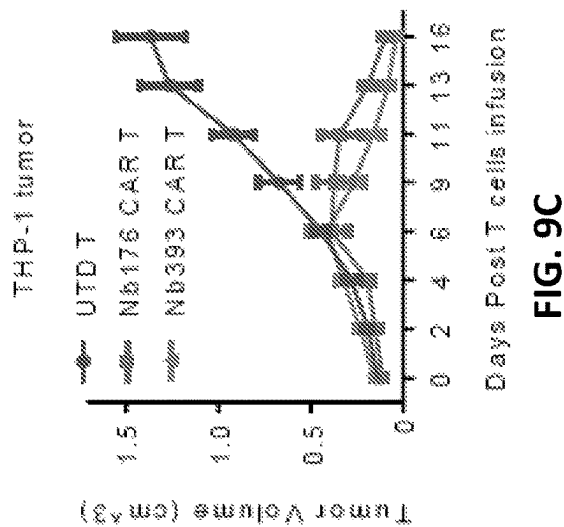


FIG. 9C

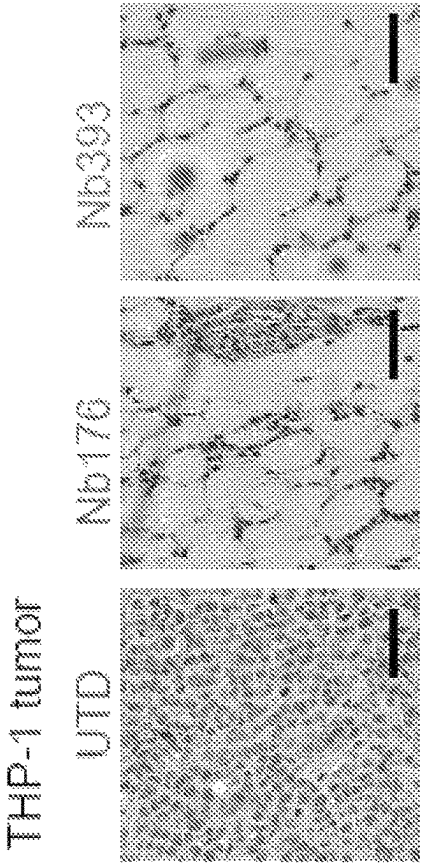


FIG. 9E

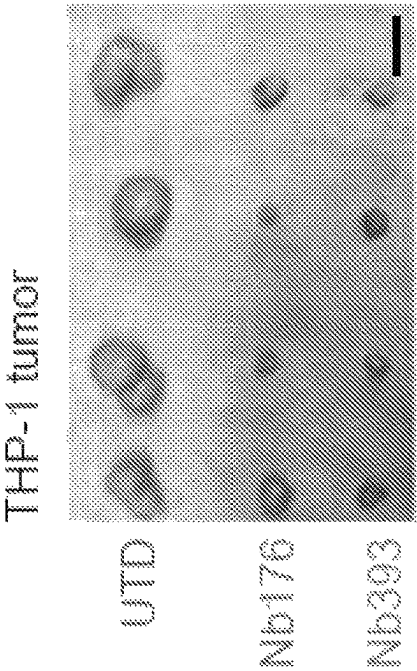
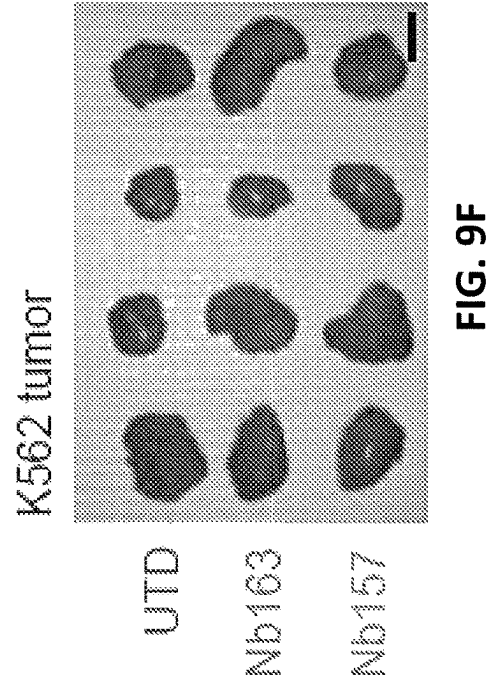
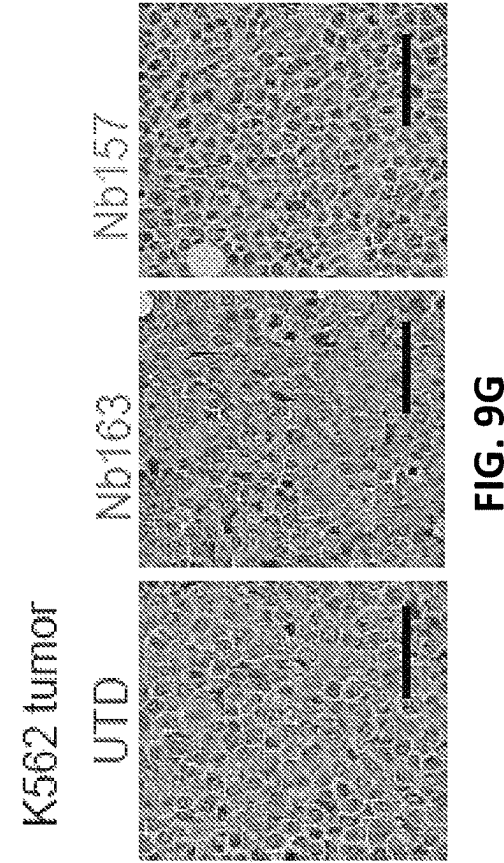


FIG. 9D



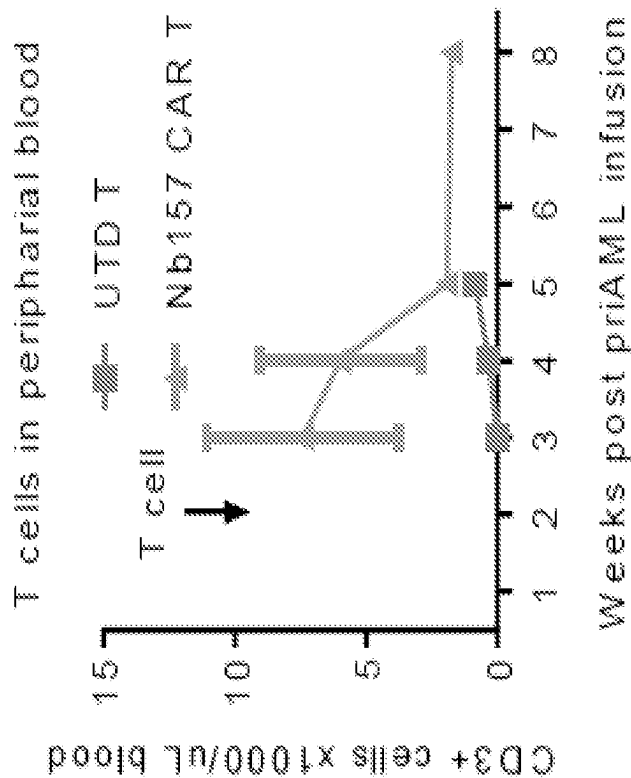


FIG. 10B

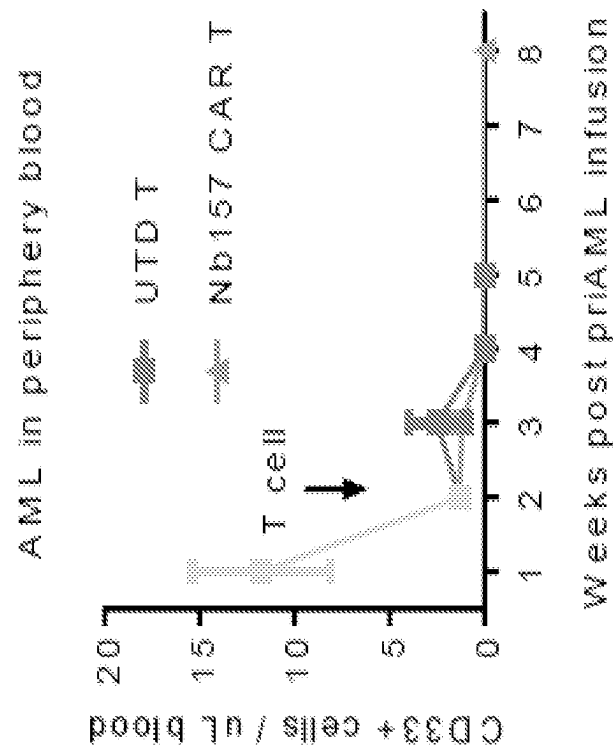
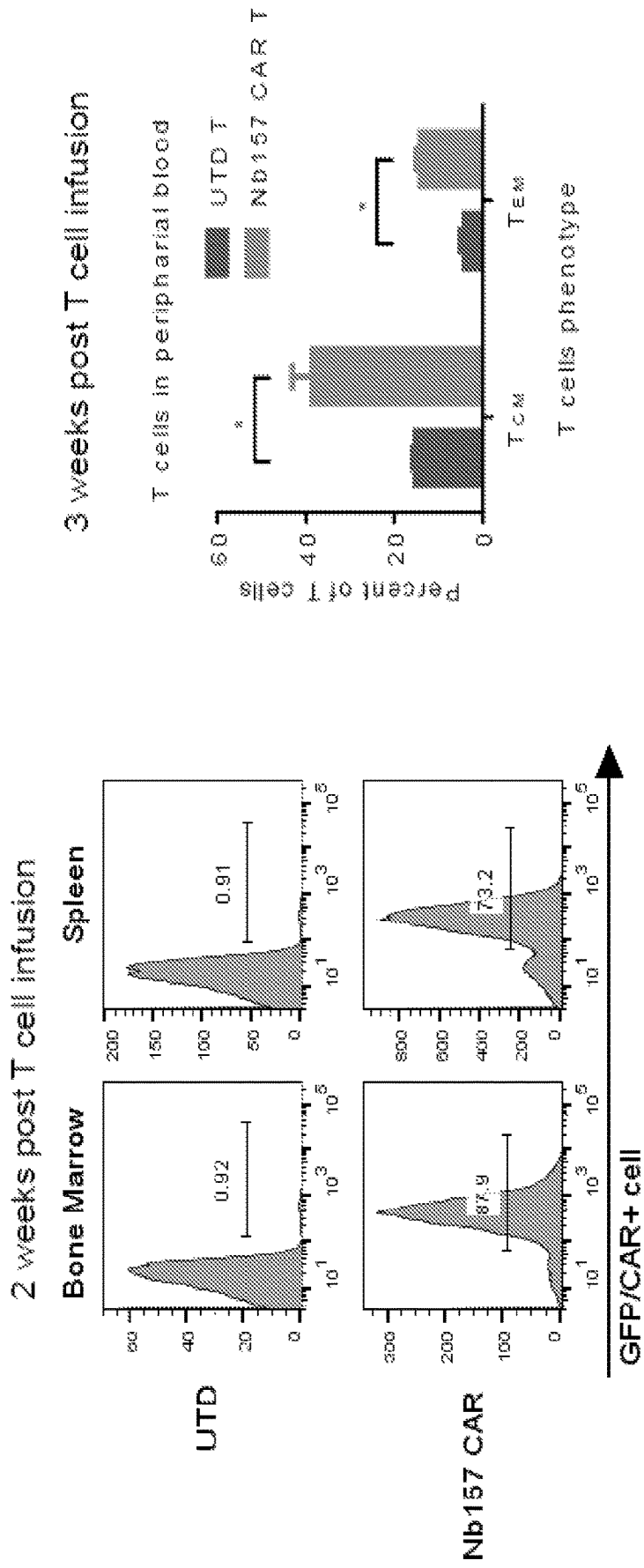


FIG. 10A



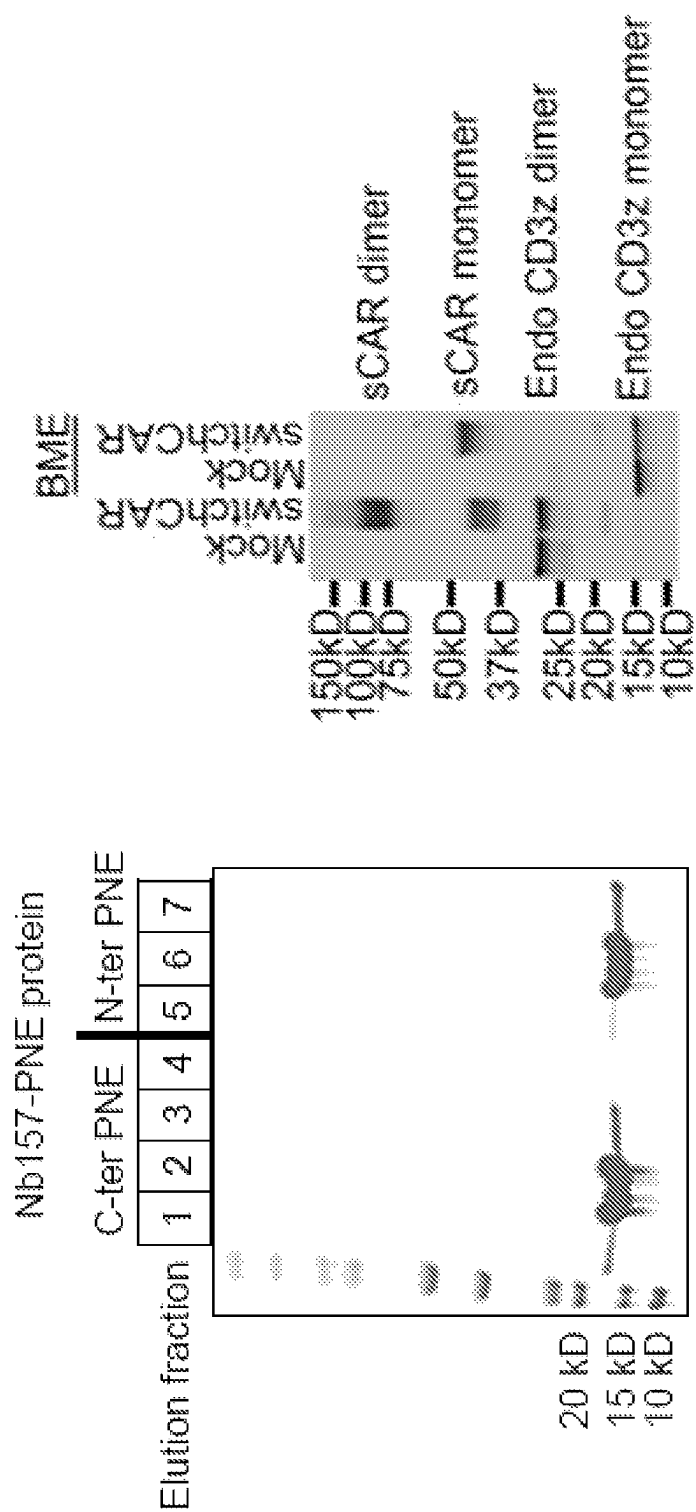
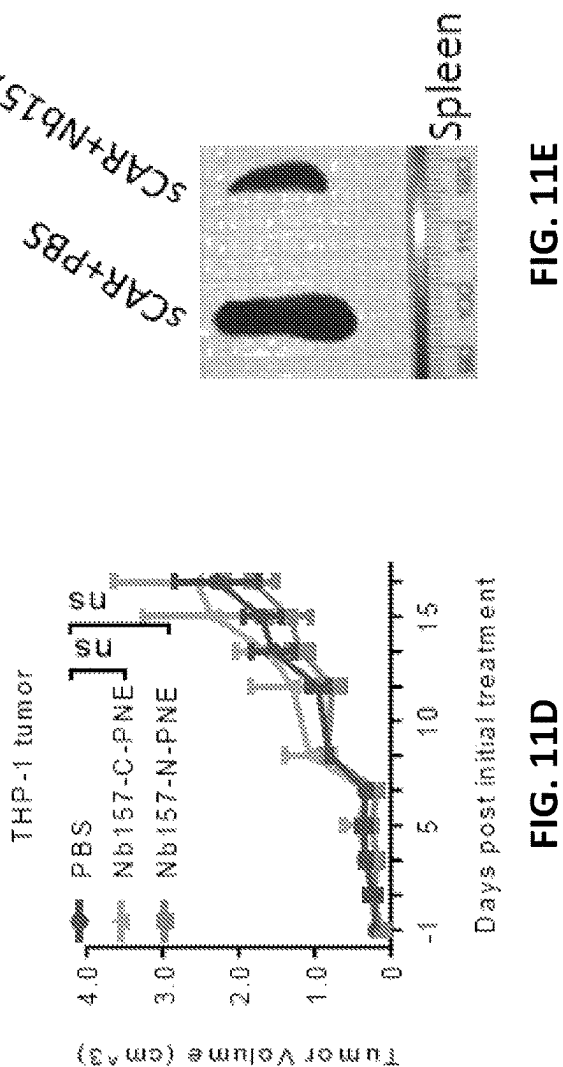
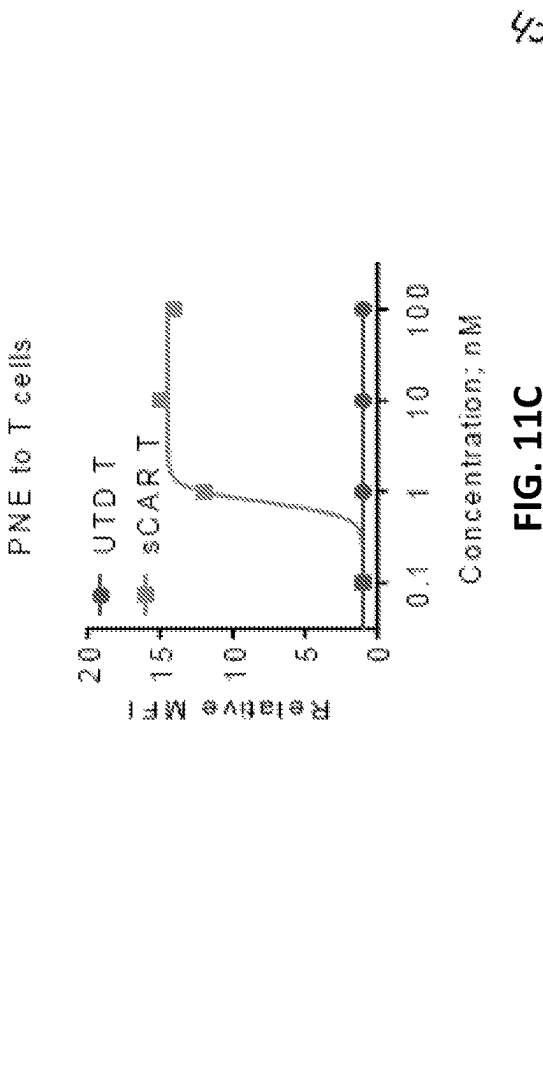


FIG. 11A

FIG. 11B



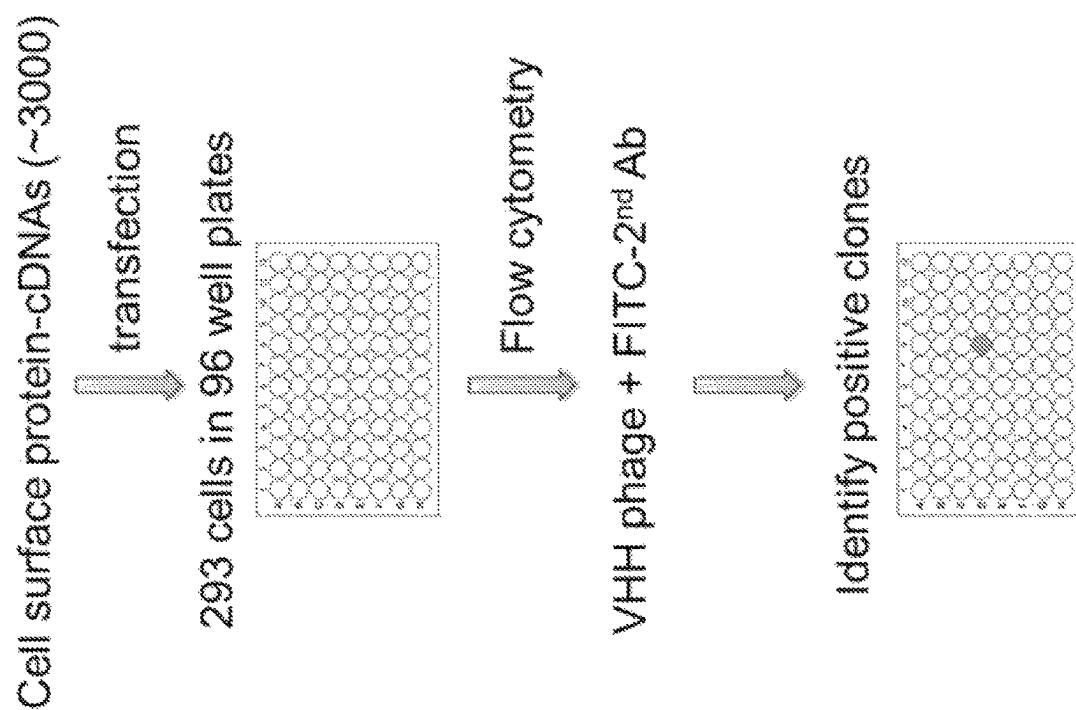
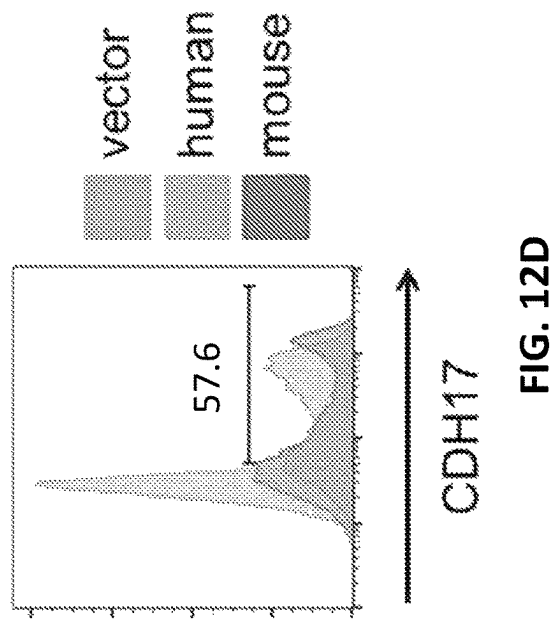
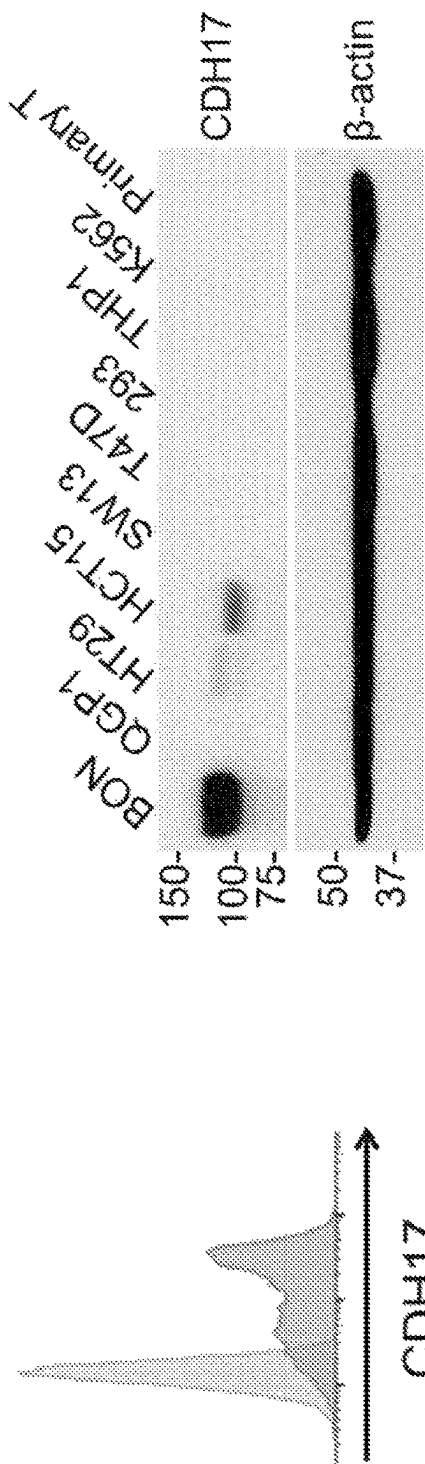


FIG. 12A



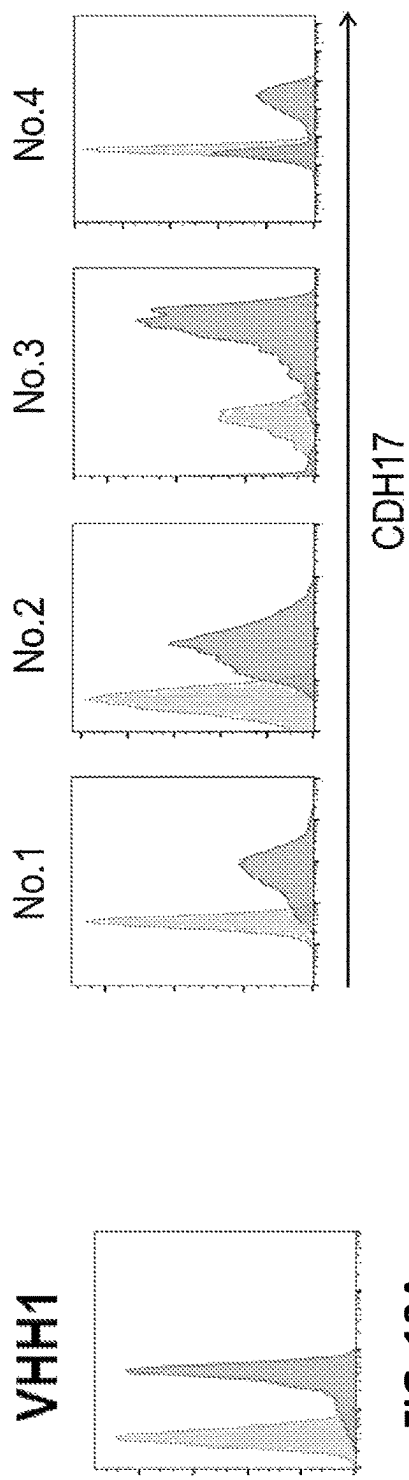


FIG. 13B

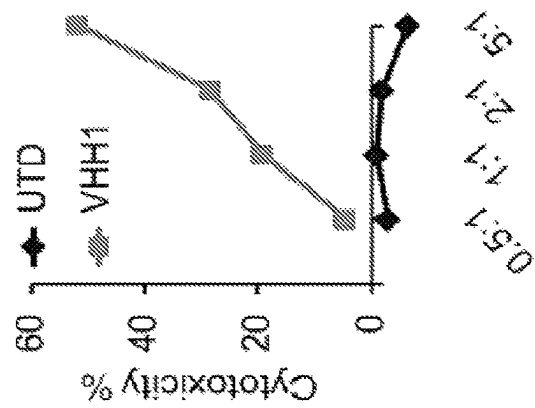


FIG. 13C

DNA seq (SEQ ID NO: 1)

CAGGTGCAGCTGCAGGAGTCTGGGGAGGATTGGTGCAGCCTGGGGCTCGACAAGACT
CTCCTGTGTATCGTCCCGCACCTTTAGTTATTATGACATGGGCTGGTTCCGCCAGGCTC
CAGGGAAGGAGCGTGAGTTCGTAGCACTGCTTAGTTGGAATGGGAAATGCAGAGTAT
TCAGACTCCGTGATGGGCCGTTTCACCGTCTCCCGAGGGAATACCCAGAAATTCGGTGAA
TCTGCAAAATGAACAACCTGAACCTGAGGACACGGGCATCTATTACTGCCAGTGACGC
ACGGTGGAGCGGTCGGTTCGTTCC TGGGCCAGGGACCCAGGTACCGTCTCCTCA

Protein seq (SEQ ID NO: 2)

QVQLQESGGGLVQPGGSTRLSGVSSRTFSYYDMGWERQAPGKEREFVALLSWNGENAEY

CDR1

CDR2

CDR3

SDSVMGRFTVSRGNTQNSVNLQMNNLKPEDTGLYYCAVTHGGARSVRSWGQGTQVTVSS

Specific amino acids differences between VH and VHH

FIG. 14

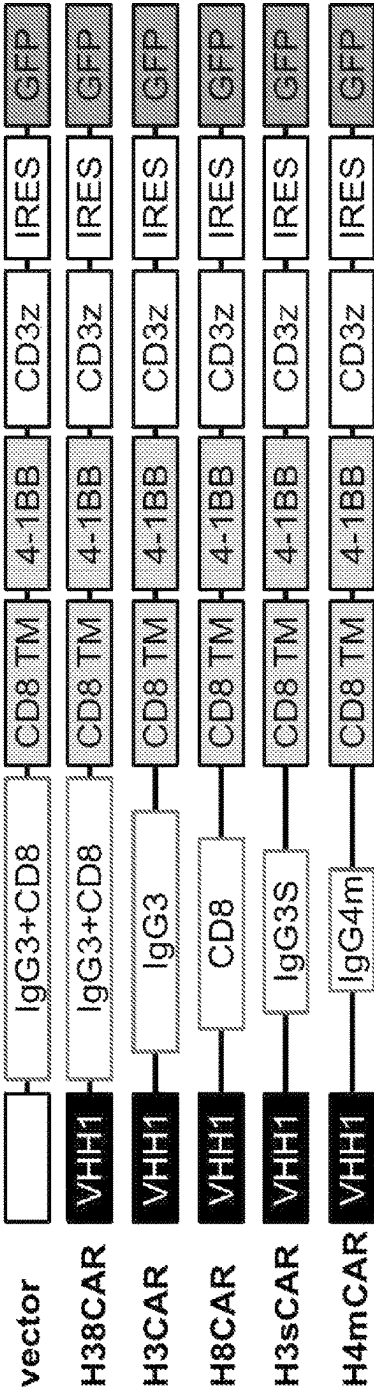


FIG. 15A

Hinge	Length (AA)
IgG3+CD8	107
IgG3	62
CD8	45
IgG3S	32
IgG4m	12

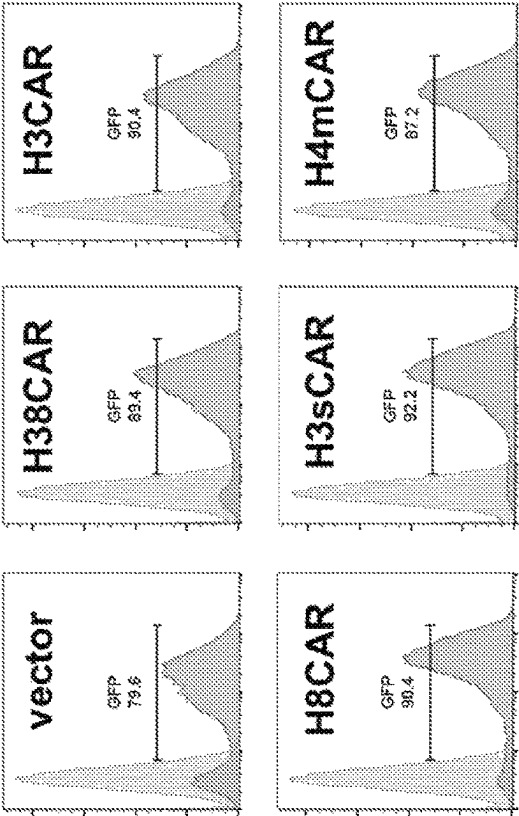


FIG. 15B

BON

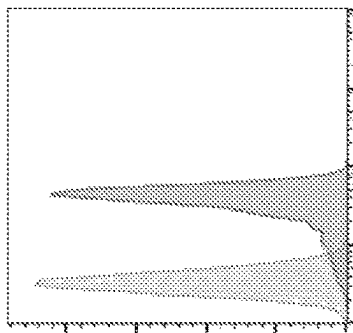


FIG. 16A

BT474

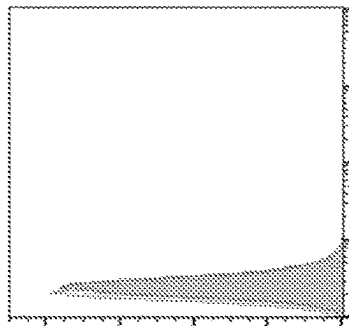


FIG. 16C

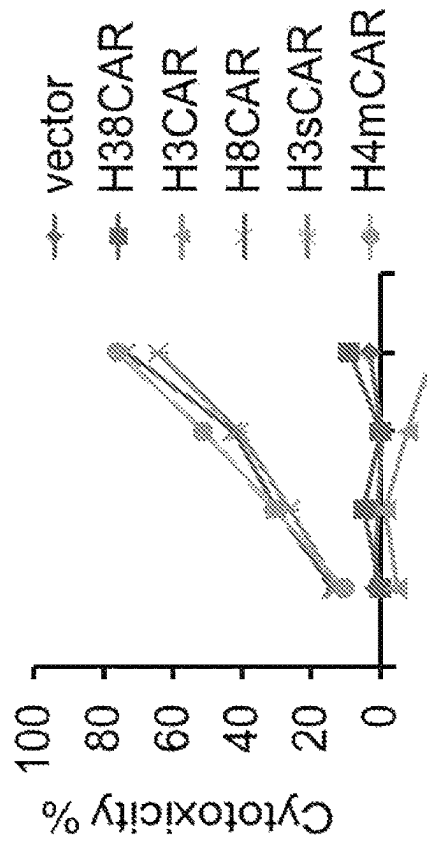


FIG. 16B

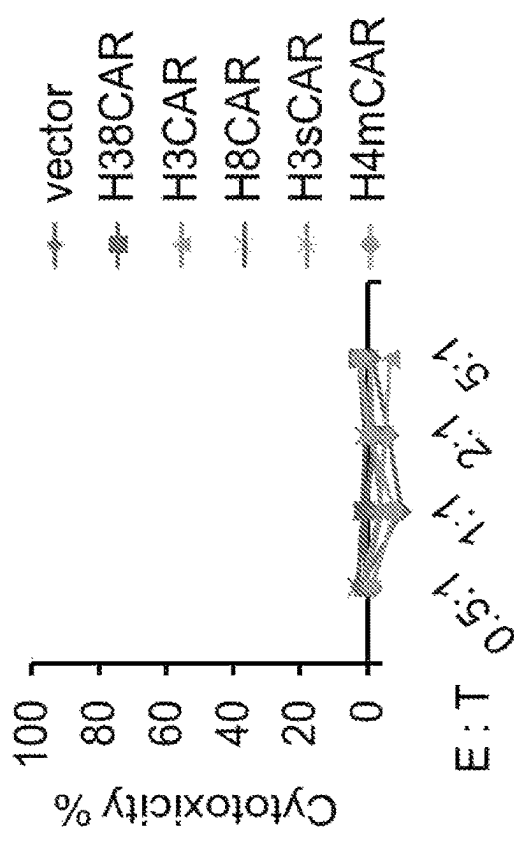


FIG. 16D

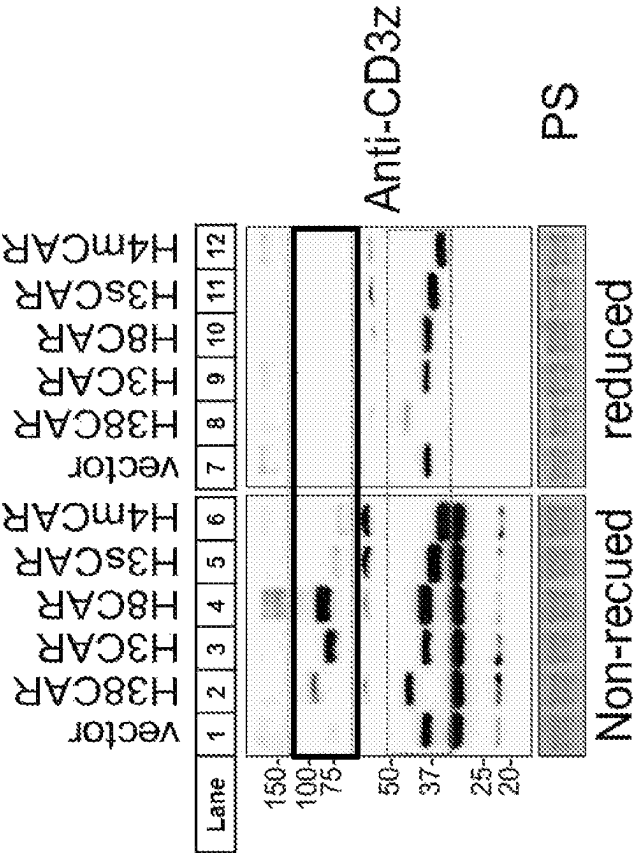


FIG. 16E

Killing	VHH1-CAR	Length of hinge (AA)
No	vector	107
No	H38CAR	107
No	H3CAR	62
Yes	H8CAR	45
Yes	H3sCAR	32
Yes	H4mCAR	12

FIG. 16F

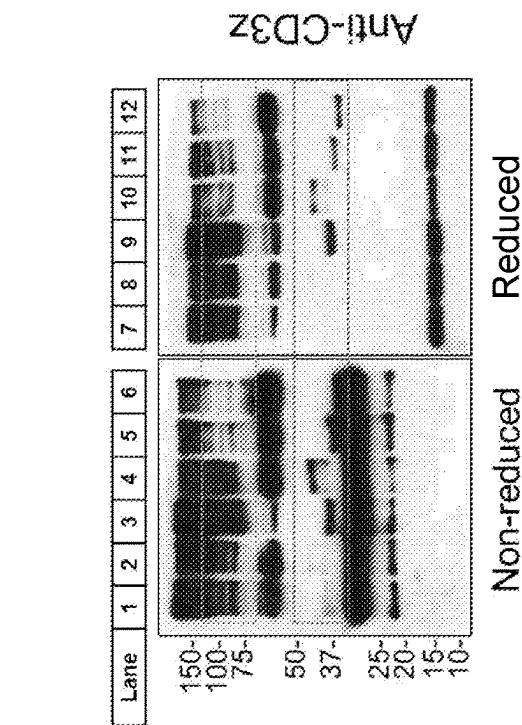


FIG. 17A

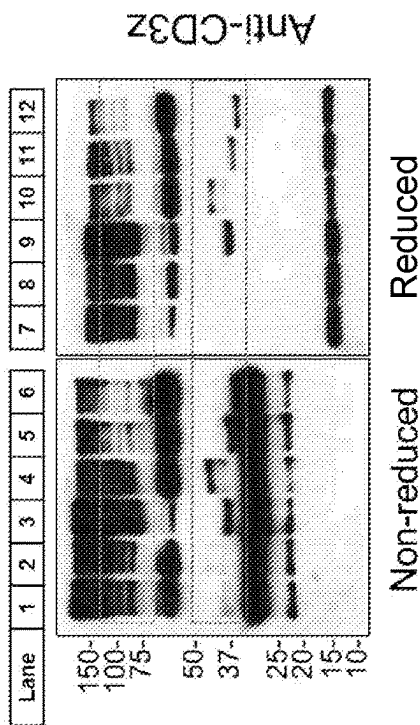


FIG. 17D

Killing	VHH1-CAR	Length of hinge (AA)
No	vector	107
No	H38CAR	107
No	H3CAR	62
Yes	H8CAR	45
No	H3sCAR	32
Yes	H4mCAR	12

FIG. 17E

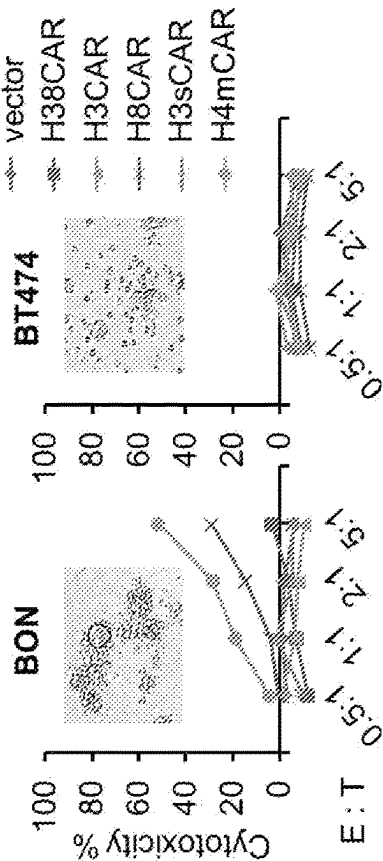


FIG. 17B

FIG. 17C

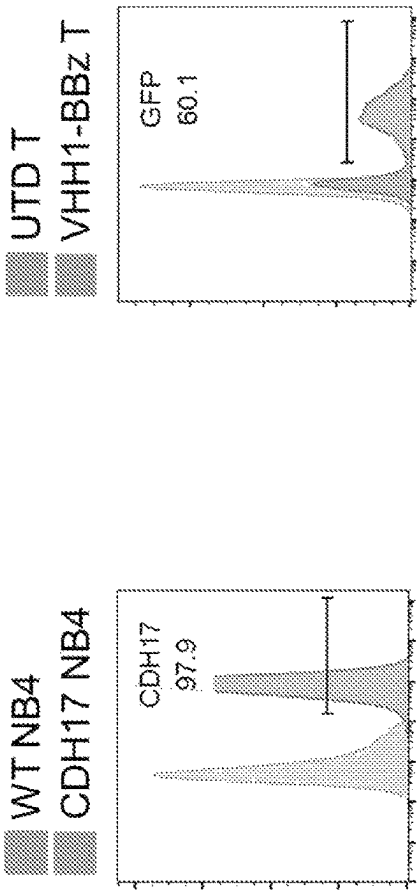


FIG. 18B

FIG. 18A

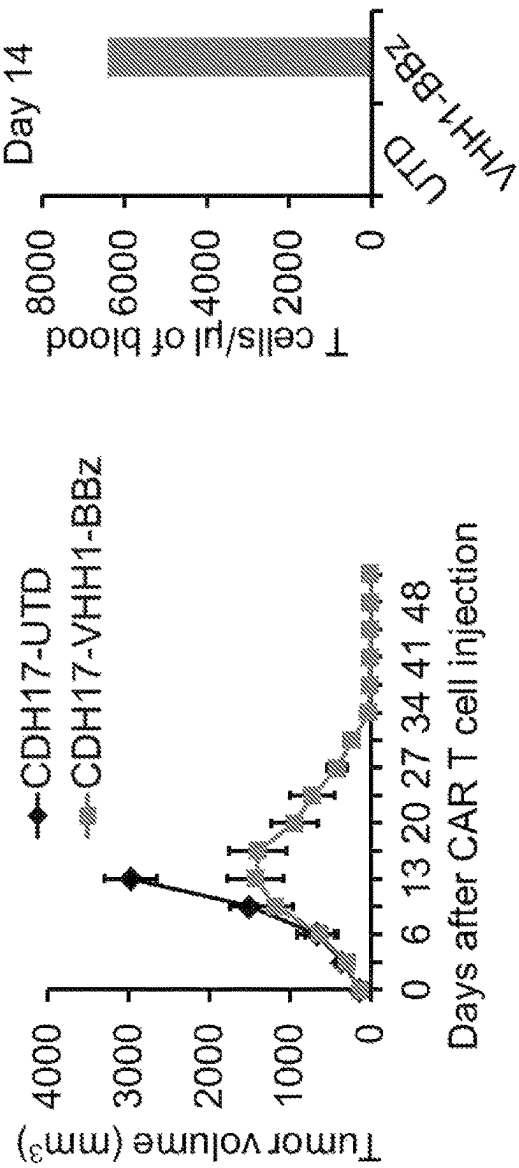


FIG. 18D

FIG. 18C

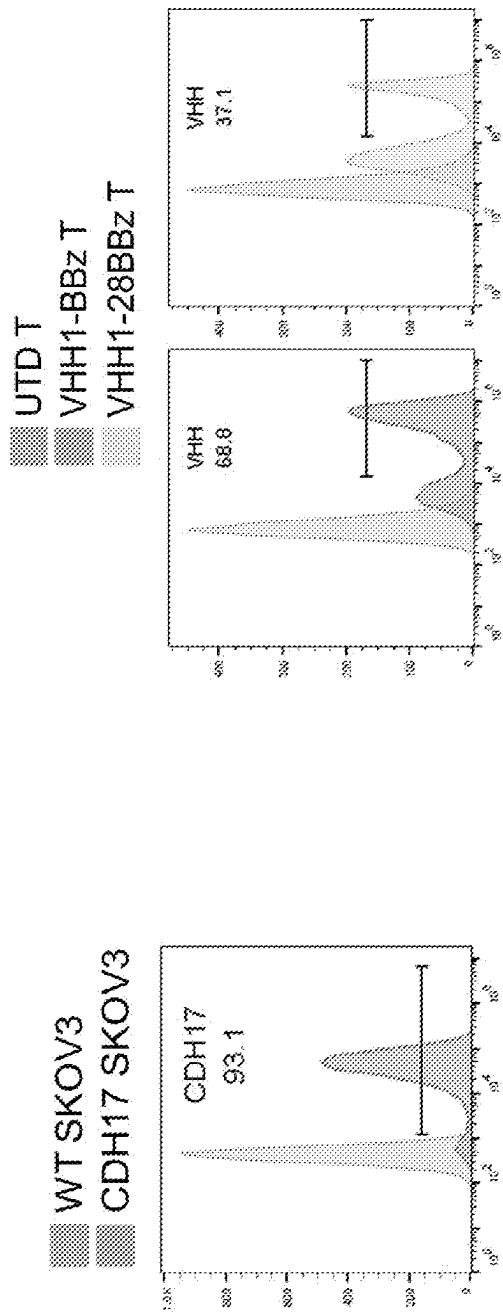


FIG. 19B

FIG. 19A

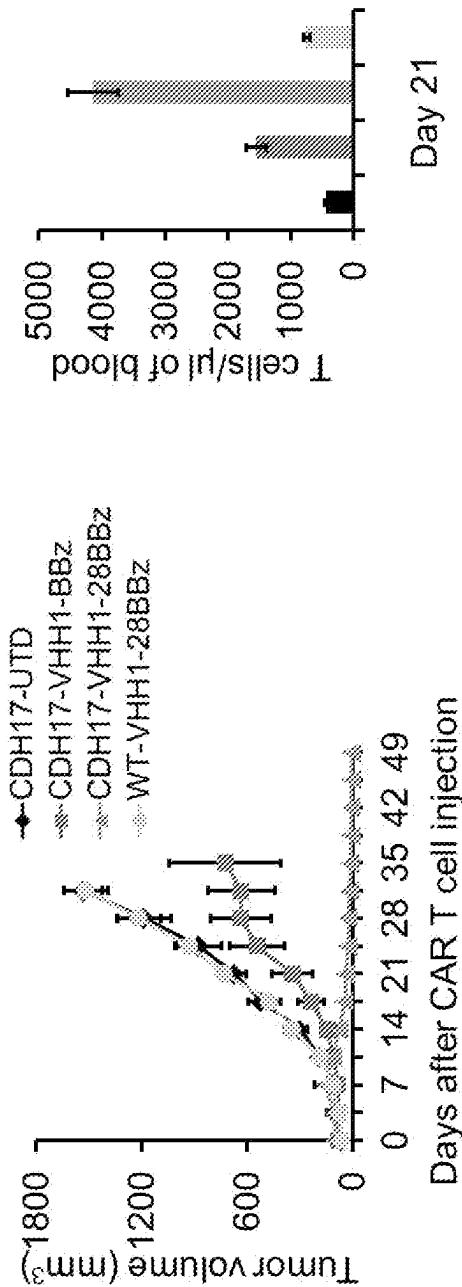


FIG. 19C

FIG. 19D

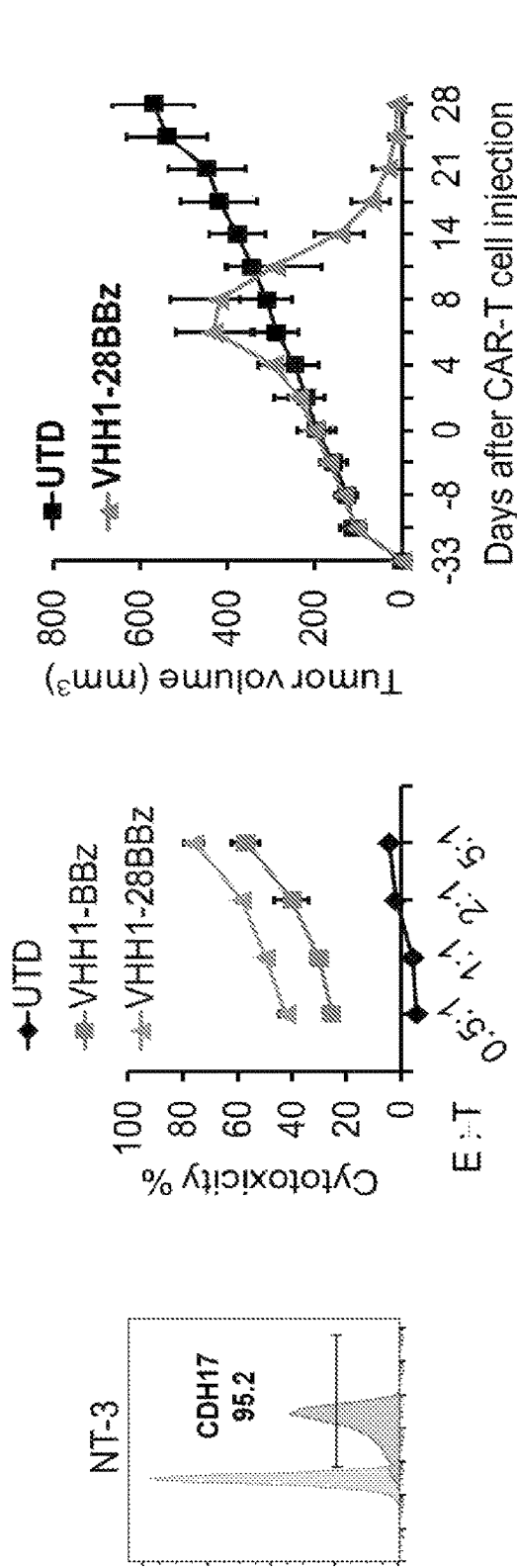


FIG. 20C

FIG. 20B

FIG. 20A

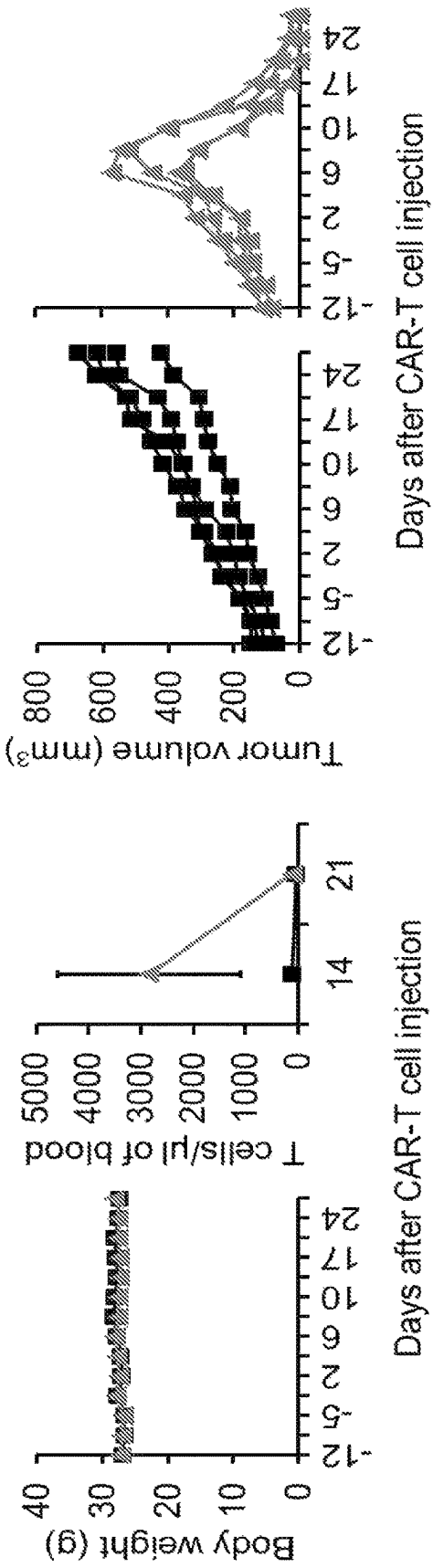


FIG. 20F

FIG. 20E

FIG. 20D

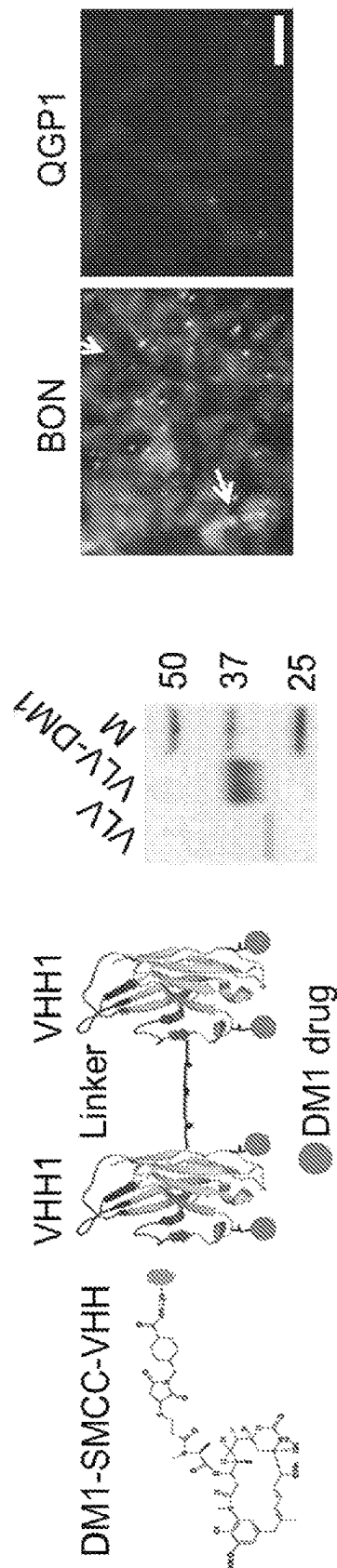


FIG. 21A

FIG. 21B

FIG. 21C

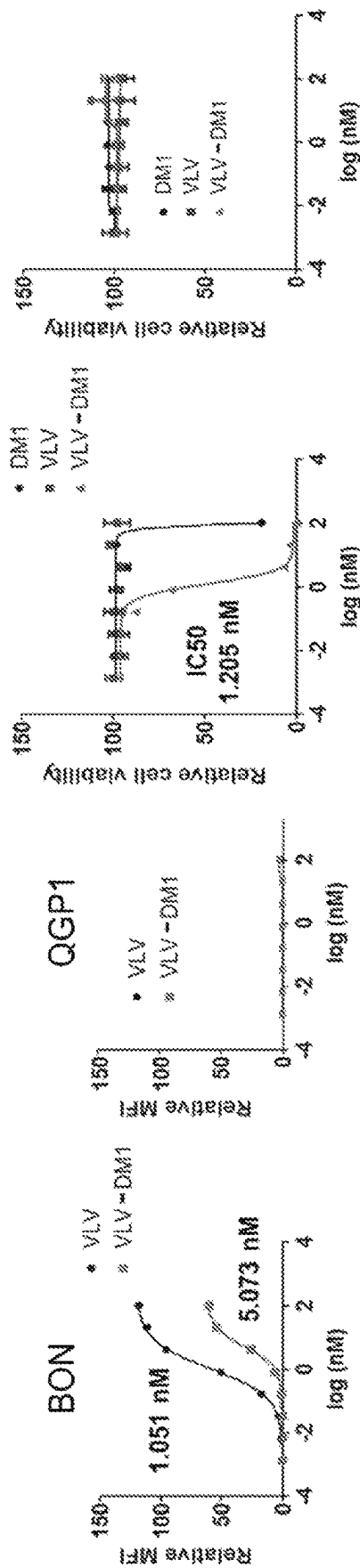


FIG. 21D

FIG. 21E

FIG. 21F

FIG. 21G

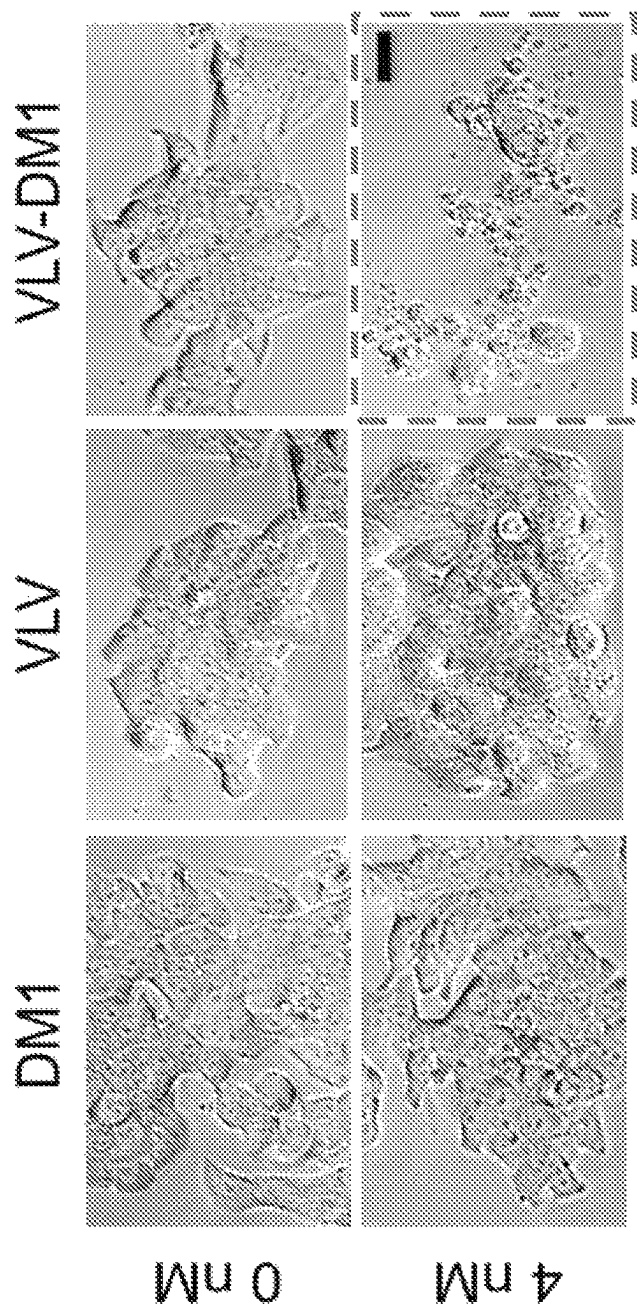


FIG. 21H

CDH17 truncations

Full	SP	EC1	EC2	EC3	EC4	EC5	EC6	EC7	TM	Cytoplasmic
Del-1	SP		EC2	EC3	EC4	EC5	EC6	EC7	TM	Cytoplasmic
Del-2	SP	EC1		EC3	EC4	EC5	EC6	EC7	TM	Cytoplasmic
Del-3	SP	EC1	EC2		EC4	EC5	EC6	EC7	TM	Cytoplasmic
Del-4	SP	EC1	EC2	EC3		EC5	EC6	EC7	TM	Cytoplasmic
Del-5	SP	EC1	EC2	EC3	EC4		EC6	EC7	TM	Cytoplasmic
Del-6	SP	EC1	EC2	EC3	EC4	EC5		EC7	TM	Cytoplasmic
Del-7	SP	EC1	EC2	EC3	EC4	EC5	EC6		TM	Cytoplasmic

FIG. 22A

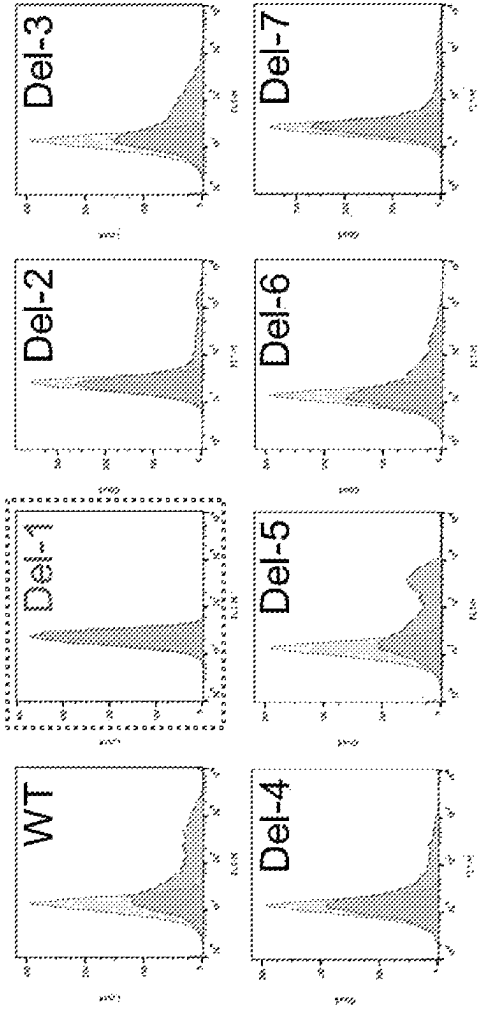


FIG. 22B

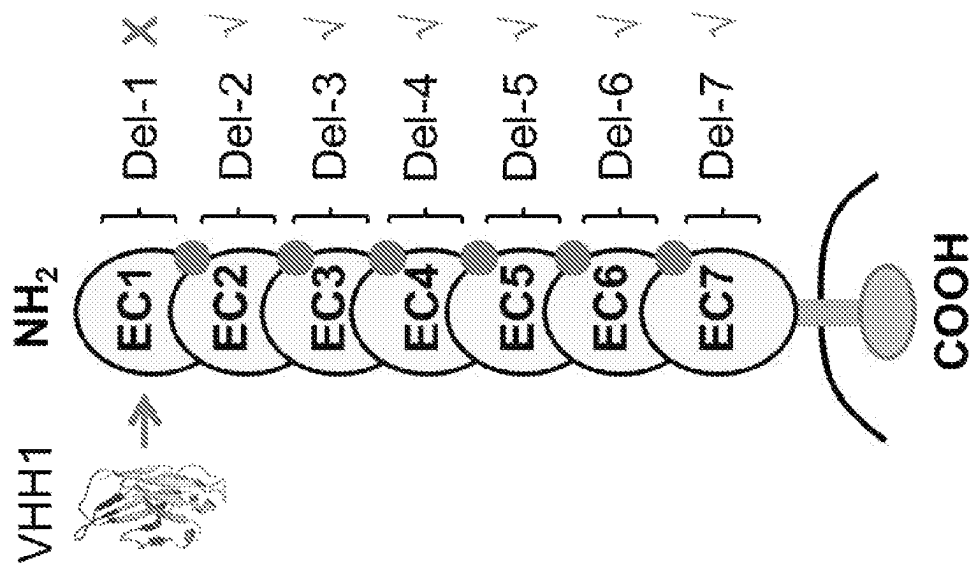


FIG. 22C

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COMPOSITIONS AND METHODS FOR RETRIEVING TUMOR-RELATED ANTIBODIES AND ANTIGENS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a 35 U.S.C. § 371 national phase application from, and claims priority to, International Application No. PCT/US2019/029333, filed Apr. 26, 2019, and published under PCT Article 21(2) in English, which is entitled to priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/663,074 filed Apr. 26, 2018, all of which applications are incorporated herein by reference in their entireties.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

This invention was made with government support under CA178856 awarded by the National Institutes of Health. The government has certain rights in the invention.

BACKGROUND OF THE INVENTION

Cancer immunotherapy has made striking progress and changed the course of cancer therapy. Adoptive T cell cancer therapy (ACT) using chimeric antigen receptor (CAR)-expressing T cells can eradicate relapsed or refractory B-cell lymphoma or B-cell lymphocytic leukemia through targeting CD19. The CAR construct has an ectodomain, generally consisting of a single-chain variable fragment (scFv) derived from a monoclonal antibody (mAb), anchored to the cells via a transmembrane domain, followed by the intracellular costimulatory 4-1BB and/or CD28 endodomains, and CD3 zeta signaling domain. CAR T cells act like “serial killers” to eliminate the cancer cells. Despite the remarkable success in CAR T therapy targeting the B cell-specific marker CD19, this success has not yet been applied to other types of cancers such as acute myeloid leukemia (AML) and neuroendocrine tumors (NETs). One big hurdle in expanding the CAR T approach to achieve this goal, among other factors such as the repressive microenvironment in solid tumors, is often lack of choices of the mAbs that can both bind the TAAs and enable CAR T cells to eliminate the targeted cancer cells.

The extracellular domain of a cell surface protein with cancer-specific mutations or overexpression could be targeted by CAR T technology. However, besides the difficulties in identifying the remedial marker on tumor cell surface, the availability of mAbs suitable for developing CAR T therapy against many potential targets is very limited. Moreover, many mAbs, when incorporated into the CAR, are not capable of endowing T cells with cytotoxicity, which requires the appropriate engagement between the T cell and target cell to elicit a productive immunological synapse to kill the cancer cells. Furthermore, the high heterogeneity within cancers and high homogeneity between tumors and normal tissues make single antibody/target-based therapy even less impressive. Therefore, it is imperative to effectively generate diverse antibodies that can redirect CAR T cells to specifically kill the cancer cells. Conventional antibodies cannot bind certain antigen surfaces due to the large size of the tetrameric variable heavy chain and light chain (VH and VL) in an antibody, coupled with the possible challenge in generating the optimal scFv. In this regard, the camelid family of animals like llamas can produce heavy

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chain-only antibodies (VHH), with the small size of 15 kD in the single domain (aka nanobodies or Nbs), that bind various epitopes including small cavities. To expedite the development of CAR T cells targeting tumor cell surface proteins systematically, it is ideal to generate numerous tumor-associated and CAR-compatible mAbs and to identify the recognized antigens, which would both expand the CAR T choices and uncover previously unappreciated cell surface antigens/targets to develop potent cancer immunotherapy. However, such a system remains to be established.

Chemotherapy-resistant Acute Myeloid Leukemia (AML) is highly aggressive with few choices of effective therapy, and thus faces poor prognosis. CAR T cells targeting CD33, a cell surface lectin, and CD123, a subunit of IL3 receptor, were tested for suppressing AML in clinical relevant models, but the clinical application was hindered by the side effects on normal hematopoietic stem cells (HSC) and other normal tissues.

A need exists for methods of isolating potent cancer killing or diagnostic antibodies. More specifically, there is an urgent need to develop potent antibodies against AML-specific and NET-specific surface targets to improve AML and NET therapy without causing devastating side effects. The present invention satisfies these needs.

SUMMARY OF THE INVENTION

In one aspect, the invention includes a chimeric antigen receptor (CAR) comprising a nanobody, a transmembrane domain, and an intracellular signaling domain.

In another aspect, the invention includes a switchable CAR system comprising a nanobody fused to a peptide neoepitope (PNE) molecule and a CAR. The CAR comprises a PNE-specific scFv, a transmembrane domain, and an intracellular signaling domain.

In yet another aspect, the invention includes a modified T cell or precursor thereof, comprising any of the CARs or switchable CAR systems disclosed herein.

In still another aspect, the invention includes a composition comprising an antibody drug conjugate (ADC) comprising a nanobody conjugated to a drug or a toxin or a radioisotope.

Another aspect of the invention includes a method for treating cancer in a subject in need thereof. The method comprises administering to the subject a modified T cell or precursor thereof comprising any of the CARs, or switchable CAR systems, or compositions comprising an ADC disclosed herein.

Yet another aspect of the invention includes a method for generating a plurality of tumor-specific and CAR T cell-compatible nanobodies. The method comprises immunizing a camelid animal with a tumor cell line, isolating PBMCs from the animal, performing phage display, selecting the tumor specific nanobodies, inserting the selected nanobodies into a CAR expressing vector thereby generating a nanobody CAR library, transducing human primary T cells with the nanobody CAR library, injecting the library into an animal and selecting the nanobodies that cause T cell enrichment in the tumor in vivo.

In various embodiments of the above aspects or any other aspect of the invention delineated herein, the nanobody is retrieved by a sequential tumor-related antibody and antigen retrieving (STAR) method.

In certain embodiments, the nanobody specifically binds to CD13. In certain embodiments, the nanobody specifically binds to CDH17. In certain embodiments, the nanobody specifically binds to the first domain of CDH17.

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In certain embodiments, the nanobody is selected from the group consisting of VH157, VH163, and VHH1.

In certain embodiments, the nanobody comprises the amino acid sequence of any one of SEQ ID NOs: 2, 19, or 24. In certain embodiments, the nanobody is encoded by the nucleotide sequence of SEQ ID NO: 1. In certain embodiments, the nanobody comprises a CDR1 region comprising the amino acid sequence of any one of SEQ ID NOs: 3, 25, or 28. In certain embodiments, the nanobody comprises a CDR2 region comprising the amino acid sequence of any one of SEQ ID NOs: 4, 26, or 29. In certain embodiments, the nanobody comprises a CDR3 region comprising the amino acid sequence of any one of SEQ ID NOs: 5, 27, or 30.

In certain embodiments, the CAR further comprises a hinge domain selected from the group consisting of a CD8 hinge, an IgG3s hinge, and an IgG4m hinge. In certain embodiments, the hinge domain comprises the amino acid sequence of any one of SEQ ID NOs: 20 or 38.

In certain embodiments, the transmembrane domain is selected from the group consisting of CD8, CD28, and ICOS. In certain embodiments, the transmembrane domain comprises SEQ ID NO: 21.

In certain embodiments, the intracellular signaling domain comprises 4-1BB and CD3 zeta. In certain embodiments, the intracellular signaling domain comprises SEQ ID NO: 22.

In certain embodiments, the CAR comprises the amino acid sequence of any one of SEQ ID NOs: 17, 23, 34 or 36. In certain embodiments, the CAR is encoded by the nucleotide sequence of any one of SEQ ID NOs: 33 or 35.

In certain embodiments, the nanobody is fused to the N-terminal region of the PNE. In certain embodiments, the nanobody is fused to the C-terminal region of the PNE.

In certain embodiments, the drug or toxin or radioisotope is selected from the group consisting of maytansinoid (DM1), SSTR2-binding octreotide, a toxin, paclitaxel, auristatin, MMAE, MMAF, dauxrubicin, duocarmycin A, 5-fluoruracil, methotrexate, tubulin polymerization inhibitors, ravtansine (DM4), Ricin A, 90Y, 177Lu, and 111In.

In certain embodiments, the ADC comprises nanobody VHH1 conjugated to DM1. In certain embodiments, the ADC comprises a first VHH1 nanobody linked to a second VHH1 nanobody (VLV) conjugated to DM1.

In certain embodiments, the cancer is acute myeloid leukemia (AML). In certain embodiments, the cancer is a neuroendocrine tumor (NET). In certain embodiments, the cancer is colorectal cancer.

BRIEF DESCRIPTION OF THE DRAWINGS

The following detailed description of specific embodiments of the invention will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there are shown in the drawings exemplary embodiments. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities of the embodiments shown in the drawings.

FIGS. 1A-1C are series of images and graphs depicting the steps for generating nanobodies that both differentially bind tumor cells and empower CAR T cells to kill the tumor cells. FIG. 1A: Flow chart of AML specific CAR compatible nanobodies in vivo screening. Llama was immunized with AML cell line THP-1. Nanobody library was generated from the llama PMBC including B cells by molecular cloning. Two rounds of conventional cell-based phage display were

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applied, which took T-ALL cell line Jurkat or chronic myelogenous leukemia (CML) cell line K-562 as negative absorption, followed by one round of count-selection. The resultant THP-1 specific nanobodies were inserted into a chimeric antigen receptor expressing lenti-vector to generate Nb-CAR library. Human primary T cells were transduced by the Nb-CAR library and injected into THP-1 or K562 borne NSG mice to perform the in vivo selection. Nbs that can adapt T cells to enrich in the tumor were PCR-amplified and sequenced. FIG. 1B: 10 million THP-1 cells or 5 million K562 cells were transplanted into NSG mice subcutaneously, followed by treatment of untransduced (UTD) T cells or Nb-lib-CAR T cells. 2 weeks later, Nbs from tumor infiltrated T cells were isolated and identified by PCR amplification. FIG. 1C: Top 5 frequent Nbs in the THP-1 tumor were listed and tested for their recognition to the THP-1 cells, Jurkat cells, or K-562 cells.

FIGS. 2A-2K are series of a diagram and graphs depicting the finding that all the nanobodies isolated by the STAR system empower CAR T cells to potentially kill AML cells in vitro. FIG. 2A: Schematic diagram of Nb CAR structure, including signal peptide (SP), IgG4 mutant (IgG4m) hinge, CD8 transmembrane domain (TM), 4-1BB and CD3zeta domain. FIGS. 2B-2D: Nb CAR showed potent and specific cytotoxicity against THP-1 or HL60 cells in a dose dependent manner, but not to K562 or Jurkat cells. Untransduced (UTD) T cells did not exert obvious killing. (One-way ANOVA: *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$, ns, $p > 0.05$). FIGS. 2E-2F: THP-1 cells stimulated Nb157 or Nb163 CAR T cells, but not UTD T cells, to release cytokines including $\text{TNF}\alpha$ and $\text{IFN}\gamma$. FIG. 2G: Only THP-1 cells induced the Nb157 or Nb163 CAR T cells to degranulate, i.e. CD107a localization to the cell membrane, after 4 hour co-culture. FIGS. 2H-2K: THP-1 cells potentially stimulated the Nb157 or Nb163 CAR T cells to proliferate after 4 days co-culture.

FIGS. 3A-3F are series of images and graphs demonstrating that nanobody-adapted CAR T cells potentially suppress AML tumors in vivo. FIGS. 3A-3B: 10 million THP-1 cells were transplanted into NSG mice subcutaneously. The tumor reached 150 mm³ after about 14 days. 3 million Nb157, Nb163 CAR or UTD T cells were injected intravenously into the mice separately. The tumor engraftments were monitored every other day. $n=4$. Scale bar was 10 mm. FIG. 3C: Hematoxylin and eosin stain of THP-1 xenograft after Nb157, Nb163 CAR or UTD T cells treatment. Scale bar was 100 μm . FIG. 3D: 3 million Nb157 CAR or UTD T cells were injected intravenously into NSG mice bearing HL60 tumor separately. The tumor engraftments were monitored every other day. ($n=4$. One-way ANOVA: ***, $p < 0.001$). FIG. 3E: 5 million K562 cells were transplanted into NSG mice subcutaneously. The tumor reached 150 mm³ after about 10 days. 3 million Nb157, Nb163 CAR or UTD T cells were injected into NSG mice separately. The tumor engraftments were monitored every other day. ($n=4$. One-way ANOVA: ns, $p > 0.05$). FIG. 3F: 1.5 million Nb157, Nb163 CAR or UTD T cells were injected intravenously into NSG mice bearing THP-1 tumor separately. The tumor engraftments were monitored every other day until the tumors were gone completely. $n=4$.

FIGS. 4A-4F are series of images and graphs illustrating the identification of CD13 as a target to kill AML cells by CAR T cells. FIG. 4A: Experiment schema: About 3000 cell membrane protein cDNAs were purified and transfected into HEK293T cells separately, followed by flow analysis with the nanobodies expressing phage and FITC-labeled secondary antibody against phage M13 protein. FIG. 4B: Flow analysis of nanobodies binding to the HEK293T cells with

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CD13 overexpression. FIG. 4C: CD13 cDNA expression confirmed in HEK293T cells by western blot. FIG. 4D: Western blot to confirm the gRNA/CRISPR guided CD13 knockout effect in THP-1 cells. Three independent gRNAs were transduced into THP-1 separately, followed by puromycin selection and single individual clone expansion. FIG. 4E: Flow analysis of Nb157 or Nb163 binding CD13 knockout THP-1 cells. FIG. 4F: Killing assay of Nbs CAR T cells to the THP-1 wild type (wt) or THP-1 cd13-ko.

FIGS. 5A-5K are series of images and graphs depicting that Nb157 CAR T cells exhibit antitumor activity in patient-derived AML cells in a NSG mouse model. FIGS. 5A-5B: Both Nb157 and Nb163 recognized the patient derived AML cells by flow analysis. FIGS. 5C-5D: Both Nb157 and Nb163 CAR T cells specifically killed the patient derived AML cells in vitro in a dose dependent manner. FIG. 5E: Nb157 CAR T cells efficiently prolonged the survival of NSG mice bearing patient derived AML. In brief, 20 million of Patient derived AML cells were injected into NSG mouse, followed by 3 million of Nb157 CAR T cells or UTD T cells treatment. The survival of mice was monitored. Each group n=10. FIGS. 5F-5I: Patient derived AML in NSG bone marrow and spleen were monitored after Nb157 CAR T cell treatment by staining with anti-human CD45/CD3/CD33, followed by flow cytometry analysis. FIGS. 5J-5K: At the end points of each group of experiment, the mice spleens were harvested and fixed by paraformaldehyde, followed by immunofluorescence staining of the anti-human CD3/CD33 and DAPI (Nuclear).

FIGS. 6A-6H are series of images and graphs depicting that Nb switch-activated CAR T cells eradicate AML in vivo in a controllable manner. FIGS. 6A-6B: Schematic diagram of switchable CAR T system. The Nb157 was expressed and purified by fusing with PNE, a fourteen amino acid peptide, either at N terminal or C terminal of the Nanobody. Primary human T cells were engineered by lentivirus transduction of scFv CAR that recognized PNE specifically. FIG. 6C: Test the affinity of Nb157, Nb157-N-PNE or Nb157-C-PNE against THP-1 cells by flow cytometry analysis. FIG. 6D: Killing assay of Nb157-PNE-mediated sCAR system against THP-1 cells in series of switch concentrations. The ratio of sCAR T to THP-1 was 2:1 and assay was performed after 16 hours incubation. FIG. 6E: 10 million THP-1 cells were transplanted into NSG mice subcutaneously to form the 150 mm³ of tumors. 5 million sCAR or UTD T cells were injected intravenously into NSG mice separately, followed by treatment of 0.25 mg/kg of Nb-C-PNE protein or PBS every other day. The engraftment volumes were monitored by measuring the lengths and widths of the tumors. FIG. 6F: sCAR potently improved the survival of NSG mice bearing the patient derived AML cells. Each group n=6. FIGS. 6G-6H: 20 million patient derived AML cells were injected into NSG mouse, followed by sCAR T cells treatment 2 weeks after AML infusion. Nb157-PNE or PBS were injected i.v. every other day for 7 weeks. The primary leukemia or CAR T cells in the mouse peripheral blood were monitored by staining with CD33 or CD3 fluorescence labeled antibodies.

FIG. 7 is series of a table and graphs depicting the affinity test of VHHs against variant of AML cell lines and patient derived AMLs. The four THP-1 specific Nbs were tested for their recognition to other AML cell lines. + was <0.5 log shift, ++ was <1.0 log shift, +++ was <1.5 log shift, - was no binding.

FIGS. 8A-8C are series of an image and graphs depicting the expression of the Nb CARs protein in human primary T cells. FIG. 8A: Western blot with anti-human CD3z to show

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the Nb CAR expression in primary T cells transduced by the lentivirus. Beta-Mercaptoethanol (BME) is suitable for reducing protein disulfide bonds, to make the dimer protein to monomer. FIG. 8B: Cytometry flow showed GFP expression in primary T cells transduced with Nb CAR lentivirus. FIG. 8C: HL60 cells, not Jurkat cell, stimulated Nb157 CAR T cell to secrete IFN-gamma. Primary T cells were activated and transduced by the Nb157 CAR lentivirus, followed by incubation with HL60 cells or Jurkat cells in different E/T ratios. 16 hour later, IFN-gamma in the supernatant was detected by ELISA.

FIGS. 9A-9G are series of images and graphs showing that Nb176 and Nb393 CAR T cells performed potent suppression against THP-1 tumor in vitro and in vivo. FIGS. 9A-9B: Nb176 and Nb393 CARs showed potent and specific cytotoxicity against THP-1 cells in a dose dependent manner, but not to K562 cells. Untransduced (UTD) T cells do not exert obvious cytotoxicity. FIGS. 9C-9D: 10 million THP-1 cells were transplanted into NSG mice subcutaneously. The tumor reached 150 mm³ after about 14 days. 3 million Nb176, Nb393 CAR or UTD T cells were injected intravenously into NSG mice separately. The tumor engraftments were monitored every other day. Scale bar was 10 mm. FIG. 9E: Hematoxylin and eosin stain of THP-1 xenograft after Nb176, Nb393 CAR or untransduced T cell treatment. Scale bar was 100 um. FIG. 9F: 5 million K562 cells were transplanted into NSG mice subcutaneously. The tumor reached 150 mm³ after about 14 days. 3 million Nb157, Nb163 CAR or UTD T cells were injected intravenously into NSG mice separately. The tumor engraftments were monitored every other day. 9 days after T cells treatment, the K562 tumor tissues were harvested and recorded. Scale bar was 10 mm. FIG. 9G: Hematoxylin and eosin stain of K562 xenograft after Nb157, Nb163 CAR or untransduced T cell treatment. Scale bar was 200 um.

FIGS. 10A-10D are series of graphs illustrating experiments in patient derived AMLs and Nb CAR T cells cells. FIGS. 10A-10B: Patient derived AMLs and Nb CAR T cells in mouse peripheral blood were monitored weekly by staining with anti-human CD45/CD33 or CD45/CD3. FIG. 10C: CAR positive T cells in mouse bone marrow and spleen were detected by tracking GFP from the hCD45+/hCD3+/hCD33- subset two weeks after T cells infusion. FIG. 10D: Memory T cells in mouse peripheral blood were detected 3 weeks after T cells infusion. CD45RA-CD62L+ are the central memory T cells, CD45RA-CD62L- are the effector memory T cells. N=3 and Student t test was applied, *, p<0.05.

FIGS. 11A-11E are series of images and graphs depicting experiments in PNEs. FIG. 11A: VHH157-N-PNE and VHH157-C-PNE fusion protein were purified from TOP10 through IPTG induction, Ni-chelating affinity and imidazole elution. Elution samples were analyzed by SDS-PAGE and Coomassie blue staining. FIG. 11B: Western blot of primary T cells transduced by sCAR lentivirus or vehicle, with anti-human CD3z. Beta-Mercaptoethanol (BME) which can break the disulfide bonds between proteins, to make dimer protein into monomer. FIG. 11C: sCAR T cells were prepared by transducing primary T cells with lentivirus. PNE peptide was synthesized with FITC to label. sCAR or UTD T cells were incubated with different concentrations of PNE-FITC, followed by flow cytometry analysis. FIG. 11D: 10 million THP-1 cells were transplanted into NSG mice subcutaneously. The tumor reached 150 mm³ after about 14 days. 0.25 mg/kg of VHH157-C-PNE, VHH157-N-PNE or control PBS was administrated into THP-1 tumor bearing NSG mice intravenously every other day. Tumor size was

monitored every other day. N=4 and ns was not significant. FIG. 11E: Mice borne patient derived AML were sacrificed 3 weeks after sCAR T with or without Nb157-C-PNE adjuvant and spleens were harvested.

FIGS. 12A-12D are a series of graphs and images depicting isolation of VHH1 from a llama-derived signal domain antibody phage display naive library by panning to BON cells, as described in FIG. 1A. VHH1 was isolated from phage library and specifically bound human neuroendocrine tumor (NET) BON cells in vitro, but not other tumor cells such as breast cancer cells or leukemia cells, by positive or negative pannings, as illustrated in FIG. 1. FIG. 12A shows about 3000 cell surface protein cDNAs were transfected into 293 cells, followed by flow analysis with VHH1. FIG. 12B depicts the antigen of VHH1 is CDH17 (through DNA sequencing individual phage clones that specifically bound BON cells), as identified by screening the transfected cells. Flow analysis of VHH1 binding to CDH17 overexpressed in 293 cells is shown. FIG. 12C shows Western blot analysis of CDH17 protein expression in NET BON cells but not in breast cancer cell line T47D and leukemia cell lines such as THP-1 cells. FIG. 12D shows human and mouse CDH17 cDNAs were transfected into 293 cells, followed by flow analysis with VHH1. The result indicates that VHH1 can bind both human and mouse CDH17 on cell surface.

FIGS. 13A-13D show a series of images and graphs depicting that VHH1 bound to BON cells as well as patient-derived metastatic pancreatic neuroendocrine tumors (PNETs), can direct CAR T cells to kill BON cells in vitro. FIG. 13A: Flow cytometry analysis showed that VHH1 antibody bound BON cells. FIG. 13B shows that CDH17 is highly expressed on the cell surface of tumor cells from PNET patients, using flow cytometry analysis of VHH1 binding to 4 human PNET samples taken via biopsy from liver, indicating CDH17 as a PNET cell surface marker vulnerable to CAR T attack for therapy. FIG. 13C shows a schematic diagram of VHH-CAR structure, consisting of VHH1 sequence (as detailed in FIG. 14), with an optimized linker, IgG4m hinge (as detailed in FIGS. 15-16), CD8 transmembrane domain (TM), 4-1BB and CD3zeta domain. To determine the impact of the VHH1 CAR T cells on killing the target cells, human primary T cells were transduced with either vector CAR or the VHH1-CAR-expressing lentiviruses. Positive VHH1-CAR T cells were determined by flow cytometry analysis of GFP expression, and used to kill BON cells with the LDH release assay. The results showed that the VHH1 CAR Ts, but not untransduced control T cells (UTD), substantially killed the target cells (FIG. 13D).

FIG. 14 depicts the DNA (SEQ ID NO: 1) and protein (SEQ ID NO: 2) sequences of the VHH1 nanobody. CDR domains of VHH1 are shown in the boxes (SEQ ID NOs: 3-5). Specific amino acid differences between VH and VHH are underlined.

FIGS. 15A-15B depict construction of VHH1-41BB-CD3z-CARs with different hinge lengths. FIG. 15A is a schematic of VHH1-CAR structures with different lengths of hinges. The VHH1-CAR includes different hinge lengths, CD8 transmembrane domain (TM), 4-1BB and CD3zeta domain, followed by IRES-GFP as a marker to demonstrate the percentage of cells expressing the CAR. FIG. 15B shows JRT3 cells were transduced with either vector CAR or the VHH1-CAR lentiviruses with different hinge lengths. Positive VHH1-CAR JRT3 cells were determined by flow cytometry analysis of GFP expression.

FIGS. 16A-16F are series of images and graphs depicting VHH1-CAR JRT3 cells with short hinges specifically killed BON cells. FIGS. 16A and 16C show flow cytometry

analysis of VHH1 specifically bound to BON cells (FIG. 16A), but not BT474 breast cancer cells (FIG. 16C). FIGS. 16B and 16D show VHH1-CAR JRT3 cells with short hinge specifically killed BON cells (FIG. 16B), but not BT474 cells (FIG. 16D), using LDH release assay. FIG. 16E shows Western blot detection of VHH1-CAR expression in JRT3 cells with anti-human CD3zeta. Ponceau S (PS) was used as an endogenous control. Reduced monomer was shown in the red box. Non-reduced dimer was shown in the black box. FIG. 16F shows a summary of killing efficacy of VHH1-CAR JRT3 cells with different hinge lengths on BON cells.

FIGS. 17A-17E are series of images and graphs depicting human primary VHH1-CAR T cells with an IgG4m hinge killed BON cells most effectively. FIG. 17A shows results from human primary T cells transduced with either vector CAR or the different length hinge VHH1-CAR lentiviruses. Positive VHH1-CAR T cells were determined by flow cytometry analysis of GFP expression. FIGS. 17B-17C show VHH1-CAR T cells with CD8 or IgG4m hinges specifically killed BON cells (FIG. 17B), but not BT474 cells (FIG. 17C), measured using LDH release assay. Microscopic observation of T cell aggregates around B cells (FIG. 17B), but not BT474 cells (FIG. 17C). FIG. 17D shows Western blot detection of VHH1-CAR expression in primary T cells with anti-human CD3zeta. Reduced monomer is shown in the grey box. Non-reduced dimer is shown in the black box. FIG. 17E shows a summary of killing efficacy of VHH1-CAR T cells with different hinge lengths on BON cells.

FIGS. 18A-18D show a series of graphs depicting CDH17CAR T cells eliminated CDH17-expressed NB4 tumors in vivo. FIG. 18A shows flow analysis of VHH1 binding to WT or sorted ectopic CDH17 expressing NB4 cells. FIG. 18B shows results from human primary T cells transduced with or without VHH1-CAR lentiviruses. Positive VHH1-CAR T cells were determined by flow cytometry analysis of GFP expression. FIG. 18C shows results from NSG mice with CDH17-NB4 xenograft treated with either UTD or VHH1-BBz CAR-T cells with injection of the T cells 5 times at day, 0, 3, 7, 22 and 24, 10-18 million VHH1-BBz CAR-T cells each time. FIG. 18D shows T cell numbers per ul in peripheral blood of NSG mice with CDH17-NB4 xenograft treated with either UTD or VHH1-BBz CAR-T cells as determined by flow cytometry at Day 14 after CAR T cell injection. These results indicate the CDH17CAR T cells specifically regressed the tumor in CDH17-dependent manner.

FIGS. 19A-19D show a series of graphs depicting that CDH17CAR T cells eliminated CDH17-expressing SKOV3 tumors in vivo. FIG. 19A shows flow analysis of VHH1 binding to WT or sorted ectopic CDH17 expressing SKOV3 cells. FIG. 19B shows results from human primary T cells transduced with or without VHH1-CAR lentiviruses. Two types of CDH17 (or VHH1) CARs were generated: VHH1-BBz is a second generation of CAR including 41-BB and CD3 zeta intracellular domains (FIG. 19B, left), and VHH1-28BBz is a third generation CAR comprising the intracellular CD28, 41-BB and CD3 zeta domains (FIG. 19B, right). Positive VHH1-CAR T cells were determined by flow cytometry analysis, using a rabbit anti-VHH antibody. FIG. 19C shows results from NSG mice with WT or CDH17-SKOV3 xenograft treated with UTD, VHH1-BBz or VHH1-28BBz CAR-T cells two times at day 0 and 5, 10-12 million CAR T cells each time, indicating that the third generation CDH17 CAR is potent to regress the tumor xenograft, but the 2nd generation of CDH17 CAR T cells only suppressed the growth of the tumor in vivo. FIG. 19D shows T cell

numbers (per ul) in peripheral blood of NSG mice with WT or CDH17-SKOV3 xenograft treated with UTD, VHH1-BBz or VHH1-28BBz CAR-T cells, as determined by flow cytometry at Day 21 after CAR T cell injection. The results indicate that the 3rd generation of CDH17 CAR Ts produced a higher number of CAR T cells in peripheral blood.

FIGS. 20A-20F are series of graphs depicting that CDH17CAR T cells eliminate NET NT-3 tumors in vivo. FIG. 20A shows flow analysis of VHH1 binding to NT-3 cells, a bona fide human NET cell line showing characteristic expression of somatostatin receptor 2 (SSTR2), chromogranin A, and insulin. FIG. 20B illustrates that CDH17CAR T cells potentially kill NT-3 cells in vitro, as determined by an LDH release assay. FIG. 20C shows results from NSG mice with NT-3 xenograft treated with either UTD or VHH1-28BBz CAR-T cells two times at day, 0 and 5, 10-12 million CAR T cells each time. FIG. 20D shows body weights of treated NSG mice, and no reduction of body weight following the CAR T injection, indicating no obvious toxicities to the mice. FIG. 20E shows T cell numbers in peripheral blood of NSG mice with NT-3 xenograft treated with either UTD or VHH1-28BBz CAR-T cells, and the T cell numbers were determined by flow cytometry at Day 14 and 21 after CAR T cell injection. FIG. 20F shows tumor growth of each NSG mouse with NT-3 xenograft treated with either UTD or VHH1-28BBz CAR-T cells. This series of studies indicate that the CDH17 CARs are capable of and potent in killing and eradicating NETs in vivo and CDH17 serves as valuable target for NETs and other CDH17-expressing tumors. Further, as the CDH17 CAR Ts did not cause toxicities in mice, these findings indicate that CDH17 serves as an effective and safer target for immunotherapy against NETs and other tumors expressing CDH17.

FIGS. 21A-21H are series of graphs and images depicting that a VHH1-ADC drug specifically kills BON cells in vitro using MTT assay. FIG. 21A shows the design of VHH1-linker-VHH1 (VLV) DM1 ADC drug. FIG. 21B shows analysis of DM1 conjugated VLV by SDS-PAGE and Coomassie blue staining. FIG. 21C shows internalization of VLV-FITC into BON or QGP1 cells, under a fluorescent microscope. Scale bar: 10 μ M. FIGS. 21D and 21E show flow cytometry analysis of unconjugated VLV or VLV-DM1 binding to BON (FIG. 21D) or QGP1 (FIG. 21E) cells, indicating the VLV-DM1 ADC can specifically kill CDH17-expressing BON cells, but not CDH17-negative QGP1 cells. FIGS. 21F and 21G show in vitro killing by free DM1, unconjugated VLV or VLV-DM1 of BON (FIG. 21F) or QGP1 (FIG. 21G) cells, demonstrating the potent killing of BON cells, but not QGP1 cells (CDH17-negative), by VLV-DM1 ADC (but not the control VLV (IC50+1.2 nM)). FIG. 21H shows microscopic observation of lysed BON cells by VLV-DM1 treatment at 4 nM. Scale bar: 10 μ M. Collectively, this series of experiments indicate that CDH17 targeting VHH1-DM1 ADC can potentially and specifically kill NET cells.

FIGS. 22A-22C are series of images and graphs depicting that the VHH1 binds to the EC1 domain of CDH17. FIG. 22A is a schema showing human CDH17 truncations. FIG. 22B shows WT CDH17 or CDH17 truncations were transfected into 293 cells, followed by flow analysis with VHH1. FIG. 22C is a schematic diagram of CDH17 structure showing VHH1 binds to the N-terminal EC1 domain of CDH17. These studies demonstrate that VHH1 directly binds CDH17 via an extracellular domain.

DETAILED DESCRIPTION

Definitions

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention pertains. Although any methods and materials similar or equivalent to those described herein can be used in the practice for testing of the present invention, the preferred materials and methods are described herein. In describing and claiming the present invention, the following terminology will be used.

It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

The articles "a" and "an" are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, "an element" means one element or more than one element.

"About" as used herein when referring to a measurable value such as an amount, a temporal duration, and the like, is meant to encompass variations of 20% or $\pm 10\%$, more preferably $\pm 5\%$, even more preferably $\pm 1\%$, and still more preferably $\pm 0.1\%$ from the specified value, as such variations are appropriate to perform the disclosed methods.

"Activation," as used herein, refers to the state of a T cell that has been sufficiently stimulated to induce detectable cellular proliferation. Activation can also be associated with induced cytokine production, and detectable effector functions. The term "activated T cells" refers to, among other things, T cells that are undergoing cell division.

As used herein, to "alleviate" a disease means reducing the severity of one or more symptoms of the disease.

"Allogeneic" refers to a graft derived from a different animal of the same species.

"Alloantigen" refers to an antigen present only in some individuals of a species and capable of inducing the production of an alloantibody by individuals which lack it.

The term "antibody," as used herein, refers to an immunoglobulin molecule which specifically binds with an antigen. Antibodies can be intact immunoglobulins derived from natural sources or from recombinant sources and can be immunoreactive portions of intact immunoglobulins. Antibodies are typically tetramers of immunoglobulin molecules. The antibodies in the present invention may exist in a variety of forms including, for example, polyclonal antibodies, monoclonal antibodies, Fv, Fab and F(ab)₂, as well as single chain antibodies (scFv) and humanized antibodies (Harlow et al., 1999, In: Using Antibodies: A Laboratory Manual, Cold Spring Harbor Laboratory Press, NY; Harlow et al., 1989, In: Antibodies: A Laboratory Manual, Cold Spring Harbor, New York; Houston et al., 1988, Proc. Natl. Acad. Sci. USA 85:5879-5883; Bird et al., 1988, Science 242:423-426).

The term "antibody fragment" refers to a portion of an intact antibody and refers to the antigenic determining variable regions of an intact antibody. Examples of antibody fragments include, but are not limited to, Fab, Fab', F(ab')₂, and Fv fragments, linear antibodies, scFv antibodies, and multispecific antibodies formed from antibody fragments.

An "antibody heavy chain," as used herein, refers to the larger of the two types of polypeptide chains present in all antibody molecules in their naturally occurring conformations.

An "antibody light chain," as used herein, refers to the smaller of the two types of polypeptide chains present in all

antibody molecules in their naturally occurring conformations. α and β light chains refer to the two major antibody light chain isotypes.

The term “antigen” or “Ag” as used herein is defined as a molecule that provokes an immune response. This immune response may involve either antibody production, or the activation of specific immunologically-competent cells, or both. The skilled artisan will understand that any macromolecule, including virtually all proteins or peptides, can serve as an antigen. Furthermore, antigens can be derived from recombinant or genomic DNA. A skilled artisan will understand that any DNA, which comprises a nucleotide sequences or a partial nucleotide sequence encoding a protein that elicits an immune response therefore encodes an “antigen” as that term is used herein. Furthermore, one skilled in the art will understand that an antigen need not be encoded solely by a full length nucleotide sequence of a gene. It is readily apparent that the present invention includes, but is not limited to, the use of partial nucleotide sequences of more than one gene and that these nucleotide sequences are arranged in various combinations to elicit the desired immune response. Moreover, a skilled artisan will understand that an antigen need not be encoded by a “gene” at all. It is readily apparent that an antigen can be generated synthesized or can be derived from a biological sample. Such a biological sample can include, but is not limited to a tissue sample, a tumor sample, a cell or a biological fluid.

As used herein, the term “autologous” is meant to refer to any material derived from the same individual to which it is later to be re-introduced into the individual.

“Allogeneic” refers to any material derived from a different animal of the same species.

The term “chimeric antigen receptor” or “CAR,” as used herein, refers to an artificial T cell receptor that is engineered to be expressed on an immune effector cell and specifically bind an antigen. CARs may be used as a therapy with adoptive cell transfer. T cells are removed from a patient and modified so that they express the receptors specific to a particular form of antigen. In some embodiments, the CAR has specificity to a selected target, for example a tumor antigen. CARs may also comprise an intracellular activation domain, a transmembrane domain and an extracellular domain comprising an antigen binding region.

“Co-stimulatory ligand,” as the term is used herein, includes a molecule on an antigen presenting cell (e.g., an aAPC, dendritic cell, B cell, and the like) that specifically binds a cognate co-stimulatory molecule on a T cell, thereby providing a signal which, in addition to the primary signal provided by, for instance, binding of a TCR/CD3 complex with an MHC molecule loaded with peptide, mediates a T cell response, including, but not limited to, proliferation, activation, differentiation, and the like. A co-stimulatory ligand can include, but is not limited to, CD7, B7-1 (CD80), B7-2 (CD86), PD-L1, PD-L2, 4-1BBL, OX40L, inducible costimulatory ligand (ICOS-L), intercellular adhesion molecule (ICAM), CD30L, CD40, CD70, CD83, HLA-G, MICA, MICB, HVEM, lymphotoxin beta receptor, 3/TR6, ILT3, ILT4, HVEM, an agonist or antibody that binds Toll ligand receptor and a ligand that specifically binds with B7-H3. A co-stimulatory ligand also encompasses, inter alia, an antibody that specifically binds with a co-stimulatory molecule present on a T cell, such as, but not limited to, CD27, CD28, 4-1BB, OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, and a ligand that specifically binds with CD83.

A “co-stimulatory molecule” refers to the cognate binding partner on a T cell that specifically binds with a co-stimulatory ligand, thereby mediating a co-stimulatory response by the T cell, such as, but not limited to, proliferation.

Co-stimulatory molecules include, but are not limited to an MHC class I molecule, BTLA and a Toll ligand receptor.

A “co-stimulatory signal”, as used herein, refers to a signal, which in combination with a primary signal, such as TCR/CD3 ligation, leads to T cell proliferation and/or upregulation or downregulation of key molecules.

A “disease” is a state of health of an animal wherein the animal cannot maintain homeostasis, and wherein if the disease is not ameliorated then the animal’s health continues to deteriorate. In contrast, a “disorder” in an animal is a state of health in which the animal is able to maintain homeostasis, but in which the animal’s state of health is less favorable than it would be in the absence of the disorder. Left untreated, a disorder does not necessarily cause a further decrease in the animal’s state of health.

The term “downregulation” as used herein refers to the decrease or elimination of gene expression of one or more genes.

“Effective amount” or “therapeutically effective amount” are used interchangeably herein, and refer to an amount of a compound, formulation, material, or composition, as described herein effective to achieve a particular biological result or provides a therapeutic or prophylactic benefit. Such results may include, but are not limited to an amount that when administered to a mammal, causes a detectable level of immune suppression or tolerance compared to the immune response detected in the absence of the composition of the invention. The immune response can be readily assessed by a plethora of art-recognized methods. The skilled artisan would understand that the amount of the composition administered herein varies and can be readily determined based on a number of factors such as the disease or condition being treated, the age and health and physical condition of the mammal being treated, the severity of the disease, the particular compound being administered, and the like.

“Encoding” refers to the inherent property of specific sequences of nucleotides in a polynucleotide, such as a gene, a cDNA, or an mRNA, to serve as templates for synthesis of other polymers and macromolecules in biological processes having either a defined sequence of nucleotides (i.e., rRNA, tRNA and mRNA) or a defined sequence of amino acids and the biological properties resulting therefrom. Thus, a gene encodes a protein if transcription and translation of mRNA corresponding to that gene produces the protein in a cell or other biological system. Both the coding strand, the nucleotide sequence of which is identical to the mRNA sequence and is usually provided in sequence listings, and the non-coding strand, used as the template for transcription of a gene or cDNA, can be referred to as encoding the protein or other product of that gene or cDNA.

As used herein “endogenous” refers to any material from or produced inside an organism, cell, tissue or system.

The term “epitope” as used herein is defined as a small chemical molecule on an antigen that can elicit an immune response, inducing B and/or T cell responses. An antigen can have one or more epitopes. Most antigens have many epitopes; i.e., they are multivalent. In general, an epitope is roughly about 10 amino acids and/or sugars in size. Preferably, the epitope is about 4-18 amino acids, more preferably about 5-16 amino acids, and even more most preferably 6-14 amino acids, more preferably about 7-12, and most preferably about 8-10 amino acids. One skilled in the art under-

stands that generally the overall three-dimensional structure, rather than the specific linear sequence of the molecule, is the main criterion of antigenic specificity and therefore distinguishes one epitope from another. Based on the present disclosure, a peptide of the present invention can be an epitope.

As used herein, the term “exogenous” refers to any material introduced from or produced outside an organism, cell, tissue or system.

The term “expand” as used herein refers to increasing in number, as in an increase in the number of T cells. In one embodiment, the T cells that are expanded *ex vivo* increase in number relative to the number originally present in the culture. In another embodiment, the T cells that are expanded *ex vivo* increase in number relative to other cell types in the culture. The term “*ex vivo*,” as used herein, refers to cells that have been removed from a living organism, (e.g., a human) and propagated outside the organism (e.g., in a culture dish, test tube, or bioreactor).

The term “expression” as used herein is defined as the transcription and/or translation of a particular nucleotide sequence driven by its promoter.

“Expression vector” refers to a vector comprising a recombinant polynucleotide comprising expression control sequences operatively linked to a nucleotide sequence to be expressed. An expression vector comprises sufficient cis-acting elements for expression; other elements for expression can be supplied by the host cell or in an *in vitro* expression system. Expression vectors include all those known in the art, such as cosmids, plasmids (e.g., naked or contained in liposomes) and viruses (e.g., Sendai viruses, lentiviruses, retroviruses, adenoviruses, and adeno-associated viruses) that incorporate the recombinant polynucleotide.

“Identity” as used herein refers to the subunit sequence identity between two polymeric molecules particularly between two amino acid molecules, such as, between two polypeptide molecules. When two amino acid sequences have the same residues at the same positions; e.g., if a position in each of two polypeptide molecules is occupied by an arginine, then they are identical at that position. The identity or extent to which two amino acid sequences have the same residues at the same positions in an alignment is often expressed as a percentage. The identity between two amino acid sequences is a direct function of the number of matching or identical positions; e.g., if half (e.g., five positions in a polymer ten amino acids in length) of the positions in two sequences are identical, the two sequences are 50% identical; if 90% of the positions (e.g., 9 of 10), are matched or identical, the two amino acids sequences are 90% identical.

The term “immunoglobulin” or “Ig,” as used herein is defined as a class of proteins, which function as antibodies. Antibodies expressed by B cells are sometimes referred to as the BCR (B cell receptor) or antigen receptor. The five members included in this class of proteins are IgA, IgG, IgM, IgD, and IgE. IgA is the primary antibody that is present in body secretions, such as saliva, tears, breast milk, gastrointestinal secretions and mucus secretions of the respiratory and genitourinary tracts. IgG is the most common circulating antibody. IgM is the main immunoglobulin produced in the primary immune response in most subjects. It is the most efficient immunoglobulin in agglutination, complement fixation, and other antibody responses, and is important in defense against bacteria and viruses. IgD is the immunoglobulin that has no known antibody function, but may serve as an antigen receptor. IgE is the immunoglobulin

that mediates immediate hypersensitivity by causing release of mediators from mast cells and basophils upon exposure to allergen.

The term “immune response” as used herein is defined as a cellular response to an antigen that occurs when lymphocytes identify antigenic molecules as foreign and induce the formation of antibodies and/or activate lymphocytes to remove the antigen.

The term “immunostimulatory” is used herein to refer to increasing overall immune response.

The term “immunosuppressive” is used herein to refer to reducing overall immune response.

“Isolated” means altered or removed from the natural state. For example, a nucleic acid or a peptide naturally present in a living animal is not “isolated,” but the same nucleic acid or peptide partially or completely separated from the coexisting materials of its natural state is “isolated.” An isolated nucleic acid or protein can exist in substantially purified form, or can exist in a non-native environment such as, for example, a host cell.

A “lentivirus” as used herein refers to a genus of the Retroviridae family. Lentiviruses are unique among the retroviruses in being able to infect non-dividing cells; they can deliver a significant amount of genetic information into the DNA of the host cell, so they are one of the most efficient methods of a gene delivery vector. HIV, SIV, and FIV are all examples of lentiviruses. Vectors derived from lentiviruses offer the means to achieve significant levels of gene transfer *in vivo*.

The term “limited toxicity” as used herein, refers to the peptides, polynucleotides, cells and/or antibodies of the invention manifesting a lack of substantially negative biological effects, anti-tumor effects, or substantially negative physiological symptoms toward a healthy cell, non-tumor cell, non-diseased cell, non-target cell or population of such cells either *in vitro* or *in vivo*.

By the term “modified” as used herein, is meant a changed state or structure of a molecule or cell of the invention. Molecules may be modified in many ways, including chemically, structurally, and functionally. Cells may be modified through the introduction of nucleic acids.

By the term “modulating,” as used herein, is meant mediating a detectable increase or decrease in the level of a response in a subject compared with the level of a response in the subject in the absence of a treatment or compound, and/or compared with the level of a response in an otherwise identical but untreated subject. The term encompasses perturbing and/or affecting a native signal or response thereby mediating a beneficial therapeutic response in a subject, preferably, a human.

In the context of the present invention, the following abbreviations for the commonly occurring nucleic acid bases are used. “A” refers to adenosine, “C” refers to cytosine, “G” refers to guanosine, “T” refers to thymidine, and “U” refers to uridine.

Unless otherwise specified, a “nucleotide sequence encoding an amino acid sequence” includes all nucleotide sequences that are degenerate versions of each other and that encode the same amino acid sequence. The phrase nucleotide sequence that encodes a protein or an RNA may also include introns to the extent that the nucleotide sequence encoding the protein may in some version contain an intron(s).

“Octretotide” is an octapeptide that mimics natural somatostatin. It is a long-acting analog of somatostatin. It is sold under the brand name Sandostatin (Novartis Pharmaceuticals). d-Phe-Cys-Phe-d-Trp-Lys-Thr-Cys-Thr-ol

"Parenteral" administration of an immunogenic composition includes, e.g., subcutaneous (s.c.), intravenous (i.v.), intramuscular (i.m.), or intrasternal injection, or infusion techniques.

The term "polynucleotide" as used herein is defined as a chain of nucleotides. Furthermore, nucleic acids are polymers of nucleotides. Thus, nucleic acids and polynucleotides as used herein are interchangeable. One skilled in the art has the general knowledge that nucleic acids are polynucleotides, which can be hydrolyzed into the monomeric "nucleotides." The monomeric nucleotides can be hydrolyzed into nucleosides. As used herein polynucleotides include, but are not limited to, all nucleic acid sequences which are obtained by any means available in the art, including, without limitation, recombinant means, i.e., the cloning of nucleic acid sequences from a recombinant library or a cell genome, using ordinary cloning technology and PCR™, and the like, and by synthetic means.

As used herein, the terms "peptide," "polypeptide," and "protein" are used interchangeably, and refer to a compound comprised of amino acid residues covalently linked by peptide bonds. A protein or peptide must contain at least two amino acids, and no limitation is placed on the maximum number of amino acids that can comprise a protein's or peptide's sequence. Polypeptides include any peptide or protein comprising two or more amino acids joined to each other by peptide bonds. As used herein, the term refers to both short chains, which also commonly are referred to in the art as peptides, oligopeptides and oligomers, for example, and to longer chains, which generally are referred to in the art as proteins, of which there are many types. "Polypeptides" include, for example, biologically active fragments, substantially homologous polypeptides, oligopeptides, homodimers, heterodimers, variants of polypeptides, modified polypeptides, derivatives, analogs, fusion proteins, among others. The polypeptides include natural peptides, recombinant peptides, synthetic peptides, or a combination thereof.

The term "self-antigen" as used herein is defined as an antigen that is expressed by a host cell or tissue. Self-antigens may be tumor antigens, but in certain embodiments, are expressed in both normal and tumor cells. A skilled artisan would readily understand that a self-antigen may be overexpressed in a cell.

By the term "specifically binds," as used herein with respect to an antibody, is meant an antibody which recognizes a specific antigen, but does not substantially recognize or bind other molecules in a sample. For example, an antibody that specifically binds to an antigen from one species may also bind to that antigen from one or more species. But, such cross-species reactivity does not itself alter the classification of an antibody as specific. In another example, an antibody that specifically binds to an antigen may also bind to different allelic forms of the antigen. However, such cross reactivity does not itself alter the classification of an antibody as specific. In some instances, the terms "specific binding" or "specifically binding," can be used in reference to the interaction of an antibody, a protein, or a peptide with a second chemical species, to mean that the interaction is dependent upon the presence of a particular structure (e.g., an antigenic determinant or epitope) on the chemical species; for example, an antibody recognizes and binds to a specific protein structure rather than to proteins generally. If an antibody is specific for epitope "A", the presence of a molecule containing epitope A (or free, unlabeled

A), in a reaction containing labeled "A" and the antibody, will reduce the amount of labeled A bound to the antibody.

By the term "stimulation," is meant a primary response induced by binding of a stimulatory molecule (e.g., a TCR/CD3 complex) with its cognate ligand thereby mediating a signal transduction event, such as, but not limited to, signal transduction via the TCR/CD3 complex. Stimulation can mediate altered expression of certain molecules, such as downregulation of TGF-beta, and/or reorganization of cytoskeletal structures, and the like.

A "stimulatory molecule," as the term is used herein, means a molecule on a T cell that specifically binds with a cognate stimulatory ligand present on an antigen presenting cell.

A "stimulatory ligand," as used herein, means a ligand that when present on an antigen presenting cell (e.g., an aAPC, a dendritic cell, a B-cell, and the like) can specifically bind with a cognate binding partner (referred to herein as a "stimulatory molecule") on a T cell, thereby mediating a primary response by the T cell, including, but not limited to, activation, initiation of an immune response, proliferation, and the like. Stimulatory ligands are well-known in the art and encompass, inter alia, an MHC Class I molecule loaded with a peptide, an anti-CD3 antibody, a superagonist anti-CD28 antibody, and a superagonist anti-CD2 antibody.

The term "subject" is intended to include living organisms in which an immune response can be elicited (e.g., mammals). A "subject" or "patient," as used therein, may be a human or non-human mammal. Non-human mammals include, for example, livestock and pets, such as ovine, bovine, porcine, canine, feline and murine mammals. Preferably, the subject is human.

As used herein, a "substantially purified" cell is a cell that is essentially free of other cell types. A substantially purified cell also refers to a cell which has been separated from other cell types with which it is normally associated in its naturally occurring state. In some instances, a population of substantially purified cells refers to a homogenous population of cells. In other instances, this term refers simply to cell that have been separated from the cells with which they are naturally associated in their natural state. In some embodiments, the cells are cultured in vitro. In other embodiments, the cells are not cultured in vitro.

As used herein, a switchable CAR (sCAR) refers to a CAR comprising a Peptide-Neo-Epitope (PNE) binding domain, a transmembrane domain, and an intracellular signaling domain.

A "target site" or "target sequence" refers to a genomic nucleic acid sequence that defines a portion of a nucleic acid to which a binding molecule may specifically bind under conditions sufficient for binding to occur.

As used herein, the term "T cell receptor" or "TCR" refers to a complex of membrane proteins that participate in the activation of T cells in response to the presentation of antigen. The TCR is responsible for recognizing antigens bound to major histocompatibility complex molecules. TCR is composed of a heterodimer of an alpha (α) and beta (β) chain, although in some cells the TCR consists of gamma and delta (γ/δ) chains. TCRs may exist in alpha/beta and gamma/delta forms, which are structurally similar but have distinct anatomical locations and functions. Each chain is composed of two extracellular domains, a variable and constant domain. In some embodiments, the TCR may be modified on any cell comprising a TCR, including, for

example, a helper T cell, a cytotoxic T cell, a memory T cell, regulatory T cell, natural killer T cell, and gamma delta T cell.

The term “therapeutic” as used herein means a treatment and/or prophylaxis. A therapeutic effect is obtained by suppression, remission, or eradication of a disease state.

“Transplant” refers to a biocompatible lattice or a donor tissue, organ or cell, to be transplanted. An example of a transplant may include but is not limited to skin cells or tissue, bone marrow, and solid organs such as heart, pancreas, kidney, lung and liver. A transplant can also refer to any material that is to be administered to a host. For example, a transplant can refer to a nucleic acid or a protein.

The term “transfected” or “transformed” or “transduced” as used herein refers to a process by which exogenous nucleic acid is transferred or introduced into the host cell. A “transfected” or “transformed” or “transduced” cell is one which has been transfected, transformed or transduced with exogenous nucleic acid. The cell includes the primary subject cell and its progeny.

To “treat” a disease as the term is used herein, means to reduce the frequency or severity of at least one sign or symptom of a disease or disorder experienced by a subject.

A “vector” is a composition of matter which comprises an isolated nucleic acid and which can be used to deliver the isolated nucleic acid to the interior of a cell. Numerous vectors are known in the art including, but not limited to, linear polynucleotides, polynucleotides associated with ionic or amphiphilic compounds, plasmids, and viruses. Thus, the term “vector” includes an autonomously replicating plasmid or a virus. The term should also be construed to include non-plasmid and non-viral compounds which facilitate transfer of nucleic acid into cells, such as, for example, polylysine compounds, liposomes, and the like. Examples of viral vectors include, but are not limited to, Sendai viral vectors, adenoviral vectors, adeno-associated virus vectors, retroviral vectors, lentiviral vectors, and the like.

“Xenogeneic” refers to any material derived from an animal of a different species.

Ranges: throughout this disclosure, various aspects of the invention can be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the invention. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 2.7, 3, 4, 5, 5.3, and 6. This applies regardless of the breadth of the range.

Description

The present invention includes compositions and methods for retrieving tumor-related antibodies and antigens.

In one aspect, the invention includes a method for Sequential Tumor-related Antibody and antigen Retrieving (STAR) which directly and efficiently identifies potent antibodies that can specifically bind to tumor-related antigens on the tumor cell surface. In another aspect, the invention includes a method for generating STAR-selected antibodies chimeric antigen receptor (CAR) T cells that specifically target tumor cell antigens. In another aspect, the invention includes a CAR comprising a nanobody, a transmembrane

domain, an intracellular signaling domain. In certain embodiments, the nanobody is retrieved by a STAR method. In yet another aspect, the invention includes compositions and methods for treating acute myeloid leukemia (AML) or neuroendocrine tumors (NETs).

Chimeric Antigen Receptor

Certain embodiments of the invention include chimeric antigen receptors (CARs) comprising the following components: An antigen binding domain, a transmembrane domain, a hinge domain, and an intracellular signaling domain.

a) Antigen Binding Domain

In one embodiment, the CAR of the invention comprises an antigen binding domain that is a variable domain heavy-chain camelid antibody (VHH), also referred to as a nanobody. In another embodiment, the CAR comprises an antigen binding domain that binds to Peptide-Neo-Epitope (PNE). The choice of antigen binding domain depends upon the type and number of antigens that are present on the surface of a target cell. For example, the antigen binding domain may be chosen to recognize an antigen that acts as a cell surface marker on a target cell associated with a particular disease state.

The antigen binding domain can include any domain that binds to the antigen and may include, but is not limited to, a monoclonal antibody, a polyclonal antibody, a synthetic antibody, a human antibody, a humanized antibody, a non-human antibody, and any fragment thereof. Thus, in one embodiment, the antigen binding domain portion comprises a mammalian antibody or a fragment thereof.

In some instances, the antigen binding domain may be derived from the same species in which the CAR will ultimately be used. For example, for use in humans, the antigen binding domain of the CAR may comprise a human antibody as described elsewhere herein, or a fragment thereof.

The antigen binding domain may be operably linked to another domain of the CAR, such as the transmembrane domain or the intracellular domain, both described elsewhere herein, for expression in the cell. In one embodiment, a nucleic acid encoding the antigen binding domain is operably linked to a nucleic acid encoding a transmembrane domain and a nucleic acid encoding an intracellular domain.

The antigen binding domains described herein can be combined with any of the transmembrane domains described herein, any of the intracellular domains or cytoplasmic domains described herein, or any of the other domains described herein that may be included in the CAR.

In certain embodiments, the antigen binding domain (e.g. nanobody) is encoded by the nucleotide sequence of SEQ ID NO: 1. In certain embodiments, the antigen binding domain comprises the amino acid sequence of SEQ ID NO: 2. In certain embodiments, the antigen binding domain comprises a CDR1 sequence comprising the amino acid sequence YYDMG (SEQ ID NO: 3). In certain embodiments, the antigen binding domain comprises a CDR2 sequence comprising the amino acid sequence LLSWNGENAEY-SDSVMGR (SEQ ID NO: 4). In certain embodiments, the antigen binding domain comprises a CDR3 sequence comprising the amino acid sequence AVTHGGARSVRS (SEQ ID NO: 5).

In certain embodiments, the antigen binding domain comprises the amino acid sequence of SEQ ID NO: 19. In certain embodiments, the antigen binding domain comprises a CDR1 sequence comprising the amino acid sequence of SEQ ID NO: 25. In certain embodiments, the antigen binding domain comprises a CDR2 sequence comprising the

(CD229), CD160 (BY55), PSGL1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Ly108), SLAM (SLAMF1, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD 162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, NKp44, NKp30, NKp46, NKG2D, Toll-like receptor 1 (TLR1), TLR2, TLR3, TLR4, TLR5, TLR6, TLR7, TLR8, TLR9, other co-stimulatory molecules described herein, any derivative, variant, or fragment thereof, any synthetic sequence of a co-stimulatory molecule that has the same functional capability, and any combination thereof.

In certain embodiments, the intracellular signaling domain of the CAR comprises 4-1BB and CD3 zeta.

The intracellular signaling domains described herein can be combined with any of the antigen binding domains described herein, any of the transmembrane domains described herein, or any of the other domains described herein that may be included in the CAR.

In certain embodiments, the CAR comprises a signal peptide, an IgG4 mutant (IgG4m) hinge region, a CD8 transmembrane domain (TM), a 4-1BB intracellular domain and CD3zeta intracellular domain.

In certain embodiments, the CAR comprises the amino acid sequence of any one of SEQ ID NOs: 17, 23, 34 or 36. In certain embodiments, the CAR is encoded by the nucleotide sequence of any one of SEQ ID NOs: 33 or 35. Tolerable variations of the CAR sequences will be known to those of skill in the art. For example, in some embodiments the CAR comprises an amino acid sequence that has at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to any of the amino acid sequences set forth in SEQ ID NOs: 17, 23, 34 or 36. In some embodiments the CAR is encoded by a nucleic acid sequence that has at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to the nucleic acid sequence set forth in SEQ ID NO: 33 or 35.

As used herein, a switchable CAR (sCAR) refers to a CAR comprising a Peptide-Neo-Epitope (PNE) binding domain, a transmembrane domain, and an intracellular signaling domain. The switchable CAR can be used in conjunction with a molecule comprising a nanobody fused to a PNE molecule (e.g. a switchable CAR system). When the nanobody-PNE molecule comes into contact with the sCAR, the "switch" is turned on and the CAR T cell is activated.

The nanobody can be fused to the C-terminal region of the PNE or the N-terminal region of the PNE. In certain embodiments, nanobody Nb157 (VHH157) is fused to the C-terminal region of PNE. In certain embodiments, nanobody Nb157 is fused to the N-terminal region of PNE. In certain embodiments, nanobody Nb163 (VHH163) is fused to the C-terminal region of PNE. In certain embodiments, nanobody Nb163 is fused to the N-terminal region of PNE.

Included in the invention are isolated polypeptides comprising CARs, isolated nucleic acids comprising CARs, vectors comprising nucleic acids comprising CARs, and modified cells (e.g. T cells) comprising CARs, nucleic acids encoding CARs, or vectors comprising CARs.

Methods of Treatment

The modified cells (e.g., CAR T cells, or cells comprising a switchable CAR system) described herein, may be

included in a composition for immunotherapy. The composition may include a pharmaceutical composition and further include a pharmaceutically acceptable carrier. A therapeutically effective amount of the pharmaceutical composition comprising the modified T cells may be administered.

In one aspect, the invention includes a method for adoptive cell transfer therapy comprising administering to a subject in need thereof a modified T cell of the present invention. In another aspect, the invention includes a method of treating a disease or condition in a subject comprising administering to a subject in need thereof a population of modified T cells. In certain embodiments, the disease to be treated is cancer.

Methods for administration of immune cells for adoptive cell therapy are known and may be used in connection with the provided methods and compositions. For example, adoptive T cell therapy methods are described, e.g., in US Patent Application Publication No. 2003/0170238 to Gruenberg et al; U.S. Pat. No. 4,690,915 to Rosenberg; Rosenberg (2011) *Nat Rev Clin Oncol.* 8(10):577-85). See, e.g., Themeli et al. (2013) *Nat Biotechnol.* 31(10): 928-933; Tsukahara et al. (2013) *Biochem Biophys Res Commun* 438(1): 84-9; Davila et al. (2013) *PLoS ONE* 8(4): e61338. In some embodiments, the cell therapy, e.g., adoptive T cell therapy is carried out by autologous transfer, in which the cells are isolated and/or otherwise prepared from the subject who is to receive the cell therapy, or from a sample derived from such a subject. Thus, in some aspects, the cells are derived from a subject, e.g., patient, in need of a treatment and the cells, following isolation and processing are administered to the same subject.

In some embodiments, the cell therapy, e.g., adoptive T cell therapy, is carried out by allogeneic transfer, in which the cells are isolated and/or otherwise prepared from a subject other than a subject who is to receive or who ultimately receives the cell therapy, e.g., a first subject. In such embodiments, the cells then are administered to a different subject, e.g., a second subject, of the same species. In some embodiments, the first and second subjects are genetically identical. In some embodiments, the first and second subjects are genetically similar. In some embodiments, the second subject expresses the same HLA class or supertype as the first subject.

In some embodiments, the subject has been treated with a therapeutic agent targeting the disease or condition, e.g. the tumor, prior to administration of the cells or composition containing the cells. In some aspects, the subject is refractory or non-responsive to the other therapeutic agent. In some embodiments, the subject has persistent or relapsed disease, e.g., following treatment with another therapeutic intervention, including chemotherapy, radiation, and/or hematopoietic stem cell transplantation (HSCT), e.g., allogeneic HSCT. In some embodiments, the administration effectively treats the subject despite the subject having become resistant to another therapy.

In some embodiments, the subject is responsive to the other therapeutic agent, and treatment with the therapeutic agent reduces disease burden. In some aspects, the subject is initially responsive to the therapeutic agent, but exhibits a relapse of the disease or condition over time. In some embodiments, the subject has not relapsed. In some such embodiments, the subject is determined to be at risk for relapse, such as at a high risk of relapse, and thus the cells are administered prophylactically, e.g., to reduce the likelihood of or prevent relapse. In some aspects, the subject has not received prior treatment with another therapeutic agent.

The modified immune cells of the present invention can be administered to an animal, preferably a mammal, even more preferably a human, to treat a cancer. In addition, the cells of the present invention can be used for the treatment of any condition related to a cancer, especially a cell-mediated immune response against a tumor cell(s), where it is desirable to treat or alleviate the disease. The types of cancers to be treated with the modified cells or pharmaceutical compositions of the invention include, acute myeloid leukemia, chronic myeloid leukemia, pancreatic neuroendocrine tumor (PNETs), gastrointestinal NETs, and lung and prostate cancer NETs, carcinoma, blastoma, and sarcoma, and certain leukemia or lymphoid malignancies, benign and malignant tumors, and malignancies e.g., sarcomas, carcinomas, and melanomas. Other exemplary cancers include but are not limited breast cancer, prostate cancer, ovarian cancer, cervical cancer, skin cancer, pancreatic cancer, colorectal cancer, renal cancer, liver cancer, brain cancer, lymphoma, leukemia, lung cancer, thyroid cancer, and the like. The cancers may be non-solid tumors (such as hematological tumors) or solid tumors. Adult tumors/cancers and pediatric tumors/cancers are also included. In one embodiment, the cancer is a solid tumor or a hematological tumor. In one embodiment, the cancer is a carcinoma. In one embodiment, the cancer is a sarcoma. In one embodiment, the cancer is a leukemia. In one embodiment the cancer is a solid tumor. In one embodiment, the cancer is ovarian cancer. In one embodiment, the cancer is endometrial cancer.

Solid tumors are abnormal masses of tissue that usually do not contain cysts or liquid areas. Solid tumors can be benign or malignant. Different types of solid tumors are named for the type of cells that form them (such as sarcomas, carcinomas, and lymphomas). Examples of solid tumors, such as sarcomas and carcinomas, include fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteosarcoma, and other sarcomas, synovioma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon carcinoma, lymphoid malignancy, pancreatic cancer, breast cancer, lung cancers, ovarian cancer, prostate cancer, hepatocellular carcinoma, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, medullary thyroid carcinoma, papillary thyroid carcinoma, pheochromocytomas sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, choriocarcinoma, Wilms' tumor, cervical cancer, testicular tumor, seminoma, bladder carcinoma, melanoma, and CNS tumors (such as a glioma (such as brainstem glioma and mixed gliomas), glioblastoma (also known as glioblastoma multiforme) astrocytoma, CNS lymphoma, germinoma, medulloblastoma, Schwannoma cranioopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, meningioma, neuroblastoma, retinoblastoma and brain metastases).

Carcinomas that can be amenable to therapy by a method disclosed herein include, but are not limited to, esophageal carcinoma, hepatocellular carcinoma, basal cell carcinoma (a form of skin cancer), squamous cell carcinoma (various tissues), bladder carcinoma, including transitional cell carcinoma (a malignant neoplasm of the bladder), bronchogenic carcinoma, colon carcinoma, colorectal carcinoma, gastric carcinoma, lung carcinoma, including small cell carcinoma and non-small cell carcinoma of the lung, adrenocortical carcinoma, thyroid carcinoma, pancreatic carcinoma, breast carcinoma, ovarian carcinoma, prostate carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocar-

cinoma, cystadenocarcinoma, medullary carcinoma, renal cell carcinoma, ductal carcinoma in situ or bile duct carcinoma, choriocarcinoma, seminoma, embryonal carcinoma, Wilms' tumor, cervical carcinoma, uterine carcinoma, testicular carcinoma, osteogenic carcinoma, epithelial carcinoma, and nasopharyngeal carcinoma, ovarian cancer, endometrial cancer, uterine sarcoma, cervical carcinoma, breast cancer, lung cancer, prostate cancer, ocular melanoma, and any MISIR-expressing tumor.

Sarcomas that can be amenable to therapy by a method disclosed herein include, but are not limited to, fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, chordoma, osteogenic sarcoma, osteosarcoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovioma, mesothelioma, Ewing's sarcoma, leiomyosarcoma, rhabdomyosarcoma, and other soft tissue sarcomas.

In certain exemplary embodiments, the modified immune cells of the invention are used to treat a myeloma, or a condition related to myeloma. Examples of myeloma or conditions related thereto include, without limitation, light chain myeloma, non-secretory myeloma, monoclonal gammopathy of undetermined significance (MGUS), plasmacytoma (e.g., solitary, multiple solitary, extramedullary plasmacytoma), amyloidosis, and multiple myeloma. In one embodiment, a method of the present disclosure is used to treat multiple myeloma. In one embodiment, a method of the present disclosure is used to treat refractory myeloma. In one embodiment, a method of the present disclosure is used to treat relapsed myeloma.

In certain exemplary embodiments, the modified immune cells of the invention are used to treat a melanoma, or a condition related to melanoma. Examples of melanoma or conditions related thereto include, without limitation, superficial spreading melanoma, nodular melanoma, lentigo maligna melanoma, acral lentiginous melanoma, amelanotic melanoma, or melanoma of the skin (e.g., cutaneous, eye, vulva, vagina, rectum melanoma). In one embodiment, a method of the present disclosure is used to treat cutaneous melanoma. In one embodiment, a method of the present disclosure is used to treat refractory melanoma. In one embodiment, a method of the present disclosure is used to treat relapsed melanoma.

In yet other exemplary embodiments, the modified immune cells of the invention are used to treat a sarcoma, or a condition related to sarcoma. Examples of sarcoma or conditions related thereto include, without limitation, angiosarcoma, chondrosarcoma, Ewing's sarcoma, fibrosarcoma, gastrointestinal stromal tumor, leiomyosarcoma, liposarcoma, malignant peripheral nerve sheath tumor, osteosarcoma, pleomorphic sarcoma, rhabdomyosarcoma, and synovial sarcoma. In one embodiment, a method of the present disclosure is used to treat synovial sarcoma. In one embodiment, a method of the present disclosure is used to treat liposarcoma such as myxoid/round cell liposarcoma, differentiated/dedifferentiated liposarcoma, and pleomorphic liposarcoma. In one embodiment, a method of the present disclosure is used to treat myxoid/round cell liposarcoma. In one embodiment, a method of the present disclosure is used to treat a refractory sarcoma. In one embodiment, a method of the present disclosure is used to treat a relapsed sarcoma.

The cells of the invention to be administered may be autologous, with respect to the subject undergoing therapy.

The administration of the cells of the invention may be carried out in any convenient manner known to those of skill in the art. The cells of the present invention may be

administered to a subject by aerosol inhalation, injection, ingestion, transfusion, implantation or transplantation. The compositions described herein may be administered to a patient transarterially, subcutaneously, intradermally, intratumorally, intranodally, intramedullary, intramuscularly, by intravenous (i.v.) injection, or intraperitoneally. In other instances, the cells of the invention are injected directly into a site of inflammation in the subject, a local disease site in the subject, a lymph node, an organ, a tumor, and the like.

In some embodiments, the cells are administered at a desired dosage, which in some aspects includes a desired dose or number of cells or cell type(s) and/or a desired ratio of cell types. Thus, the dosage of cells in some embodiments is based on a total number of cells (or number per kg body weight) and a desired ratio of the individual populations or sub-types, such as the CD4⁺ to CD8⁺ ratio. In some embodiments, the dosage of cells is based on a desired total number (or number per kg of body weight) of cells in the individual populations or of individual cell types. In some

embodiments, the dosage is based on a combination of such features, such as a desired number of total cells, desired ratio, and desired total number of cells in the individual populations.

In some embodiments, the populations or sub-types of cells, such as CD8⁺ and CD4⁺ T cells, are administered at or within a tolerated difference of a desired dose of total cells, such as a desired dose of T cells. In some aspects, the desired dose is a desired number of cells or a desired number of cells per unit of body weight of the subject to whom the cells are administered, e.g., cells/kg. In some aspects, the desired dose is at or above a minimum number of cells or minimum number of cells per unit of body weight. In some aspects, among the total cells, administered at the desired dose, the individual populations or sub-types are present at or near a desired output ratio (such as CD4⁺ to CD8⁺ ratio), e.g., within a certain tolerated difference or error of such a ratio.

In some embodiments, the cells are administered at or within a tolerated difference of a desired dose of one or more of the individual populations or sub-types of cells, such as a desired dose of CD4⁺ cells and/or a desired dose of CD8⁺ cells. In some aspects, the desired dose is a desired number of cells of the sub-type or population, or a desired number of such cells per unit of body weight of the subject to whom the cells are administered, e.g., cells/kg. In some aspects, the desired dose is at or above a minimum number of cells of the population or subtype, or minimum number of cells of the population or sub-type per unit of body weight. Thus, in some embodiments, the dosage is based on a desired fixed dose of total cells and a desired ratio, and/or based on a desired fixed dose of one or more, e.g., each, of the individual sub-types or sub-populations. Thus, in some embodiments, the dosage is based on a desired fixed or minimum dose of T cells and a desired ratio of CD4⁺ to CD8⁺ cells, and/or is based on a desired fixed or minimum dose of CD4⁺ and/or CD8⁺ cells.

In certain embodiments, the cells, or individual populations of sub-types of cells, are administered to the subject at a range of about one million to about 100 billion cells, such as, e.g., 1 million to about 50 billion cells (e.g., about 5 million cells, about 25 million cells, about 500 million cells, about 1 billion cells, about 5 billion cells, about 20 billion cells, about 30 billion cells, about 40 billion cells, or a range defined by any two of the foregoing values), such as about 10 million to about 100 billion cells (e.g., about 20 million cells, about 30 million cells, about 40 million cells, about 60 million cells, about 70 million cells, about 80 million cells, about 90 million cells, about 10 billion cells, about 25 billion

cells, about 50 billion cells, about 75 billion cells, about 90 billion cells, or a range defined by any two of the foregoing values), and in some cases about 100 million cells to about 50 billion cells (e.g., about 120 million cells, about 250 million cells, about 350 million cells, about 450 million cells, about 650 million cells, about 800 million cells, about 900 million cells, about 3 billion cells, about 30 billion cells, about 45 billion cells) or any value in between these ranges.

In some embodiments, the dose of total cells and/or dose of individual sub-populations of cells is within a range of between at or about 1×10^5 cells/kg to about 1×10^{11} cells/kg 10^4 and at or about 10^{11} cells/kilograms (kg) body weight, such as between 10^5 and 10^6 cells/kg body weight, for example, at or about 1×10^5 cells/kg, 1.5×10^5 cells/kg, 2×10^5 cells/kg, or 1×10^6 cells/kg body weight. For example, in some embodiments, the cells are administered at, or within a certain range of error of, between at or about 10^4 and at or about 10^9 T cells/kilograms (kg) body weight, such as between 10^5 and 10^6 T cells/kg body weight, for example, at or about 1×10^5 T cells/kg, 1.5×10^5 T cells/kg, 2×10^5 T cells/kg, or 1×10^6 T cells/kg body weight. In other exemplary embodiments, a suitable dosage range of modified cells for use in a method of the present disclosure includes, without limitation, from about 1×10^5 cells/kg to about 1×10^6 cells/kg, from about 1×10^6 cells/kg to about 1×10^7 cells/kg, from about 1×10^7 cells/kg about 1×10^8 cells/kg, from about 1×10^8 cells/kg about 1×10^9 cells/kg, from about 1×10^9 cells/kg about 1×10^{10} cells/kg, from about 1×10^{10} cells/kg about 1×10^{11} cells/kg. In an exemplary embodiment, a suitable dosage for use in a method of the present disclosure is about 1×10^8 cells/kg. In an exemplary embodiment, a suitable dosage for use in a method of the present disclosure is about 1×10^7 cells/kg. In other embodiments, a suitable dosage is from about 1×10^7 total cells to about 5×10^7 total cells. In some embodiments, a suitable dosage is from about 1×10^8 total cells to about 5×10^8 total cells. In some embodiments, a suitable dosage is from about 1.4×10^7 total cells to about 1.1×10^9 total cells. In an exemplary embodiment, a suitable dosage for use in a method of the present disclosure is about 7×10^9 total cells.

In some embodiments, the cells are administered at or within a certain range of error of between at or about 10^4 and at or about 10^9 CD4⁺ and/or CD8⁺ cells/kilograms (kg) body weight, such as between 10^5 and 10^6 CD4⁺ and/or CD8⁺ cells/kg body weight, for example, at or about 1×10^5 CD4⁺ and/or CD8⁺ cells/kg, 1.5×10^5 CD4⁺ and/or CD8⁺ cells/kg, 2×10^5 CD4⁺ and/or CD8⁺ cells/kg, or 1×10^6 CD4⁺ and/or CD8⁺ cells/kg body weight. In some embodiments, the cells are administered at or within a certain range of error of, greater than, and/or at least about 1×10^6 , about 2.5×10^6 , about 5×10^6 , about 7.5×10^6 , or about 9×10^6 CD4⁺ cells, and/or at least about 1×10^6 , about 2.5×10^6 , about 5×10^6 , about 7.5×10^6 , or about 9×10^6 CD8⁺ cells, and/or at least about 1×10^6 , about 2.5×10^6 , about 5×10^6 , about 7.5×10^6 , or about 9×10^6 T cells. In some embodiments, the cells are administered at or within a certain range of error of between about 10^8 and 10^{12} or between about 10^{10} and 10^{11} T cells, between about 10^8 and 10^{12} or between about 10^{10} and 10^{11} CD4⁺ cells, and/or between about 10^8 and 10^{12} or between about 10^{10} and 10^{11} CD8⁺ cells.

In some embodiments, the cells are administered at or within a tolerated range of a desired output ratio of multiple cell populations or sub-types, such as CD4⁺ and CD8⁺ cells or sub-types. In some aspects, the desired ratio can be a specific ratio or can be a range of ratios, for example, in some embodiments, the desired ratio (e.g., ratio of CD4⁺ to CD8⁺ cells) is between at or about 5:1 and at or about 5:1

(or greater than about 1:5 and less than about 5:1), or between at or about 1:3 and at or about 3:1 (or greater than about 1:3 and less than about 3:1), such as between at or about 2:1 and at or about 1:5 (or greater than about 1:5 and less than about 2:1, such as at or about 5:1, 4.5:1, 4:1, 3.5:1, 3:1, 2.5:1, 2:1, 1.9:1, 1.8:1, 1.7:1, 1.6:1, 1.5:1, 1.4:1, 1.3:1, 1.2:1, 1.1:1, 1:1, 1:1.1, 1:1.2, 1:1.3, 1:1.4, 1:1.5, 1:1.6, 1:1.7, 1:1.8, 1:1.9:1:2, 1:2.5, 1:3, 1:3.5, 1:4, 1:4.5, or 1:5. In some aspects, the tolerated difference is within about 1%, about 2%, about 3%, about 4% about 5%, about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50% of the desired ratio, including any value in between these ranges.

In some embodiments, a dose of modified cells is administered to a subject in need thereof, in a single dose or multiple doses. In some embodiments, a dose of modified cells is administered in multiple doses, e.g., once a week or every 7 days, once every 2 weeks or every 14 days, once every 3 weeks or every 21 days, once every 4 weeks or every 28 days. In an exemplary embodiment, a single dose of modified cells is administered to a subject in need thereof. In an exemplary embodiment, a single dose of modified cells is administered to a subject in need thereof by rapid intravenous infusion.

For the prevention or treatment of disease, the appropriate dosage may depend on the type of disease to be treated, the type of cells or recombinant receptors, the severity and course of the disease, whether the cells are administered for preventive or therapeutic purposes, previous therapy, the subject's clinical history and response to the cells, and the discretion of the attending physician. The compositions and cells are in some embodiments suitably administered to the subject at one time or over a series of treatments.

In some embodiments, the cells are administered as part of a combination treatment, such as simultaneously with or sequentially with, in any order, another therapeutic intervention, such as an antibody or engineered cell or receptor or agent, such as a cytotoxic or therapeutic agent. The cells in some embodiments are co-administered with one or more additional therapeutic agents or in connection with another therapeutic intervention, either simultaneously or sequentially in any order. In some contexts, the cells are co-administered with another therapy sufficiently close in time such that the cell populations enhance the effect of one or more additional therapeutic agents, or vice versa. In some embodiments, the cells are administered prior to the one or more additional therapeutic agents. In some embodiments, the cells are administered after the one or more additional therapeutic agents. In some embodiments, the one or more additional agents include a cytokine, such as IL-2, for example, to enhance persistence. In some embodiments, the methods comprise administration of a chemotherapeutic agent.

In certain embodiments, the modified cells of the invention (e.g., a modified cell comprising a CAR) may be administered to a subject in combination with an immune checkpoint antibody (e.g., an anti-PD1, anti-CTLA-4, or anti-PDL1 antibody). For example, the modified cell may be administered in combination with an antibody or antibody fragment targeting, for example, PD-1 (programmed death 1 protein). Examples of anti-PD-1 antibodies include, but are not limited to, pembrolizumab (KEYTRUDA®, formerly lambrolizumab, also known as MK-3475), and nivolumab (BMS-936558, MDX-1106, ONO-4538, OPDIVA®) or an antigen-binding fragment thereof. In certain embodiments, the modified cell may be administered in combination with an anti-PD-L1 antibody or antigen-binding fragment

thereof. Examples of anti-PD-L1 antibodies include, but are not limited to, BMS-936559, MPDL3280A (TECENTRIQ®, Atezolizumab), and MEDI4736 (Durvalumab, Imfinzi). In certain embodiments, the modified cell may be administered in combination with an anti-CTLA-4 antibody or antigen-binding fragment thereof. An example of an anti-CTLA-4 antibody includes, but is not limited to, Ipilimumab (trade name Yervoy). Other types of immune checkpoint modulators may also be used including, but not limited to, small molecules, siRNA, miRNA, and CRISPR systems. Immune checkpoint modulators may be administered before, after, or concurrently with the modified cell comprising the CAR. In certain embodiments, combination treatment comprising an immune checkpoint modulator may increase the therapeutic efficacy of a therapy comprising a modified cell of the present invention.

Following administration of the cells, the biological activity of the engineered cell populations in some embodiments is measured, e.g., by any of a number of known methods. Parameters to assess include specific binding of an engineered or natural T cell or other immune cell to antigen, in vivo, e.g., by imaging, or ex vivo, e.g., by ELISA or flow cytometry. In certain embodiments, the ability of the engineered cells to destroy target cells can be measured using any suitable method known in the art, such as cytotoxicity assays described in, for example, Kochenderfer et al., J. Immunotherapy, 32(7): 689-702 (2009), and Herman et al. J. Immunological Methods, 285(1): 25-40 (2004). In certain embodiments, the biological activity of the cells is measured by assaying expression and/or secretion of one or more cytokines, such as CD 107a, IFN γ , IL-2, and TNF. In some aspects the biological activity is measured by assessing clinical outcome, such as reduction in tumor burden or load.

In certain embodiments, the subject is provided a secondary treatment. Secondary treatments include but are not limited to chemotherapy, radiation, surgery, and medications.

In some embodiments, the subject can be administered a conditioning therapy prior to CAR T cell therapy. In some embodiments, the conditioning therapy comprises administering an effective amount of cyclophosphamide to the subject. In some embodiments, the conditioning therapy comprises administering an effective amount of fludarabine to the subject. In preferred embodiments, the conditioning therapy comprises administering an effective amount of a combination of cyclophosphamide and fludarabine to the subject. Administration of a conditioning therapy prior to CAR T cell therapy may increase the efficacy of the CAR T cell therapy. Methods of conditioning patients for T cell therapy are described in U.S. Pat. No. 9,855,298, which is incorporated herein by reference in its entirety.

In some embodiments, a specific dosage regimen of the present disclosure includes a lymphodepletion step prior to the administration of the modified T cells. In an exemplary embodiment, the lymphodepletion step includes administration of cyclophosphamide and/or fludarabine.

In some embodiments, the lymphodepletion step includes administration of cyclophosphamide at a dose of between about 200 mg/m²/day and about 2000 mg/m²/day (e.g., 200 mg/m²/day, 300 mg/m²/day, or 500 mg/m²/day). In an exemplary embodiment, the dose of cyclophosphamide is about 300 mg/m²/day. In some embodiments, the lymphodepletion step includes administration of fludarabine at a dose of between about 20 mg/m²/day and about 900 mg/m²/day (e.g., 20 mg/m²/day, 25 mg/m²/day, 30 mg/m²/day, or 60 mg/m²/day). In an exemplary embodiment, the dose of fludarabine is about 30 mg/m²/day.

In some embodiment, the lymphodepletion step includes administration of cyclophosphamide at a dose of between about 200 mg/m²/day and about 2000 mg/m²/day (e.g., 200 mg/m²/day, 300 mg/m²/day, or 500 mg/m²/day), and fludarabine at a dose of between about 20 mg/m²/day and about 900 mg/m²/day (e.g., 20 mg/m²/day, 25 mg/m²/day, 30 mg/m²/day, or 60 mg/m²/day). In an exemplary embodiment, the lymphodepletion step includes administration of cyclophosphamide at a dose of about 300 mg/m²/day, and fludarabine at a dose of about 30 mg/m²/day.

In an exemplary embodiment, the dosing of cyclophosphamide is 300 mg/m²/day over three days, and the dosing of fludarabine is 30 mg/m²/day over three days.

Dosing of lymphodepletion chemotherapy may be scheduled on Days -6 to -4 (with a -1 day window, i.e., dosing on Days -7 to -5) relative to T cell (e.g., CAR-T, TCR-T, a modified T cell, etc.) infusion on Day 0.

In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including 300 mg/m² of cyclophosphamide by intravenous infusion 3 days prior to administration of the modified T cells. In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including 300 mg/m² of cyclophosphamide by intravenous infusion for 3 days prior to administration of the modified T cells.

In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including fludarabine at a dose of between about 20 mg/m²/day and about 900 mg/m²/day (e.g., 20 mg/m²/day, 25 mg/m²/day, 30 mg/m²/day, or 60 mg/m²/day). In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including fludarabine at a dose of 30 mg/m² for 3 days.

In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including cyclophosphamide at a dose of between about 200 mg/m²/day and about 2000 mg/m²/day (e.g., 200 mg/m²/day, 300 mg/m²/day, or 500 mg/m²/day), and fludarabine at a dose of between about 20 mg/m²/day and about 900 mg/m²/day (e.g., 20 mg/m²/day, 25 mg/m²/day, 30 mg/m²/day, or 60 mg/m²/day). In an exemplary embodiment, for a subject having cancer, the subject receives lymphodepleting chemotherapy including cyclophosphamide at a dose of about 300 mg/m²/day, and fludarabine at a dose of 30 mg/m² for 3 days.

Cells of the invention can be administered in dosages and routes and at times to be determined in appropriate pre-clinical and clinical experimentation and trials. Cell compositions may be administered multiple times at dosages within these ranges. Administration of the cells of the invention may be combined with other methods useful to treat the desired disease or condition as determined by those of skill in the art.

It is known in the art that one of the adverse effects following infusion of CAR T cells is the onset of immune activation, known as cytokine release syndrome (CRS). CRS is immune activation resulting in elevated inflammatory cytokines. CRS is a known on-target toxicity, development of which likely correlates with efficacy. Clinical and laboratory measures range from mild CRS (constitutional symptoms and/or grade-2 organ toxicity) to severe CRS (sCRS; grade ≥3 organ toxicity, aggressive clinical intervention, and/or potentially life threatening). Clinical features include: high fever, malaise, fatigue, myalgia, nausea, anorexia, tachycardia/hypotension, capillary leak, cardiac dysfunction, renal impairment, hepatic failure, and disseminated

intravascular coagulation. Dramatic elevations of cytokines including interferon-gamma, granulocyte macrophage colony-stimulating factor, IL-10, and IL-6 have been shown following CAR T-cell infusion. One CRS signature is elevation of cytokines including IL-6 (severe elevation), IFN-gamma, TNF-alpha (moderate), and IL-2 (mild). Elevations in clinically available markers of inflammation including ferritin and C-reactive protein (CRP) have also been observed to correlate with the CRS syndrome. The presence of CRS generally correlates with expansion and progressive immune activation of adoptively transferred cells. It has been demonstrated that the degree of CRS severity is dictated by disease burden at the time of infusion as patients with high tumor burden experience a more sCRS.

Accordingly, the invention provides for, following the diagnosis of CRS, appropriate CRS management strategies to mitigate the physiological symptoms of uncontrolled inflammation without dampening the antitumor efficacy of the engineered cells (e.g., CAR T cells). CRS management strategies are known in the art. For example, systemic corticosteroids may be administered to rapidly reverse symptoms of sCRS (e.g., grade 3 CRS) without compromising initial antitumor response.

In some embodiments, an anti-IL-6R antibody may be administered. An example of an anti-IL-6R antibody is the Food and Drug Administration-approved monoclonal antibody tocilizumab, also known as atilizumab (marketed as Actemra, or RoActemra). Tocilizumab is a humanized monoclonal antibody against the interleukin-6 receptor (IL-6R). Administration of tocilizumab has demonstrated near-immediate reversal of CRS.

CRS is generally managed based on the severity of the observed syndrome and interventions are tailored as such. CRS management decisions may be based upon clinical signs and symptoms and response to interventions, not solely on laboratory values alone.

Mild to moderate cases generally are treated with symptom management with fluid therapy, non-steroidal anti-inflammatory drug (NSAID) and antihistamines as needed for adequate symptom relief. More severe cases include patients with any degree of hemodynamic instability; with any hemodynamic instability, the administration of tocilizumab is recommended. The first-line management of CRS may be tocilizumab, in some embodiments, at the labeled dose of 8 mg/kg IV over 60 minutes (not to exceed 800 mg/dose); tocilizumab can be repeated Q8 hours. If suboptimal response to the first dose of tocilizumab, additional doses of tocilizumab may be considered. Tocilizumab can be administered alone or in combination with corticosteroid therapy. Patients with continued or progressive CRS symptoms, inadequate clinical improvement in 12-18 hours or poor response to tocilizumab, may be treated with high-dose corticosteroid therapy, generally hydrocortisone 100 mg IV or methylprednisolone 1-2 mg/kg. In patients with more severe hemodynamic instability or more severe respiratory symptoms, patients may be administered high-dose corticosteroid therapy early in the course of the CRS. CRS management guidance may be based on published standards (Lee et al. (2019) *Biol Blood Marrow Transplant*, doi.org/10.1016/j.bbmt.2018.12.758; Neelapu et al. (2018) *Nat Rev Clin Oncology*, 15:47; Teachey et al. (2016) *Cancer Discov*, 6(6):664-679).

Features consistent with Macrophage Activation Syndrome (MAS) or Hemophagocytic lymphohistiocytosis (HLH) have been observed in patients treated with CAR-T therapy (Henter, 2007), coincident with clinical manifestations of the CRS. MAS appears to be a reaction to immune

activation that occurs from the CRS, and should therefore be considered a manifestation of CRS. MAS is similar to HLH (also a reaction to immune stimulation). The clinical syndrome of MAS is characterized by high grade non-remitting fever, cytopenias affecting at least two of three lineages, and hepatosplenomegaly. It is associated with high serum ferritin, soluble interleukin-2 receptor, and triglycerides, and a decrease of circulating natural killer (NK) activity.

In one aspect, the invention includes a method of treating cancer in a subject in need thereof, comprising administering to the subject any one of the modified immune or precursor cells disclosed herein. Yet another aspect of the invention includes a method of treating cancer in a subject in need thereof, comprising administering to the subject a modified immune or precursor cell generated by any one of the methods disclosed herein.

In one aspect, the invention includes a method for treating cancer in a subject in need thereof, the method comprising administering to the subject a modified cell comprising a CAR, wherein the CAR comprises a nanobody retrieved by the sequential tumor-related antibody and antigen retrieving (STAR) method, a transmembrane domain, and an intracellular signaling domain.

In another aspect, the invention provides a method for treating cancer in a subject in need thereof, the method comprising administering to the subject a modified cell comprising a CAR, wherein the CAR comprises a CD13-specific nanobody, a transmembrane domain, and an intracellular signaling domain.

A method for treating cancer in a subject in need thereof, the method comprising administering to the subject a modified cell comprising a CAR, wherein the CAR comprises a CDH17-specific nanobody, a transmembrane domain, and an intracellular signaling domain.

In certain embodiments, the T cell is a human cell. In certain embodiments, the T cell is autologous.

Nanobodies and Antibody Drug Conjugates (ADCs)

Also provided in the invention are nanobodies. In certain embodiments, the nanobody specifically binds to CD13. In certain embodiments, the nanobody specifically binds to CDH17. In certain embodiments, the nanobody specifically binds to the first domain of CDH17. In certain embodiments, the nanobody is encoded by the nucleotide sequence of SEQ ID NO: 1. In certain embodiments, the nanobody comprises the amino acid sequence of SEQ ID NO: 2. In certain embodiments, the nanobody comprises a CDR1 sequence comprising the amino acid sequence of SEQ ID NO: 3. In certain embodiments, the nanobody comprises a CDR2 sequence comprising the amino acid sequence of SEQ ID NO: 4. In certain embodiments, the nanobody comprises a CDR3 sequence comprising the amino acid sequence of ID NO: 5.

In certain embodiments, the nanobody comprises the amino acid sequence of SEQ ID NO: 19. In certain embodiments, the nanobody comprises a CDR1 sequence comprising the amino acid sequence of SEQ ID NO: 25. In certain embodiments, the nanobody comprises a CDR2 sequence comprising the amino acid sequence of SEQ ID NO: 26. In certain embodiments, the nanobody comprises a CDR3 sequence comprising the amino acid sequence of SEQ ID NO: 27.

In certain embodiments, the nanobody comprises the amino acid sequence of SEQ ID NO: 23. In certain embodiments, the nanobody comprises a CDR1 sequence comprising the amino acid sequence of SEQ ID NO: 28. In certain embodiments, the nanobody comprises a CDR2 sequence comprising the amino acid sequence of SEQ ID NO: 29. In

certain embodiments, the nanobody comprises a CDR3 sequence comprising the amino acid sequence of SEQ ID NO: 30.

Also provided in the invention are antibody drug conjugates (ADCs), comprising any of the nanobodies disclosed herein linked/conjugated to a drug (e.g. a cancer cell-killing drug), or a toxin, or a radioisotope. In certain embodiments, NETs can be specifically targeted via CAR T cells or a NET-specific ADC to effectively eradicate NET cells. In certain embodiments, specific cell surface tumor-associated antigens and corresponding antibodies (nanobodies) are identified used the STAR system described elsewhere herein and conjugated to a drug/toxin/radioisotope. In certain embodiments, the invention includes an ADC comprising nanobody VH157 linked to a drug/toxin/radioisotope. In certain embodiments, the ADC comprises nanobody VH163 linked to a drug/toxin/radioisotope. In certain embodiments, the ADC comprises nanobody VHH1 linked to a drug/toxin/radioisotope. In certain embodiments, the ADC comprises a CD13-specific nanobody linked to a drug/toxin/radioisotope. In certain embodiments, the ADC comprises a CDH17-specific nanobody linked to a drug/toxin/radioisotope.

In certain embodiments, the ADC comprises a nanobody comprising the amino acid sequence of any one of SEQ ID NOs: 2, 19, or 24, linked to a drug/toxin/radioisotope.

The antibodies (nanobodies) can be linked/conjugated to any drug or cancer targeting agent known to one of skill in the art, including but not limited to maytansinoid (DM1), or SSTR2-binding octreotide, or toxin, including but not limited to paclitaxel, auristatin (MMAE and MMAF), dauxrubicin, duocarmycin A, 5-fluorouracil, methotrexate, tubulin polymerization inhibitors, ravtansine (DM4), Ricin A. The nanobodies may also be linked to radioactive isotopes, including but not limited to 90Y and 177Lu, 111In, and such ADC can also be used for imaging applications (e.g. imaging cancer cells).

In certain embodiments, the ADC comprises nanobody VHH1 conjugated to DM1. In certain embodiments, the ADC comprises a first VHH1 nanobody linked to a second VHH1 nanobody (VLV) conjugated to DM1. In certain embodiments, the VHH1 or VLV is linked to DM1 and another agent including, but not limited to, any of the above described drugs, toxins, and radioactive isotopes.

Vectors

A vector may be used to introduce the CAR into a T cell as described elsewhere herein. In certain aspects, the invention includes vectors comprising nucleic acid sequences encoding a CAR. The vector can comprise a plasmid vector, viral vector, retrotransposon (e.g. piggyback, sleeping beauty), site directed insertion vector (e.g. CRISPR, Zn finger nucleases, TALEN), suicide expression vector, lentiviral vector, RNA vector, or other known vector in the art.

The production of any of the molecules described herein can be verified by sequencing. Expression of the full length proteins may be verified using immunoblot, immunohistochemistry, flow cytometry or other technology well known and available in the art.

The present invention also provides a vector in which DNA of the present invention is inserted. Vectors, including those derived from retroviruses such as lentivirus, are suitable tools to achieve long-term gene transfer since they allow long-term, stable integration of a transgene and its propagation in daughter cells. Lentiviral vectors have the added advantage over vectors derived from onco-retroviruses, such as murine leukemia viruses, in that they can transduce non-proliferating cells, such as hepatocytes. They

also have the added advantage of resulting in low immunogenicity in the subject into which they are introduced.

The expression of natural or synthetic nucleic acids is typically achieved by operably linking a nucleic acid or portions thereof to a promoter, and incorporating the construct into an expression vector. The vector is one generally capable of replication in a mammalian cell, and/or also capable of integration into the cellular genome of the mammal. Typical vectors contain transcription and translation terminators, initiation sequences, and promoters useful for regulation of the expression of the desired nucleic acid sequence.

The nucleic acid can be cloned into any number of different types of vectors. For example, the nucleic acid can be cloned into a vector including, but not limited to a plasmid, a phagemid, a phage derivative, an animal virus, and a cosmid. Vectors of particular interest include expression vectors, replication vectors, probe generation vectors, and sequencing vectors.

The expression vector may be provided to a cell in the form of a viral vector. Viral vector technology is well known in the art and is described, for example, in Sambrook et al., 2012, *MOLECULAR CLONING: A LABORATORY MANUAL*, volumes 1-4, Cold Spring Harbor Press, NY), and in other virology and molecular biology manuals. Viruses, which are useful as vectors include, but are not limited to, retroviruses, adenoviruses, adeno-associated viruses, herpes viruses, and lentiviruses. In general, a suitable vector contains an origin of replication functional in at least one organism, a promoter sequence, convenient restriction endonuclease sites, and one or more selectable markers, (e.g., WO 01/96584; WO 01/29058; and U.S. Pat. No. 6,326,193).

Additional promoter elements, e.g., enhancers, regulate the frequency of transcriptional initiation. Typically, these are located in the region 30-110 bp upstream of the start site, although a number of promoters have recently been shown to contain functional elements downstream of the start site as well. The spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. In the thymidine kinase (tk) promoter, the spacing between promoter elements can be increased to 50 bp apart before activity begins to decline. Depending on the promoter, it appears that individual elements can function either cooperatively or independently to activate transcription.

An example of a promoter is the immediate early cytomegalovirus (CMV) promoter sequence. This promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. However, other constitutive promoter sequences may also be used, including, but not limited to the simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, the EF-1 alpha promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter. Further, the invention should not be limited to the use of constitutive promoters. Inducible promoters are also contemplated as part of the invention. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression

when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionein promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

In order to assess expression of a polypeptide or portions thereof, the expression vector to be introduced into a cell can also contain either a selectable marker gene or a reporter gene or both to facilitate identification and selection of expressing cells from the population of cells sought to be transfected or infected through viral vectors. In other aspects, the selectable marker may be carried on a separate piece of DNA and used in a co-transfection procedure. Both selectable markers and reporter genes may be flanked with appropriate regulatory sequences to enable expression in the host cells. Useful selectable markers include, for example, antibiotic-resistance genes, such as neo and the like.

Reporter genes are used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a polypeptide whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assessed at a suitable time after the DNA has been introduced into the recipient cells. Suitable reporter genes may include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (e.g., Ui-Tei et al., 2000 *FEBS Letters* 479: 79-82). Suitable expression systems are well known and may be prepared using known techniques or obtained commercially. In general, the construct with the minimal 5' flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions may be linked to a reporter gene and used to evaluate agents for the ability to modulate promoter-driven transcription.

Introduction of Nucleic Acids

Methods of introducing nucleic acids into a cell include physical, biological and chemical methods. Physical methods for introducing a polynucleotide, such as RNA, into a host cell include calcium phosphate precipitation, lipofection, particle bombardment, microinjection, electroporation, and the like. RNA can be introduced into target cells using commercially available methods which include electroporation (Amaya Nucleofector-II (Amaya Biosystems, Cologne, Germany)), (ECM 830 (BTX) (Harvard Instruments, Boston, Mass.) or the Gene Pulser II (BioRad, Denver, Colo.), Multiporator (Eppendorf, Hamburg Germany). RNA can also be introduced into cells using cationic liposome mediated transfection using lipofection, using polymer encapsulation, using peptide mediated transfection, or using biolistic particle delivery systems such as "gene guns" (see, for example, Nishikawa, et al. *Hum Gene Ther.*, 12(8):861-70 (2001)).

Biological methods for introducing a polynucleotide of interest into a host cell include the use of DNA and RNA vectors. Viral vectors, and especially retroviral vectors, have become the most widely used method for inserting genes into mammalian, e.g., human cells. Other viral vectors can be derived from lentivirus, poxviruses, herpes simplex virus 1, adenoviruses and adeno-associated viruses, and the like. See, for example, U.S. Pat. Nos. 5,350,674 and 5,585,362.

Chemical means for introducing a polynucleotide into a host cell include colloidal dispersion systems, such as macromolecule complexes, nanocapsules, microspheres, beads, and lipid-based systems including oil-in-water emulsions, micelles, mixed micelles, and liposomes. An exemplary

colloidal system for use as a delivery vehicle in vitro and in vivo is a liposome (e.g., an artificial membrane vesicle).

Lipids suitable for use can be obtained from commercial sources. For example, dimyristyl phosphatidylcholine ("DMPC") can be obtained from Sigma, St. Louis, MO; 5 dicetyl phosphate ("DCP") can be obtained from K & K Laboratories (Plainview, NY); cholesterol ("Choi") can be obtained from Calbiochem-Behring; dimyristyl phosphatidylglycerol ("DMPG") and other lipids may be obtained from Avanti Polar Lipids, Inc. (Birmingham, AL). Stock solutions of lipids in chloroform or chloroform/methanol can be stored at about -20° C. Chloroform is used as the only solvent since it is more readily evaporated than methanol. "Liposome" is a generic term encompassing a variety of single and multilamellar lipid vehicles formed by the generation of enclosed lipid bilayers or aggregates. Liposomes can be characterized as having vesicular structures with a phospholipid bilayer membrane and an inner aqueous medium. Multilamellar liposomes have multiple lipid layers separated by aqueous medium. They form spontaneously when phospholipids are suspended in an excess of aqueous solution. The lipid components undergo self-rearrangement before the formation of closed structures and entrap water and dissolved solutes between the lipid bilayers (Ghosh et al., 1991 *Glycobiology* 5: 505-10). However, compositions that have different structures in solution than the normal vesicular structure are also encompassed. For example, the lipids may assume a micellar structure or merely exist as nonuniform aggregates of lipid molecules. Also contemplated are lipofectamine-nucleic acid complexes.

Regardless of the method used to introduce exogenous nucleic acids into a host cell or otherwise expose a cell to the inhibitor of the present invention, in order to confirm the presence of the nucleic acids in the host cell, a variety of assays may be performed. Such assays include, for example, "molecular biological" assays well known to those of skill in the art, such as Southern and Northern blotting, RT-PCR and PCR; "biochemical" assays, such as detecting the presence or absence of a particular peptide, e.g., by immunological means (ELISAs and Western blots) or by assays described herein to identify agents falling within the scope of the invention.

Moreover, the nucleic acids may be introduced by any means, such as transducing the expanded T cells, transfecting the expanded T cells, and electroporating the expanded T cells. One nucleic acid may be introduced by one method and another nucleic acid may be introduced into the T cell by a different method.

RNA

In one embodiment, the nucleic acids introduced into the T cell are RNA. In another embodiment, the RNA is mRNA that comprises in vitro transcribed RNA or synthetic RNA. The RNA is produced by in vitro transcription using a polymerase chain reaction (PCR)-generated template. DNA of interest from any source can be directly converted by PCR into a template for in vitro mRNA synthesis using appropriate primers and RNA polymerase. The source of the DNA can be, for example, genomic DNA, plasmid DNA, phage DNA, cDNA, synthetic DNA sequence or any other appropriate source of DNA. The desired template for in vitro transcription is a chimeric membrane protein. By way of example, the template encodes an antibody, a fragment of an antibody or a portion of an antibody. By way of another example, the template comprises an extracellular domain comprising a single chain variable domain of an antibody, such as anti-CD3, and an intracellular domain of a co-stimulatory molecule. In one embodiment, the template for

the RNA chimeric membrane protein encodes a chimeric membrane protein comprising an extracellular domain comprising an antigen binding domain derived from an antibody to a co-stimulatory molecule, and an intracellular domain derived from a portion of an intracellular domain of CD28 and 4-1BB.

PCR can be used to generate a template for in vitro transcription of mRNA which is then introduced into cells. Methods for performing PCR are well known in the art. Primers for use in PCR are designed to have regions that are substantially complementary to regions of the DNA to be used as a template for the PCR. "Substantially complementary", as used herein, refers to sequences of nucleotides where a majority or all of the bases in the primer sequence are complementary, or one or more bases are non-complementary, or mismatched. Substantially complementary sequences are able to anneal or hybridize with the intended DNA target under annealing conditions used for PCR. The primers can be designed to be substantially complementary to any portion of the DNA template. For example, the primers can be designed to amplify the portion of a gene that is normally transcribed in cells (the open reading frame), including 5' and 3' UTRs. The primers can also be designed to amplify a portion of a gene that encodes a particular domain of interest. In one embodiment, the primers are designed to amplify the coding region of a human cDNA, including all or portions of the 5' and 3' UTRs. Primers useful for PCR are generated by synthetic methods that are well known in the art. "Forward primers" are primers that contain a region of nucleotides that are substantially complementary to nucleotides on the DNA template that are upstream of the DNA sequence that is to be amplified. "Upstream" is used herein to refer to a location 5' to the DNA sequence to be amplified relative to the coding strand. "Reverse primers" are primers that contain a region of nucleotides that are substantially complementary to a double-stranded DNA template that are downstream of the DNA sequence that is to be amplified. "Downstream" is used herein to refer to a location 3' to the DNA sequence to be amplified relative to the coding strand.

Chemical structures that have the ability to promote stability and/or translation efficiency of the RNA may also be used. The RNA preferably has 5' and 3' UTRs. In one embodiment, the 5' UTR is between zero and 3000 nucleotides in length. The length of 5' and 3' UTR sequences to be added to the coding region can be altered by different methods, including, but not limited to, designing primers for PCR that anneal to different regions of the UTRs. Using this approach, one of ordinary skill in the art can modify the 5' and 3' UTR lengths required to achieve optimal translation efficiency following transfection of the transcribed RNA.

The 5' and 3' UTRs can be the naturally occurring, endogenous 5' and 3' UTRs for the gene of interest. Alternatively, UTR sequences that are not endogenous to the gene of interest can be added by incorporating the UTR sequences into the forward and reverse primers or by any other modifications of the template. The use of UTR sequences that are not endogenous to the gene of interest can be useful for modifying the stability and/or translation efficiency of the RNA. For example, it is known that AU-rich elements in 3' UTR sequences can decrease the stability of mRNA. Therefore, 3' UTRs can be selected or designed to increase the stability of the transcribed RNA based on properties of UTRs that are well known in the art.

In one embodiment, the 5' UTR can contain the Kozak sequence of the endogenous gene. Alternatively, when a 5' UTR that is not endogenous to the gene of interest is being

added by PCR as described above, a consensus Kozak sequence can be redesigned by adding the 5' UTR sequence. Kozak sequences can increase the efficiency of translation of some RNA transcripts, but does not appear to be required for all RNAs to enable efficient translation. The requirement for Kozak sequences for many mRNAs is known in the art. In other embodiments the 5' UTR can be derived from an RNA virus whose RNA genome is stable in cells. In other embodiments various nucleotide analogues can be used in the 3' or 5' UTR to impede exonuclease degradation of the mRNA.

To enable synthesis of RNA from a DNA template without the need for gene cloning, a promoter of transcription should be attached to the DNA template upstream of the sequence to be transcribed. When a sequence that functions as a promoter for an RNA polymerase is added to the 5' end of the forward primer, the RNA polymerase promoter becomes incorporated into the PCR product upstream of the open reading frame that is to be transcribed. In one embodiment, the promoter is a T7 polymerase promoter, as described elsewhere herein. Other useful promoters include, but are not limited to, T3 and SP6 RNA polymerase promoters. Consensus nucleotide sequences for T7, T3 and SP6 promoters are known in the art.

In one embodiment, the mRNA has both a cap on the 5' end and a 3' poly(A) tail which determine ribosome binding, initiation of translation and stability mRNA in the cell. On a circular DNA template, for instance, plasmid DNA, RNA polymerase produces a long concatameric product which is not suitable for expression in eukaryotic cells. The transcription of plasmid DNA linearized at the end of the 3' UTR results in normal sized mRNA which is not effective in eukaryotic transfection even if it is polyadenylated after transcription.

On a linear DNA template, phage T7 RNA polymerase can extend the 3' end of the transcript beyond the last base of the template (Schenborn and Mierendorf, *Nuc Acids Res.*, 13:6223-36 (1985); Nacheva and Berzal-Herranz, *Eur. J. Biochem.*, 270:1485-65 (2003)).

The conventional method of integration of polyA/T stretches into a DNA template is molecular cloning. However polyA/T sequence integrated into plasmid DNA can cause plasmid instability, which is why plasmid DNA templates obtained from bacterial cells are often highly contaminated with deletions and other aberrations. This makes cloning procedures not only laborious and time consuming but often not reliable. That is why a method which allows construction of DNA templates with polyA/T 3' stretch without cloning highly desirable.

The polyA/T segment of the transcriptional DNA template can be produced during PCR by using a reverse primer containing a polyT tail, such as 100T tail (size can be 50-5000 T), or after PCR by any other method, including, but not limited to, DNA ligation or in vitro recombination. Poly(A) tails also provide stability to RNAs and reduce their degradation. Generally, the length of a poly(A) tail positively correlates with the stability of the transcribed RNA. In one embodiment, the poly(A) tail is between 100 and 5000 adenosines.

Poly(A) tails of RNAs can be further extended following in vitro transcription with the use of a poly(A) polymerase, such as *E. coli* polyA polymerase (E-PAP). In one embodiment, increasing the length of a poly(A) tail from 100 nucleotides to between 300 and 400 nucleotides results in about a two-fold increase in the translation efficiency of the RNA. Additionally, the attachment of different chemical groups to the 3' end can increase mRNA stability. Such attachment can contain modified/artificial nucleotides,

aptamers and other compounds. For example, ATP analogs can be incorporated into the poly(A) tail using poly(A) polymerase. ATP analogs can further increase the stability of the RNA.

5' caps also provide stability to RNA molecules. In a preferred embodiment, RNAs produced by the methods disclosed herein include a 5' cap. The 5' cap is provided using techniques known in the art and described herein (Cougot, et al., *Trends in Biochem. Sci.*, 29:436-444 (2001); Stepinski, et al., *RNA*, 7:1468-95 (2001); Elango, et al., *Biochim. Biophys. Res. Commun.*, 330:958-966 (2005)).

The RNAs produced by the methods disclosed herein can also contain an internal ribosome entry site (IRES) sequence. The IRES sequence may be any viral, chromosomal or artificially designed sequence which initiates cap-independent ribosome binding to mRNA and facilitates the initiation of translation. Any solutes suitable for cell electroporation, which can contain factors facilitating cellular permeability and viability such as sugars, peptides, lipids, proteins, antioxidants, and surfactants can be included.

In some embodiments, the RNA is electroporated into the cells, such as in vitro transcribed RNA.

The disclosed methods can be applied to the modulation of T cell activity in basic research and therapy, in the fields of cancer, stem cells, acute and chronic infections, and autoimmune diseases, including the assessment of the ability of the genetically modified T cell to kill a target cancer cell.

The methods also provide the ability to control the level of expression over a wide range by changing, for example, the promoter or the amount of input RNA, making it possible to individually regulate the expression level. Furthermore, the PCR-based technique of mRNA production greatly facilitates the design of the mRNAs with different structures and combination of their domains.

One advantage of RNA transfection methods of the invention is that RNA transfection is essentially transient and a vector-free. A RNA transgene can be delivered to a lymphocyte and expressed therein following a brief in vitro cell activation, as a minimal expressing cassette without the need for any additional viral sequences. Under these conditions, integration of the transgene into the host cell genome is unlikely. Cloning of cells is not necessary because of the efficiency of transfection of the RNA and its ability to uniformly modify the entire lymphocyte population.

Genetic modification of T cells with in vitro-transcribed RNA (IVT-RNA) makes use of two different strategies both of which have been successively tested in various animal models. Cells are transfected with in vitro-transcribed RNA by means of lipofection or electroporation. It is desirable to stabilize IVT-RNA using various modifications in order to achieve prolonged expression of transferred IVT-RNA.

Some IVT vectors are known in the literature which are utilized in a standardized manner as template for in vitro transcription and which have been genetically modified in such a way that stabilized RNA transcripts are produced. Currently protocols used in the art are based on a plasmid vector with the following structure: a 5' RNA polymerase promoter enabling RNA transcription, followed by a gene of interest which is flanked either 3' and/or 5' by untranslated regions (UTR), and a 3' polyadenyl cassette containing 50-70 A nucleotides. Prior to in vitro transcription, the circular plasmid is linearized downstream of the polyadenyl cassette by type II restriction enzymes (recognition sequence corresponds to cleavage site). The polyadenyl cassette thus corresponds to the later poly(A) sequence in the transcript. As a result of this procedure, some nucleotides remain as part of the enzyme cleavage site after linearization and

extend or mask the poly(A) sequence at the 3' end. It is not clear, whether this nonphysiological overhang affects the amount of protein produced intracellularly from such a construct.

RNA has several advantages over more traditional plasmid or viral approaches. Gene expression from an RNA source does not require transcription and the protein product is produced rapidly after the transfection. Further, since the RNA has to only gain access to the cytoplasm, rather than the nucleus, and therefore typical transfection methods result in an extremely high rate of transfection. In addition, plasmid based approaches require that the promoter driving the expression of the gene of interest be active in the cells under study.

In another aspect, the RNA construct is delivered into the cells by electroporation. See, e.g., the formulations and methodology of electroporation of nucleic acid constructs into mammalian cells as taught in US 2004/0014645, US 2005/0052630A1, US 2005/0070841A1, US 2004/0059285A1, US 2004/0092907A1. The various parameters including electric field strength required for electroporation of any known cell type are generally known in the relevant research literature as well as numerous patents and applications in the field. See e.g., U.S. Pat. Nos. 6,678,556, 7,171,264, and 7,173,116. Apparatus for therapeutic application of electroporation are available commercially, e.g., the MedPulser™ DNA Electroporation Therapy System (Inovio/Genetronics, San Diego, Calif), and are described in patents such as U.S. Pat. Nos. 6,567,694; 6,516,223, 5,993,434, 6,181,964, 6,241,701, and 6,233,482; electroporation may also be used for transfection of cells in vitro as described e.g. in US20070128708A1. Electroporation may also be utilized to deliver nucleic acids into cells in vitro. Accordingly, electroporation-mediated administration into cells of nucleic acids including expression constructs utilizing any of the many available devices and electroporation systems known to those of skill in the art presents an exciting new means for delivering an RNA of interest to a target cell.

Sources of T Cells

Prior to expansion, a source of T cells is obtained from a subject. Non-limiting examples of subjects include humans, dogs, cats, mice, rats, and transgenic species thereof. Preferably, the subject is a human. T cells can be obtained from a number of sources, including peripheral blood mononuclear cells, bone marrow, lymph node tissue, spleen tissue, umbilical cord, and tumors. In certain embodiments, any number of T cell lines available in the art, may be used. In certain embodiments, T cells can be obtained from a unit of blood collected from a subject using any number of techniques known to the skilled artisan, such as Ficoll separation. In one embodiment, cells from the circulating blood of an individual are obtained by apheresis or leukapheresis. The apheresis product typically contains lymphocytes, including T cells, monocytes, granulocytes, B cells, other nucleated white blood cells, red blood cells, and platelets. The cells collected by apheresis may be washed to remove the plasma fraction and to place the cells in an appropriate buffer or media, such as phosphate buffered saline (PBS) or wash solution lacks calcium and may lack magnesium or may lack many if not all divalent cations, for subsequent processing steps. After washing, the cells may be resuspended in a variety of biocompatible buffers, such as, for example, Ca-free, Mg-free PBS. Alternatively, the undesirable components of the apheresis sample may be removed and the cells directly resuspended in culture media.

In another embodiment, T cells are isolated from peripheral blood by lysing the red blood cells and depleting the monocytes, for example, by centrifugation through a PERCOLL™ gradient. Alternatively, T cells can be isolated from umbilical cord. In any event, a specific subpopulation of T cells can be further isolated by positive or negative selection techniques.

The cord blood mononuclear cells so isolated can be depleted of cells expressing certain antigens, including, but not limited to, CD34, CD8, CD14, CD19 and CD56. Depletion of these cells can be accomplished using an isolated antibody, a biological sample comprising an antibody, such as ascites, an antibody bound to a physical support, and a cell bound antibody.

Enrichment of a T cell population by negative selection can be accomplished using a combination of antibodies directed to surface markers unique to the negatively selected cells. A preferred method is cell sorting and/or selection via negative magnetic immunoadherence or flow cytometry that uses a cocktail of monoclonal antibodies directed to cell surface markers present on the cells negatively selected. For example, to enrich for CD4+ cells by negative selection, a monoclonal antibody cocktail typically includes antibodies to CD14, CD20, CD11b, CD16, HLA-DR, and CD8.

For isolation of a desired population of cells by positive or negative selection, the concentration of cells and surface (e.g., particles such as beads) can be varied. In certain embodiments, it may be desirable to significantly decrease the volume in which beads and cells are mixed together (i.e., increase the concentration of cells), to ensure maximum contact of cells and beads. For example, in one embodiment, a concentration of 2 billion cells/ml is used. In one embodiment, a concentration of 1 billion cells/ml is used. In a further embodiment, greater than 100 million cells/ml is used. In a further embodiment, a concentration of cells of 10, 15, 20, 25, 30, 35, 40, 45, or 50 million cells/ml is used. In yet another embodiment, a concentration of cells from 75, 80, 85, 90, 95, or 100 million cells/ml is used. In further embodiments, concentrations of 125 or 150 million cells/ml can be used. Using high concentrations can result in increased cell yield, cell activation, and cell expansion.

T cells can also be frozen after the washing step, which does not require the monocyte-removal step. While not wishing to be bound by theory, the freeze and subsequent thaw step provides a more uniform product by removing granulocytes and to some extent monocytes in the cell population. After the washing step that removes plasma and platelets, the cells may be suspended in a freezing solution. While many freezing solutions and parameters are known in the art and will be useful in this context, in a non-limiting example, one method involves using PBS containing 20% DMSO and 8% human serum albumin, or other suitable cell freezing media. The cells are then frozen to -80° C. at a rate of 1° per minute and stored in the vapor phase of a liquid nitrogen storage tank. Other methods of controlled freezing may be used as well as uncontrolled freezing immediately at -20° C. or in liquid nitrogen.

In one embodiment, the population of T cells is comprised within cells such as peripheral blood mononuclear cells, cord blood cells, a purified population of T cells, and a T cell line. In another embodiment, peripheral blood mononuclear cells comprise the population of T cells. In yet another embodiment, purified T cells comprise the population of T cells.

In certain embodiments, T regulatory cells (Tregs) can be isolated from a sample. The sample can include, but is not limited to, umbilical cord blood or peripheral blood. In

certain embodiments, the Tregs are isolated by flow-cytometry sorting. The sample can be enriched for Tregs prior to isolation by any means known in the art. The isolated Tregs can be cryopreserved, and/or expanded prior to use. Methods for isolating Tregs are described in U.S. Pat. Nos. 7,754,482, 8,722,400, and 9,555,105, and U.S. patent application Ser. No. 13/639,927, contents of which are incorporated herein in their entirety.

Expansion of T Cells

As demonstrated by the data disclosed herein, expanding the T cells by the methods disclosed herein can be multiplied by about 10 fold, 20 fold, 30 fold, 40 fold, 50 fold, 60 fold, 70 fold, 80 fold, 90 fold, 100 fold, 200 fold, 300 fold, 400 fold, 500 fold, 600 fold, 700 fold, 800 fold, 900 fold, 1000 fold, 2000 fold, 3000 fold, 4000 fold, 5000 fold, 6000 fold, 7000 fold, 8000 fold, 9000 fold, 10,000 fold, 100,000 fold, 1,000,000 fold, 10,000,000 fold, or greater, and any and all whole or partial integers therebetween. In one embodiment, the T cells expand in the range of about 20 fold to about 50 fold.

Following culturing, the T cells can be incubated in cell medium in a culture apparatus for a period of time or until the cells reach confluency or high cell density for optimal passage before passing the cells to another culture apparatus. The culturing apparatus can be of any culture apparatus commonly used for culturing cells in vitro. Preferably, the level of confluence is 70% or greater before passing the cells to another culture apparatus. More preferably, the level of confluence is 90% or greater. A period of time can be any time suitable for the culture of cells in vitro. The T cell medium may be replaced during the culture of the T cells at any time. Preferably, the T cell medium is replaced about every 2 to 3 days. The T cells are then harvested from the culture apparatus whereupon the T cells can be used immediately or cryopreserved to be stored for use at a later time. In one embodiment, the invention includes cryopreserving the expanded T cells. The cryopreserved T cells are thawed prior to introducing nucleic acids into the T cell.

In another embodiment, the method comprises isolating T cells and expanding the T cells. In another embodiment, the invention further comprises cryopreserving the T cells prior to expansion. In yet another embodiment, the cryopreserved T cells are thawed for electroporation with the RNA encoding the chimeric membrane protein.

Another procedure for ex vivo expansion cells is described in U.S. Pat. No. 5,199,942 (incorporated herein by reference). Expansion, such as described in U.S. Pat. No. 5,199,942 can be an alternative or in addition to other methods of expansion described herein. Briefly, ex vivo culture and expansion of T cells comprises the addition to the cellular growth factors, such as those described in U.S. Pat. No. 5,199,942, or other factors, such as flt3-L, IL-1, IL-3 and c-kit ligand. In one embodiment, expanding the T cells comprises culturing the T cells with a factor selected from the group consisting of flt3-L, IL-1, IL-3 and c-kit ligand.

The culturing step as described herein (contact with agents as described herein or after electroporation) can be very short, for example less than 24 hours such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, or 23 hours. The culturing step as described further herein (contact with agents as described herein) can be longer, for example 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or more days.

Various terms are used to describe cells in culture. Cell culture refers generally to cells taken from a living organism and grown under controlled condition. A primary cell culture

is a culture of cells, tissues or organs taken directly from an organism and before the first subculture. Cells are expanded in culture when they are placed in a growth medium under conditions that facilitate cell growth and/or division, resulting in a larger population of the cells. When cells are expanded in culture, the rate of cell proliferation is typically measured by the amount of time required for the cells to double in number, otherwise known as the doubling time.

Each round of subculturing is referred to as a passage.

When cells are subcultured, they are referred to as having been passaged. A specific population of cells, or a cell line, is sometimes referred to or characterized by the number of times it has been passaged. For example, a cultured cell population that has been passaged ten times may be referred to as a P10 culture. The primary culture, i.e., the first culture following the isolation of cells from tissue, is designated P0. Following the first subculture, the cells are described as a secondary culture (P1 or passage 1). After the second subculture, the cells become a tertiary culture (P2 or passage 2), and so on. It will be understood by those of skill in the art that there may be many population doublings during the period of passaging; therefore the number of population doublings of a culture is greater than the passage number. The expansion of cells (i.e., the number of population doublings) during the period between passaging depends on many factors, including but is not limited to the seeding density, substrate, medium, and time between passaging.

In one embodiment, the cells may be cultured for several hours (about 3 hours) to about 14 days or any hourly integer value in between. Conditions appropriate for T cell culture include an appropriate media (e.g., Minimal Essential Media or RPMI Media 1640 or, X-vivo 15, (Lonza)) that may contain factors necessary for proliferation and viability, including serum (e.g., fetal bovine or human serum), interleukin-2 (IL-2), insulin, IFN-gamma, IL-4, IL-7, GM-CSF, IL-10, IL-12, IL-15, TGF-beta, and TNF- α . or any other additives for the growth of cells known to the skilled artisan. Other additives for the growth of cells include, but are not limited to, surfactant, plasmanate, and reducing agents such as N-acetyl-cysteine and 2-mercaptoethanol. Media can include RPMI 1640, AIM-V, DMEM, MEM, α -MEM, F-12, X-Vivo 15, and X-Vivo 20, Optimizer, with added amino acids, sodium pyruvate, and vitamins, either serum-free or supplemented with an appropriate amount of serum (or plasma) or a defined set of hormones, and/or an amount of cytokine(s) sufficient for the growth and expansion of T cells. Antibiotics, e.g., penicillin and streptomycin, are included only in experimental cultures, not in cultures of cells that are to be infused into a subject. The target cells are maintained under conditions necessary to support growth, for example, an appropriate temperature (e.g., 37° C.) and atmosphere (e.g., air plus 5% CO₂).

The medium used to culture the T cells may include an agent that can co-stimulate the T cells. For example, an agent that can stimulate CD3 is an antibody to CD3, and an agent that can stimulate CD28 is an antibody to CD28. This is because, as demonstrated by the data disclosed herein, a cell isolated by the methods disclosed herein can be expanded approximately 10 fold, 20 fold, 30 fold, 40 fold, 50 fold, 60 fold, 70 fold, 80 fold, 90 fold, 100 fold, 200 fold, 300 fold, 400 fold, 500 fold, 600 fold, 700 fold, 800 fold, 900 fold, 1000 fold, 2000 fold, 3000 fold, 4000 fold, 5000 fold, 6000 fold, 7000 fold, 8000 fold, 9000 fold, 10,000 fold, 100,000 fold, 1,000,000 fold, 10,000,000 fold, or greater. In one embodiment, the T cells expand in the range of about 20 fold to about 50 fold, or more by culturing the electroporated population. In one embodiment, human T regulatory cells

are expanded via anti-CD3 antibody coated KT64.86 artificial antigen presenting cells (aAPCs). Methods for expanding and activating T cells can be found in U.S. Pat. Nos. 7,754,482, 8,722,400, and 9,555,105, contents of which are incorporated herein in their entirety.

In one embodiment, the method of expanding the T cells can further comprise isolating the expanded T cells for further applications. In another embodiment, the method of expanding can further comprise a subsequent electroporation of the expanded T cells followed by culturing. The subsequent electroporation may include introducing a nucleic acid encoding an agent, such as a transducing the expanded T cells, transfecting the expanded T cells, or electroporating the expanded T cells with a nucleic acid, into the expanded population of T cells, wherein the agent further stimulates the T cell. The agent may stimulate the T cells, such as by stimulating further expansion, effector function, or another T cell function.

Pharmaceutical Compositions

Pharmaceutical compositions of the present invention may comprise the modified T cell as described herein, in combination with one or more pharmaceutically or physiologically acceptable carriers, diluents or excipients. Such compositions may comprise buffers such as neutral buffered saline, phosphate buffered saline and the like; carbohydrates such as glucose, mannose, sucrose or dextrans, mannitol; proteins; polypeptides or amino acids such as glycine; antioxidants; chelating agents such as EDTA or glutathione; adjuvants (e.g., aluminum hydroxide); and preservatives. Compositions of the present invention are preferably formulated for intravenous administration.

Pharmaceutical compositions of the present invention may be administered in a manner appropriate to the disease to be treated (or prevented). The quantity and frequency of administration will be determined by such factors as the condition of the patient, and the type and severity of the patient's disease, although appropriate dosages may be determined by clinical trials.

The cells of the invention to be administered may be autologous, allogeneic or xenogeneic with respect to the subject undergoing therapy.

Cells of the invention can be administered in dosages and routes and at times to be determined in appropriate pre-clinical and clinical experimentation and trials. Cell compositions may be administered multiple times at dosages within these ranges. Administration of the cells of the invention may be combined with other methods useful to treat the desired disease or condition as determined by those of skill in the art.

It can generally be stated that a pharmaceutical composition comprising the modified T cells described herein may be administered at a dosage of 10^4 to 10^9 cells/kg body weight, in some instances 10^5 to 10^6 cells/kg body weight, including all integer values within those ranges. T cell compositions may also be administered multiple times at these dosages. The cells can be administered by using infusion techniques that are commonly known in immunotherapy (see, e.g., Rosenberg et al., New Eng. J. of Med. 319:1676, 1988). The optimal dosage and treatment regime for a particular patient can readily be determined by one skilled in the art of medicine by monitoring the patient for signs of disease and adjusting the treatment accordingly.

The administration of the modified T cells of the invention may be carried out in any convenient manner known to those of skill in the art. The cells of the present invention may be administered to a subject by aerosol inhalation, injection, ingestion, transfusion, implantation or transplantation. The

compositions described herein may be administered to a patient transarterially, subcutaneously, intradermally, intratumorally, intranodally, intramedullary, intramuscularly, by intravenous (i.v.) injection, or intraperitoneally. In other instances, the cells of the invention are injected directly into a site of inflammation in the subject, a local disease site in the subject, a lymph node, an organ, a tumor, and the like.

It should be understood that the method and compositions that would be useful in the present invention are not limited to the particular formulations set forth in the examples. The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the cells, expansion and culture methods, and therapeutic methods of the invention, and are not intended to limit the scope of what the inventors regard as their invention.

The practice of the present invention employs, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry and immunology, which are well within the purview of the skilled artisan. Such techniques are explained fully in the literature, such as, "Molecular Cloning: A Laboratory Manual", fourth edition (Sambrook, 2012); "Oligonucleotide Synthesis" (Gait, 1984); "Culture of Animal Cells" (Freshney, 2010); "Methods in Enzymology" "Handbook of Experimental Immunology" (Weir, 1997); "Gene Transfer Vectors for Mammalian Cells" (Miller and Calos, 1987); "Short Protocols in Molecular Biology" (Ausubel, 2002); "Current Protocols in Immunology" (Coligan, 2002). These techniques are applicable to the production of the polynucleotides and polypeptides of the invention, and, as such, may be considered in making and practicing the invention. Particularly useful techniques for particular embodiments will be discussed in the sections that follow.

EXPERIMENTAL EXAMPLES

The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only, and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather, should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Without further description, it is believed that one of ordinary skill in the art can, using the preceding description and the following illustrative examples, make and utilize the compounds of the present invention and practice the claimed methods. The following working examples therefore, specifically point out the preferred embodiments of the present invention, and are not to be construed as limiting in any way the remainder of the disclosure.

The Materials and Methods used in the performance of the experiments disclosed herein are now described. Nanobody Phage Library Panning

The titer of concentrated phage was generally 1×10^{13} to 1×10^{14} cfu/mL. In the first round of panning, the phage was applied to bind THP-1 directly, followed by wash, acidic glycine elution and neutralization, to enrich the tumor specific Nb-expressing phage. In the second round of panning, the elution of phage was amplified and concentrated to bind Jurkat cells as the negative selection. The unbound phage from Jurkat cells were incubated with THP-1 cells to enrich and then to be eluted with glycine solution.

Construction of Nanobody CAR Vector

Generation of the Nanobody CAR (Nb CAR) constructs in lentiviral vector was shown in FIG. 2A. In brief, the plasmid backbone is pHIV-eGFP, a third-generation self-inactivating lentiviral vector plasmid. Nb CAR constructs, including CD8 signal peptide (SP), SfiI cut sites, IgG4m hinge, 4-1BB domain and CD3z domain, were custom synthesized by IDT-DNA. IRES-GFP was used as a marker to track Nb CAR positive cells. Nb CAR expression was confirmed by Western blots using anti-CD3z (FIG. 8A) (Abcam). Nanobody fragments were cloned into in-frame from pComb3XSS into pHIV-eGFP-CAR at the two sequential SfiI sites.

In Vivo Tumor-Mediated Selection of Nanobodies from CAR T Cells

Nanobody cDNAs isolated from cell based display were amplified with PCR using primers "2nd round Forward" and "2nd round Backward", followed by SfiI digestion and ligation into pHIV-eGFP-CAR vector. The ligation product was transformed to *E. coli* competent cells, followed by lentivirus packaging and infecting activated human primary T cells to generate Nb-lib-CAR-expressing T cells. The resulting T cells or untransduced (UTD) T cells were administered intravenously into NSG mice bearing either THP-1 tumor or K562 tumor, as shown in FIG. 1B. Fourteen days later, cDNAs for nanobodies were amplified by PCR from the genomic DNA isolated from the tumor infiltrated Nb-lib-CAR T cells, with the primer Nb-Amp-Forward, 5'-ATTTCAGGTGTCGTGAGCGG (SEQ ID NO: 6); and Nb-Amp-Backward, 5'-AGGAGAAGGACCCACAAGT (SEQ ID NO: 7). The PCR product was digested with SfiI and inserted into pComb-3XSS, followed by randomly picking of individual clones and sequencing with primer Nb-Seq-Forward, 5-CAGCTATCGCGATTGCAGT (SEQ ID NO: 8).

Animals and In Vivo Models

All laboratory mice were maintained on a 12 hr light-dark cycle. NOD/Shi-scid/IL-2R γ null (NSG) mice, 8-12 weeks old, were obtained from Jackson Laboratories. NSG mice were inoculated with 1×10^7 cells of THP-1 or HL60 subcutaneously, or with 0.5×10^7 cells of K562 subcutaneously. When tumor volume reached 100 mm³ about 12 days after xenograft, Nb CAR T cells or untransduced (UTD) human T cells (1×10^7 cells) were administered via tail vein. Mice and tumors were monitored every other day. Tumor dimensions were measured with Vernier calipers and tumor volume was calculated as $\frac{1}{2}$ larger diameter $\times$ (smaller diameter).

NSG mice were conditioned by Busulfex (30 mg/kg) 24 hrs prior to tail injection with 2×10^7 of patient-derived AML cells. Two weeks later CAR or UTD T cells were transduced into the mice. The recipient mice were sacrificed at the experimental end point based on the protocol, and the long bones (femurs), spleens and livers were collected for histological analysis by H & E staining. Mice were sacrificed according to protocol when moribund or upon the development of hind-limb paralysis.

Statistical Analysis

Microsoft Excel and GraphPad Prism software was used for statistical analysis.

Student's t test was used to determine the significance of the results unless otherwise indicated. Kaplan-Meier statistical analysis was performed using the log rank test. In the figures, asterisks denote statistically significant p values (*, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$), and "ns" indicates lack of statistical significance ($p > 0.05$).

Cell Lines, Cell Culture, Plasmids and Antibodies

The THP-1, Jurkat, K562, HL60, U937, MV4-11, NB4 and SKOV3 cell lines were obtained from the ATCC and maintained in RPMI1640 with 10% FBS and 1% penicillin/streptomycin (R10 medium) and maintained at 37 C and 5% CO₂. HEK293T cells were obtained from ATCC and cultured in DMEM supplemented with 10% fetal bovine serum (FBS). NET NT-3 cell line was cultured in RPMI medium supplemented with 10% FBS, penicillin/streptomycin, HEPES, EGF (20 ng/mL), and FGF2 (10 ng/mL). Deidentified Patient derived AML cells were obtained from the University of Pennsylvania Stem Cell and Xenograft Core facility, and maintained in the R10 medium. Normal donor total T cells were obtained from the Human Immunology Core at University of Pennsylvania, and maintained in the R10 medium.

pComb3XSS was a gift from Carlos Barbas (Addgene plasmid #63890). pHIV-EGFP was a gift from Bryan Welm & Zena Werb (Addgene plasmid #21373). lentiCRISPR v2 was a gift from Feng Zhang (Addgene plasmid #52961). Human membrane protein cDNA library were provided by High-throughput Screen Core at University of Pennsylvania. Nanobody Phage Library Construction from the THP-1 Cell-Immunized Llama

Llama was immunized with 2×10^7 of THP-1 cells monthly for 3 times (Caprologics, MA). Peripheral blood mononuclear cells (PBMCs) isolation, RNA extraction and cDNA synthesis were performed as previously described (Zhang, X. et al. (2016). *J Immunol* 196, 1591-1603). In brief, PBMCs were isolated from whole blood of immunized llama through Ficoll gradient centrifugation (GE). Total RNA was extracted by the RNeasy Mini Kit (QIAGEN) as the manufacturer's instructions, followed by cDNA synthesis from SuperScript™ III First-Strand Synthesis System (Invitrogen).

Nanobody encoding fragments were amplified from llama PBMCs cDNA by two rounds of PCR as instructed in "Phage Display, a laboratory Manual", with two pairs of nesting primers containing two flanking SfiI (GGCCNNNNNGGCC; SEQ ID NO: 9) sites as listed. 1st round Forward, 5'-GTCCTGGCTGCTCTTCTACAAGG (SEQ ID NO: 10); Backward, 5'-GGTACGTGCTGTTGAAGTGTTC (SEQ ID NO: 11). 2nd round Forward, 5'-GAGGAGGAGGAGGAG-GAGGCGGGGCCAGGCGGCCAGGTG CAGCTGCAGGAGTCTGGRGGAGG (SEQ ID NO: 12); Backward, 5'-GAGGAG GAGGAGGAG-GAGCCTGGCCGGCCTGGC-CACTAGTGGCGGCCGCTGAGGAGA CCGTGACCTGGGT (SEQ ID NO: 13). The PCR product was digested with SfiI enzymes, followed by cloning into the phagemid system pComb3XSS. The ligation product was transformed into *E. coli* competent cells, yielding $\sim 10^9$ single clones/ μ g ligated DNA in transformation. The resulting phage library was generated by infection with wild type M13 helper phage, and concentrated from supernatant for binding cancer cell surface.

In Vitro Analysis of T Cell Function:

For T-cell transduction, HEK293T cells were co-transfected with lenti-vector plasmid, pSPAX2 and VSV-G plasmids DNA to produce the lentivirus 48h after transfection. Normal donor T cells were positively selected from leukapheresis packs using anti-CD4 and CD8 microbeads (Miltenyi), expanded in vitro with anti-CD3/CD28 beads (Invitrogen) for up to 12 days. Total T cells were transduced with lentiviral 24 hours after activation. The resulting virus

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from the supernatant were concentrated via ultracentrifugation at 25,000 g for 2.5 h at 4 C.

Killing assays were performed as previously described (Cao, L. F. et al. (2010) *Cytometry A* 77, 534-545). In brief, target cells were labeled by anti-CD33 (BD) for detecting cell number with flow cytometry analysis or labeled by CellTrace Far Red for tracing cell division. Target cells were incubated with effector T cells for 16 hrs at a series of ratios. Cells were then harvested, washed, and stained by Propidium Iodide prior to flow cytometry analysis. Quantification were calculated by either Countbright beads or volume.

To detect cytokine secretion, effector and target cells were incubated at a 1:1 ratio in R10 medium for 16 hrs as indicated. Supernatant was analyzed using Human TNF-alpha or IFN-gamma DuoSet ELISA kits according to the manufacturer's instructions (R&D System).

To detect cell degranulation, activated and Nb CAR transduced or untransduced T cells (1×10^5 cells) were co-cultured with THP-1 or K526 cells at a 1:1 ratio in 96-well plates for 4 hrs, in the presence of APC-conjugated anti-CD107a antibody, followed by wash and flow cytometry analysis.

To monitor cell proliferation assay, T cells were labeled by CellTrace™ Far Red Cell Proliferation Kit (Invitrogen) as the manufacturer's instructions. The reaction was quenched with R10 medium, and the cells were washed twice. T cells were incubated at a 1:1 ratio with heat-inactivated target cells for 96 hrs.

THP-1 CD13 Knockout Cell Line:

To knockout CD13 in THP-1 cell line, LentiCRISPRv2.0 vector was applied according to the protocol (Sanjana, N. E. et al, (2014) *Nature methods* 11, 783-784). sgRNA targeting human CD13 were listed as below, sgCD13-1 ATGGCCGGCTCATCGAAGCA (SEQ ID NO: 14), sgCD13-2 CTTCCCATGCTTCGATGAGC (SEQ ID NO: 15), sgCD13-3 CTTCATGGGGCCATAGACCT (SEQ ID NO: 16). Plasmids were confirmed by sequencing and packaged into lentivirus, followed by infecting THP-1 cells. Single clones were randomly picked up from each group after puromycin selection (2.5 ug/mL) for 4 days.

Switchable CAR (sCAR) T System:

Anti-PNE single chain variable fragment (scFv) was custom synthesized by GeneArt (Rodgers, D. T. et al. (2016) *Proceedings of the National Academy of Sciences of the United States of America* 113, E459-468), followed by insertion into pHIV-41BB-CD3z vector. sCAR lentivirus was packaged and used to transduce human T cells. sCAR expression was detected by flow cytometry and western blot. Nb157 with C-terminal PNE (Nb157-C-PNE) or with N-terminal PNE (Nb157-N-PNE) were constructed by molecular cloning, followed by prokaryotic expression and purification via Ni-NTA affinity (QIAGEN) in TOP10 (Invitrogen) induced by isopropyl-β-D-thiogalactoside (IPTG). Development of Nanobody (NB)-Driven CAR T Cells for AML Therapy.

After preparing the variable domains heavy-chain camelid antibodies (VHH) CAR expressing library, the lentivirus vectors were packaged and the primary human T cells were transformed to prepare the VHH CAR expressing T cell library, followed by treatment the NSG mice with THP-1 tumor or K562 tumor. After two weeks, the tumor was removed and a PCR was done on the VHH and sequence.

Through this method, VHH157 and VHH163 were identified as THP-1 specific nanobodies. VHH157 CAR and VHH163 CAR were constructed with the structure as shown below.

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VHH157 CAR (SEQ ID NO: 17):

MALPVTALLPLALLHAARPGSAAQAAQVQLQESGGGLVQPGGSLSL
CTASGFTFSSYSMAWVRQAPGKGPWVSGIYPSDGKTRYADFKGRFSI
SRDNAKNMPLYLQMNLEPEDTALYYCARGITGLGPRGQGTQVTVSSAA
*TSGQAGQSGESKYGPCCPCPAS*IWAPLAGTCGVLLSLVITLY**KRG**
RKKLLYIFKQPFMRPVQTTQEEEDGCSRFPEEEGGCELVRKFSRSADA
PAYKQGNQLYNELNLGRREEYDVLDKRRGRDPGEMGKPRKPNQEGLY
NELQDKMAEAYSEIGMKGERRRGKGDGLYQGLSTATKDTYDALHMQA
LPFR

Bold is signal peptide (SEQ ID NO: 18)

Underlined is VHH157 (SEQ ID NO: 19)

Italicized is IgG4mh (SEQ ID NO: 20)

Dotted underlined is CD8 transmembrane (SEQ ID NO: 21)

Double underlined is 4-1BB and CD3z (SEQ ID NO: 22)

SYSMA is the CDR1 (SEQ ID NO: 25)

GIYPSDGKTRYADFKGR is the CDR2 (SEQ ID NO: 26)

ARGITGLGP is the CDR3 (SEQ ID NO: 27)

VHH163 CAR (SEQ ID NO: 23):

MALPVTALLPLALLHAARPGSAAQAAQVQLQESGGGLVQPGGSLRLS
CVPSGFTFDGYLIGWFRQAPGSEKAVSCISVNGDRNTYADSVKGRFTI
SRDNAKNVTVYLQMNLSLRPNDAIYYCATRRGNRLYNNNCPYFEYGTWQ
GTQVTVSSAAATSGQAGQSGESKYGPCCPCPAS
*YIWAPLAGTCGVLLSLVITLY***KSRGKLLYIFKQPFMRPVQTTQEEED**
GCSRFPEEEGGCELVRKFSRSADAPAYKQGNQLYNELNLGRREEYD
VLDKRRGRDPGEMGKPRKPNQEGLYNELQDKMAEAYSEIGMKGERR

RGKGDGLYQGLSTATKDTYDALHMOALPPR

Bold is signal peptide (SEQ ID NO: 18)

Underlined is VHH163 (SEQ ID NO: 24)

Italicized is IgG4mh (SEQ ID NO: 20)

Dotted underlined is CD8 transmembrane (SEQ ID NO: 21)

Double underlined is 4-1BB and CD3z (SEQ ID NO: 22)

GYLIG is the CDR1 (SEQ ID NO: 28)

CISVNGDRNTYADSVKGR is the CDR2 (SEQ ID NO: 29)

ATRRGNRLYNNNCPYFEYGT is the CDR3 (SEQ ID NO: 30)

VHH157-BBz pLenti-CAR Vector

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 3841 ccagtgaacc ccttgctcct gccgctggcc ttgctgctcc acgccgccag gccgggatcc
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 4081 gatagaacaa actatgcaga ttccgtgaag ggccgattca ccattctccag agacaacgcc
 4141 aagaacacgg tgatctgca aatgaacagc ctgagacctc acgacacagc catttattac
 4201 tgtgcgaccc gcaggggaaa tcgtctttat aataataact gccatactt tgagtatggc
 4261 acctggggcc aggggaccca ggtcaccgtc tcctcagcgg ccgccactag tggccaggcc
 4321 ggccagtcgg gagagagcaa gtacggccct ccctgcccc cttgcctcgc tagctacac
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 4501 caaactactc aagaggaaga tggctgtagc tgccgatttc cagaagaaga agaaggagga
 4561 tgtgaactga gagtgaagt cagcaggagc gcagacgccc ccgctgacaa gcagggccag
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 4681 agacgtggcc gggaccctga gatgggggga aagccgagaa ggaagaacct tcagggaaggc
 4741 ctgtacaatg aactgcagaa agataagatg gcggaggcct acagtgagat tgggatgaaa
 4801 ggcgagcgcc ggaggggcaa ggggcacgat ggctttacc agggctctcag tacagccacc
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 4921 ctccctcccc ccccctaac gttactggcc gaagccgctt ggaataaggc cgggtgtcgt
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5041 ctggccctgt cttcttgacg agcattccta ggggtctttc ccctctcgcc aaaggaatgc
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5341 tgaaggatgc ccagaaggta ccccatgtga tgggatctga tctggggcct cggtgcacat
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5581 aagttcagcg tgtccggcga gggcgagggc gatgccacct acggcaagct gacctgaag
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5701 tacggcgtgc agtgcttcag ccgctacccc gaccacatga agcagcacga cttcttcaag
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5821 tacaagaccc gcgccagagt gaagtctcag ggcgacaccc tgggtaaccg catcgagctg
5881 aaggcgatcg acttcaagga ggacggcaac atcctggggc acaagctgga gtacaactac
5941 aacagccaca acgtctatat catggccgac aagcagaaga acggcatcaa ggtgaacttc
6001 aagatccgcc acaacatcga ggacggcagc gtgcagctcg ccgaccacta ccagcagaac
6061 acccccacg gcgacggccc cgtgctgctg cccgacaacc actacctgag caccagtc
6121 gccctgagca aagaccccaa cgagaagcgc gatcacatgg tctgctgga gttcgtgacc
6181 gccgccggga tcaactctcg catggacgag ctgtacaagt aaagcggccg catcgatacc
6241 gtcgacctcg atcgagacct agaaaaacat ggagcaatca caagtagcaa tacagcagct
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6541 ccagggatca gatatccact gacctttgga tgggtgtaca agctagtacc agttgagcaa
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7321 ccggtaaagc acgacttate gccactggca gcagccactg gtaacaggat tagcagagcg
7381 aggtatgtag gcggtgctac agagtctctg aagtggtggc ctaactacgg ctacactaga

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7441 agaacagtat ttggtatctg cgctctgctg aagccagtta ccttcggaaa aagagttggt
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 7681 atcttcacct agatcctttt aaattaaaaa tgaagtttta aatcaatcta aagtatatat
 7741 gagtaaacct ggtctgacag ttaccaatgc ttaatcagt aggcacctat ctcagcgatc
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 8041 ccagttaata gtttgcgcaa cgttgttgcc attgctacag gcatcgtggt gtcacgctcg
 8101 tcgtttggta tggcttcatt cagctccggt tcccaacgat caaggcgagt tacatgatcc
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 8281 ccacccgtaa gatgcttttc tgtgactggt gagtactcaa ccaagtcatt ctgagaatag
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 8401 agcagaactt taaaagtgt catcattgga aaacgttctt cggggcgaaa actctcaagg
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 8521 gcatctttta ctttcaccag cgtttctggg tgagcaaaaa caggaaggca aaatgcgca
 8581 aaaaagggaa taaggcgac acggaaatgt tgaatactca tactcttct ttttcaatat
 8641 tattgaagca tttatcaggg ttattgtctc atgagcggat acatatttga atgtatttag
 8701 aaaaataaac aaataggggt tccgcgcaca tttccccgaa aagtgccacc tgac

VHH1-BBz CAR DNA (SEQ ID NO: 33) (as shown in FIG. 19):

ATGGCCTTACCACTGACCGCTTGCTCCTGCGCTGGCCTTGCTGCTCC
 ACGCCGCCAGGCCGGATCCGCGGCCAGGCCGCGCCAGGTGCAGCTGCA
 GGAGTCTGGGGAGGATTTGGTGCAGCCTGGGGCTCGACAAGACTCTCC
 TGTGTATCGTCCCGACCTTTAGTTATTATGACATGGGCTGGTTCGCGC
 AGGCTCCAGGGAAGGAGCGTAGTTTCGTAGCACTGCTTAGTTGGAATGG
 GGAAAATGCAGAGTATTGAGACTCCGTGATGGGCCGTTTCACCGTCTCC
 CGAGGGAATACCCAGAATTCGGTGAATCTGCAATGAACAACCTGAAAC
 CTGAGGACACGGGCATCTATTACTGCGCAGTGACGACGGTGGAGCGCG
 GTCGGTTCGTTCTCTGGGGCCAGGGGACCCAGGTACCGTCTCCTCAGCG
 GCCGCCACTAGTGGCCAGGCCGCGCAGTCCGAGAGTCTAAGTACGGCC
 CTCCCTGCCCTCCTTGCCAGTAGCTACATCTGGGCGCCCTTGCGCGG
 GACTTGTGGGTCCTTCTCCTGTCACTGGTTATCACCCTTTACTGCAAA
 CGGGGCAGAAAGAACTCCTGTATATATCAACAACCATTTATGAGAC
 CAGTACAACTACTCAAGAGGAAGATGGCTGTAGCTGCCGATTTCAGA
 AGAAGAAGAGGAGGATGTGAAGTGAAGTTCAGCAGGAGCGCA
 GACGCCCCCGCTACAAGCAGGCCAGAACAGCTCTATAACGAGCTCA

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40 ATCTAGGACGAAGAGAGGAGTACGATGTTTTGGACAAGAGACGTGGCCG
 GGACCTTGAGATGGGGGAAAGCCGAGAAGGAAGAACCCTCAGGAAGGC
 45 CTGTACAATGAACTGCAGAAAGATAAGATGGCGAGGCCCTACAGTGAGA
 TTGGGATGAAAGGCGAGCGCCGAGGGGCAAGGGGCACGATGGCCTTTA
 50 CCAGGCTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTCACATG
 CAGGCCCTGCCCCCTCGC

55 VHH1-BBz CAR protein (SEQ ID NO: 34) (as shown in FIG. 19):

MALPVTALLPLALLHAARPGSAAQAAQVLQESGGGLVQPGGSTRLS
 60 CVSSRTFSYYDMGWFRQAPGKEREFVALLSWNGENAEYSDSVMGRFTVS
RGNTQNSVNLQMNLLKPEDTGIYYCAVTHGGARSVRSWGQGTQTVSSA
AATSGAQSGESKYGPCPCPASXIWARLAGTCGVLLLSLVITLYCK
 65 RGRKLLYIFKQPFMRPVQTTQEEEDGCSRFPEEEEGGCELRVKESRSA

FVWLIVVGGVLACYSLLVTVAFIIFWVRSKRSRLHSDYMNMTPRRP.
GPTRKHYOPYAPPRDFAAYRSASKRGRKKLLYIFKOPFMRPVQTTOEED

tained the nanobodies that preferentially bind the cell surface antigens of THP-1 with high affinity.

Example 2: Identifying Nanobodies that are
Capable of Redirecting CAR T Cells to the THP-1
Tumor In Vivo

The above isolated nanobodies preferentially bound the THP-1 cell surface proteins, however, not all of these nanobodies were capable of empowering CAR T lymphocytes with antitumor activities, which requires effective antibody-antigen interaction and appropriate immunological synapse. To address these issues and screen the active Nbs, the nanobodies were fused to CAR constructs to establish Nb-lib CAR T cells via lentivirus transduction, followed by treating the NSG mice bearing THP-1 tumor or K562 tumor burdens, untransduced (UTD) T cells as a control group (FIG. 1). The CAR construct was composed of a signal peptide (SP), nanobody fragment, IgG4 mutation (IgG4m) hinge, CD8 transmembrane domain (TM), 4-1BB and CD3zeta domain, as shown in FIG. 2A.

It was hypothesized that the most suitable nanobodies that redirected the cognate CAR T cells to eradicate the tumor would be enriched along with the activation and proliferation of T cells upon encountering the targets. Moreover, the in vivo selection model could reduce and clean the noise signals of the nanobodies from Nb-lib CAR T as the non-tumor-specific Nb-lib CAR T cells may be distributed to and/or absorbed by various mouse tissues. Therefore, desirable nanobodies would guide the CAR T cells to the THP-1 tumor, but not K562 tumor, in vivo. To do this, 14 days post T cells infusion, the UTD T cells-treated THP-1 tumor, Nb-lib CAR T cells-treated THP-1 tumor, and the Nb-lib CAR T cells-treated K562 tumor tissues were collected to decode the Nbs sequences that were integrated into the genome of the CAR T cells infiltrated in the tumors. PCR results indicated that Nb sequences were only retrieved from Nb-lib CAR T cells-treated THP-1 tumor (FIG. 1B, lane 2), but not from either K562 tumor or UTD-treated THP-1 tumor (FIG. 1). About 400 individual clones from amplified nanobodies (FIG. 1B, lane 2) were sequenced. Among the most enriched five unique Nbs, four of them, i.e. Nb157, Nb163, Nb176 and Nb393, specifically bound THP-1 cells, but not Jurkat or K562 cells, as shown by flow cytometry analysis (FIG. 1C). Moreover, Nb157 and Nb163 also bound other AML cell lines like HL60, NB4, and U937 (FIG. 7). Together, these results indicated that the STAR system was capable of enriching and isolating multiple nanobodies that specifically bound AML cells.

Example 3: All the Nanobodies Isolated by the
STAR System Redirected CAR T Cells to Potently
Kill the AML Cells In Vitro

Based on the design of STAR system, most of the identified nanobodies should be able to guide CAR T cells to kill target tumor cells. To test this, individual CARs (FIG. 2A) were transduced and expressed into activated primary human T cells (FIGS. 8A-8B). In vitro cytotoxicity assays showed that Nb157 CAR T cells potently and specifically killed THP-1 AML cells, but not as much in the K562 CML cells or Jurkat T leukemia cells, in a CAR T effector to target cells ratio-dependent manner (FIGS. 2B and 2D). As a control, UTD T cells did not cause obvious cytotoxicity (FIGS. 2B-2D). Similarly, Nb163, Nb176, Nb393 CAR T cells also killed THP-1 AML specifically, but not K562 CML cells (FIG. 2C and FIGS. 9A-9B). To determine if the

Nb CAR T cells can also kill other Nb-recognizing AML cells, we incubated Nb157 CAR T with HL60 cell, another human AML cell line, and found that they also caused specific cytotoxicity against the AML cells, but not the Jurkat cells (FIG. 2D).

Activated T cells, upon encountering the targets, release various cytokines including $\text{TNF}\alpha$ and $\text{IFN}\gamma$, degranulate to deliver cytolytic proteins along with the mobilization of CD107a to the cell surface, and enhance proliferation. To determine if the target AML cells can specifically activate the Nb CAR T cells, THP-1 cells were co-incubated with either Nb157 or Nb163 CAR T. THP-1 cells, but not control K562 cells, stimulated Nb157 and Nb163 CAR T cells to release cytokines including $\text{TNF}\alpha$ and $\text{IFN}\gamma$ (FIGS. 2E-2F). As a control, neither target THP-1 cells nor K562 cells stimulated cytokine release from UTD T cells (FIGS. 2E-2F). A similar increase in cytokine release was detected in the co-incubation of HL60 with Nb157 CAR T cells (FIG. 8C). Moreover, THP-1 cells, but not K562 cells, specifically induced the CAR T degranulation as shown by increase in cell surface CD107a (FIG. 2G). Further, THP-1 cells, but not K562 cells, specifically induced proliferation of the Nb157 and Nb163 CAR T cells (FIGS. 2H, 2I). Together, these findings demonstrated that Nb157 and Nb163 CAR T cells were specifically activated by target THP-1 AML cells, leading to enhanced proliferation, cytokine release and degranulation to kill the target AML cells.

Example 4: Nb CAR T Cells Potently Induced the
AML Tumor Regression In Vivo

To determine whether the Nb CAR T cells suppressed the AML tumors in vivo, THP-1 cells were transplanted into NSG mice subcutaneously. When the tumor reached $\sim 150 \text{ mm}^3$, Nb157 CAR T, Nb163 CAR T, or control UTD T cells were injected intravenously into the mice. The tumors from the control UTD T-treated mice grew exponentially (FIGS. 3A-3B). Notably, the tumors in the Nb157 or Nb163 CAR T-treated mice failed to grow substantially, and eventually regressed (FIGS. 3A-3B). Histological studies indicated that while the control tumor contained abundant tumor cells (FIG. 3C, UTD), the Nb CAR T cells eradicated the cancer cells, leaving the fibrotic tissues in the site (FIG. 3C).

Next, the impact of Nb157 CAR on other AML cells in vivo was examined. HL60 AML cells were transplanted into NSG mice to form tumors, followed by treatment of either Nb157 CAR or UTD T cells. Compared to the UTD T cell group, Nb157 CAR T cells exhibited significant antitumor efficacy against HL60 tumor burden in vivo (FIG. 3D). As a control, both Nb157 and Nb163 CAR T cells failed to induce the regression of the non-target K562 tumors (FIG. 3E), supporting the specificity of CAR T suppressing tumors. Further, it was determined if long-term maintenance of the CAR T treated mice would show recurrence of the tumors. Low dose of Nb157 or Nb163 CAR T cells showed complete remission of THP-1 tumor without recurrence for a long time (FIG. 3F), supporting the eradication of the tumor in the mice.

To explore whether the CAR T cells guided by other STAR-isolated nanobodies were capable of killing the AML cells in vitro and in vivo, CAR constructs were generated with either Nb173 or Nb393, two additional THP-1 AML specific nanobodies. The in vitro cytotoxicity assay showed that, similar to Nb157 and Nb163 CAR T cells, Nb173 and Nb393 CAR T cells potently killed THP-1 cells, but not much K562 cells (FIGS. 9A-9B), indicating specific killing by these two types of Nb CAR T cells. Furthermore, in the

animal studies Nb176 and Nb393 also performed remarkable antitumor efficacy against THP-1 xenografts (FIGS. 9C-9D). Histological studies also showed that treatment of the Nb CAR T cells, but not UTD T cells, eradicated the tumor cells (FIG. 9E). Collectively, these results demonstrated that all the STAR-isolated nanobodies were capable of empowering the cognate CAR T cells with significant antitumor activities, highlighting an effective approach to isolate the CAR-compatible antibodies to kill cancer cells.

Example 5: Identification of CD13 as a Target to Kill AML Cells by CAR T Cells

To identify the protein antigens of the isolated nanobodies, about 3000 human cell membrane protein cDNAs were prepared and transfected into HEK293T cells individually, followed by flow cytometry detection with the nanobody-expressing phage and FITC-labeled secondary antibody against phage M13 protein (FIG. 4A). Both Nb157 and Nb163 bound the cells transfected with human full length aminopeptidase N (APN, aka CD13) (FIG. 4B), but Nb176 or Nb393 did not bind CD13 (FIG. 4B). As a confirmation, human CD13 cDNA expression was also shown by western blot (FIG. 4C). CD13 is preferentially expressed in acute myeloid blast cells. It is also expressed moderately in normal myeloid cells and in digestive tract including liver and pancreas; however, knockout of this gene does not cause lethality in mice.

To confirm that Nb157 or Nb163 CAR T cells indeed killed the AML cells by targeting CD13, three individual gRNAs/Cas9 targeting human CD13 were transduced into THP-1 cells by lentivirus, followed by selection and single cell clone expansion. The stable CD13^{-/-} THP-1 cell lines were verified by Western blot (FIG. 4D). Consistent with previous findings herein, flow cytometry analysis showed that CD13 knockout abrogated the binding of Nb157 and Nb163 (FIG. 4E, middle and right panels), as well as the killing of Nb157 CAR T cells to the target cells (FIG. 4F). CD13 knockout diminished the killing of Nb163 to the targets significantly, but cannot abrogated the effect completely, which indicated Nb163 might have off-CD13 antigen on the THP-1 cell surface (FIG. 4F). These results demonstrated that Nb157 CAR T cells performed potent antitumor efficacy against AML by specifically targeting CD13.

Example 6: Nb157 CAR T Cells Exhibit Antitumor Activity in Patient-Derived AML Cells in NSG Mouse Model

To further determine whether the Nb157 CAR T cells can also kill the patient-derived AML cells, CD13 expression of the leukemia cells from patient AML2844 was examined. Both Nb157 and Nb163 detected the CD13 expression on the primary AML cells (FIGS. 5A-5B). In vitro cytotoxicity assay also showed that both Nb157 and Nb163 CAR T cells, but not UTD T cells, potently lysed the patient-derived AML cells (FIGS. 5C-5D). Thereafter, the therapeutic effect of the CAR T cells on the mice with the patient-derived AML cells was examined. To this end, the NSG mice were transplanted with the primary AML cells, followed by treatment with either control UTD or Nb157 CAR T cells. The Kaplan Maier curve showed that the control mice died out 45 days following the primary AML infusion, however, Nb157 CAR T cells infusion significantly prolonged the life of the treated mice (FIG. 5E).

To investigate the dynamics of the leukemia and T cells in the clinical relevant mouse model, changes in AML cells and UTD/CAR T cells in bone marrow, spleen and peripheral blood were monitored. The results showed that in the first week after T cells injection, few CD33+ AML cells, but a large percentage (>90% of hCD45+) of CD3+ T cells were detectable in bone marrow in both UTD and Nb157 CAR T cells groups (FIGS. 5F-5G). Notably, in the second week after the T injection, an increasing number of CD33+ AML appeared in the bone marrow of the UTD T cell-injected recipients (FIG. 5H), but injection of the Nb157 CAR T cells markedly prevented the accumulation of the CD33+ AML cells in the bone marrow (FIG. 5I). Consistently, cell quantification in peripheral blood showed that CD33+ AML cells reduced substantially following injection of Nb157 CAR T cells (FIG. 10A). In the meantime, much more CAR T cells than the control UTD T cells were detectable in the peripheral blood after the T cell infusion (FIG. 10B). However, the CAR T cells in peripheral blood reduced in number at the third week after CAR T infusion, likely reflecting tumor regression and subsiding of the anti-tumor response.

At the end points of experiment (day 45 for UTD group, FIG. 5J; and day 90 for Nb157 CAR group, FIG. 5K), the mouse spleens were harvested and processed for immunofluorescent staining for CD3+ T cells and CD33+ AML cells. The results indicated that a large number of CD33+ AML cells were detected in the spleen of control mice injected with UTD T cells but there were no obvious T cells (FIG. 5J). In contrast, in the spleens of the CAR T-injected mice, a large number of CD3+ CAR T cells were detected, but the CD33+ AML cells were eradicated (FIG. 5K). Consistently, there was apparent enrichment for CAR T cells in the bone marrow and spleen, increases in GFP percentage from initial 30% to more than 70% in the CAR T injected mice (FIG. 8B and FIG. 10C). Meanwhile, Nb157 CAR can induce T cells persistent memory phenotypes including both the central memory and the effector memory populations (FIG. 10C), which were correlated with complete remissions in CAR T clinical therapy. Together, these results indicated that Nb157 CAR T cells effectively eliminated the patient-derived AML cells in the bone marrow and spleen of the recipient mice and remarkably prolonged the survival.

Example 7: Nanobody Switch-Activated CAR T Cells Eradicated AML In Vivo in a Controllable Manner

CAR T therapy may cause serious cytokine release syndrome and other on-target-off-tumor effects. To reduce this risk, as well as further attenuate the potential side effects of targeting moderately expressed CD13 in non-AML cells, an inducible and controllable system to manipulate the antitumor activity of Nb157 CAR T cells to AML was developed herein. Without wishing to be bound by specific theory, the CAR T cell system comprising a nanobody-switch could conditionally redirect CAR T cells to kill the tumors, but avoid permanent damage of other normal tissues to reduce the toxicity (e.g. mitigate cytokine release syndrome (CRS)).

To this end, Nb157 was fused to a peptide PNE to form an inducible nanobody switch (FIG. 6A). Nb switches were expressed in bacteria and the purified proteins were ~15 kD as expected (FIG. 11A). The PNE peptide could be recognized by an established scFv which was fused to the CAR protein to construct the switchable CAR (sCAR) T cells (FIG. 6B). The sCAR construct was expressed in human primary T cells in a dimer of ~100 kD in non-reducing

condition, which was broken down to ~50 kd in reducing condition. FITC-labeled PNE-peptide could bind sCAR T cells with EC₅₀ of 1 nM in vitro, but no binding for UTD T cells was observed (FIG. 11C). Flow cytometry analysis showed that Nb157-C-PNE and unmodified Nb157 bound THP-1 cells with a similar K_d of ~35 nM (FIG. 6C). Notably, Nb157-N-PNE slightly reduced the binding affinity to ~90 nM (FIG. 6C). In vitro cytotoxicity assay showed that addition of Nb157-C-PNE switch potently triggered the killing of THP-1 cells by the sCAR T cells, but not by the UTD T cells (FIG. 6D). Consistently, a lower affinity of Nb157-N-PNE switch slightly reduced the cytolysis-inducing activity (FIG. 6D).

To determine whether the inducible switch, in concert with the sCAR T cells, can kill tumor in mice, THP-1 cells were transplanted into NSG mice subcutaneously to form tumors. Then sCAR or UTD T cells were injected intravenously into the tumor-bearing mice separately, followed by treatment of either Nb157-C-PNE protein or PBS every other day as indicated in FIG. 6e. The results indicated that the tumors in the UTD T, or UTD T plus Nb157 switch, or sCAR T cells-treated mice all grew exponentially (FIG. 6E). Moreover, injection of the switches alone did not inhibit the tumor growth (FIG. 11C). Notably, tumors from mice treated with both sCAR T cells and Nb157-C-PNE switch significantly slowed down growth, during the first five times of switch injection (FIG. 6E). However, the tumors resumed to growth from day 10 to day 20 without the switch injection (FIG. 6E). It was determined whether resumption of the Nb157 switch injection suppress the tumor growth again. From day 21, Nb157 switch was restarted to treat the mice every other day. Interestingly, the tumor started to be shrunk and regressed from day 28. The tumor size continued to decrease until almost undetectable on day 52 along with Nb157 switch treatment for the whole process (FIG. 6E).

It was explored whether the switch CAR system can also suppress primary AML cells in mice. To this end, patient-derived AML cells were injected into the NSG mice to build leukemic mice model as determined by detection of the human CD33+ AML cells in the peripheral blood of the recipient mice (FIG. 6G). The leukemic mice were treated by sCAR T cells, followed by addition of either Nb157-C-PNE switch or PBS every other day as indicated in FIG. 6F. The appearance of CD33+ AML cells or CD3+ T cells in peripheral blood were monitored by flow cytometry analysis weekly. The results indicated that peripheral blood AML cells, following the first week of injection, gradually decreased in both sCAR T alone-injected mice and sCAR T plus Nb157 switch-injected mice (FIG. 6G), consistent with heavy leukemia infiltration in the spleen in later stage (FIG. 5H and FIG. 10A). Notably, treatment with sCAR T plus Nb157 switch, but not UTD T or sCAR T alone, increased peripheral T cells number two week after the sCAR T injection, reflecting the switch-triggered activation and proliferation of the sCAR T cells to kill the AML cells (FIG. 6H). Consistent with this observation, Kaplan Maier curve showed that treatment with sCAR T cells plus Nb157-C-PNE switch significantly protected the leukemic mice from dying compare to the treatment of sCAR T cell alone (FIG. 6F). Therefore, the results demonstrated that the Nb157-C-PNE switch targeting CD13 can effectively redirect sCAR T cells to eradicate the patient-derived AML cells in the clinical relevant models.

Example 8: Overview

Remarkable success for CAR T treatment in B cell malignancies highlights an important and promising direc-

tion to improve cancer immunotherapy. However, this kind of success has not yet expanded to other types of cancers such as AML. One rate-limiting factor, besides lack of cell type or tumor-specific surface targets, is insufficiency or unavailability of mAbs to evaluate the CARs' efficacy for cancer therapy. Moreover, many mAbs targeting cell surface proteins are not suitable for CAR T functions in vivo, failing to enable T cell trafficking, amplifying, destroying targets. As a result, novel strategies are necessary to develop multiple antibodies for multiple cell surface targets to evaluate the CAR T cell efficacies both in vitro and in vivo, accelerating the pace of developing effective cancer CAR T therapy.

In this respect, several strengths for the STAR system are noteworthy to meet this need. First, STAR system is able to isolate numerous specific antibodies against various targets that are differentially expressed on the tumor cell surface based on the phage display. Nbs that recognized T cells were also filtered out in the process of negative cell absorption. Second, among these antibodies, only the ones that can both redirect the cognate CAR T cells to the tumor site from systemic circulation, and accumulate the CAR T cells in vivo will be isolated and chosen for further studies. Third, with the Nbs that passed the above two functional screenings, their antigens can be reliably retrieved for detailed CAR T studies. Notably, all the retrieved Nbs, when assembled into the CAR T cells, were capable of potentially killing AML in vitro and empowering the cognate CAR T cells to kill the AML cells in vivo. These findings highlight the robust performance of this system to quickly uncover the new mAbs and their targets for developing novel and effective cancer CAR T therapy. Thus, the STAR system can be utilized to accelerate CAR T development by identifying effective CAR T-compatible antibodies and their antigens for various types of tumor models.

AML relapsed from chemotherapy is highly aggressive with poor prognosis. There is a strong need to develop more approaches to target AML cells via CAR T cells to improve therapy. The two of the Nbs we generated using the STAR system bound CD13 (aminopeptidase N, APN), and the Nb-driven CAR T specifically and potently killed AML cells, but not chronic leukemia cells, in vitro and in vivo. CD13 is preferentially expressed on AML cells and leukemia stem cells, and is moderately expressed in monocytic leukemia cells. Kaneko, T. et al described anti-CD13-based bi-specific T-cell engagers (BiTEs) could inhibit AML cell colony formation in culture³⁵ however, no any preclinical evaluation was further reported. CD13 is also expressed in human liver cancer stem cells, and inhibiting its enzyme activity results in suppression of liver cancer in mouse models. Together, the findings disclosed herein suggest that targeting CD13 by CAR T cells may preferentially kill the AML cells but have less impact on non-leukemic cells.

The persistence is the characteristic of CAR T cells in treated patients and may exacerbate the likelihood of CAR T on-target-off-tumor toxicity. The sCAR T system with the nanobody-driven switch developed herein allow for the temporary killing of CD13-expressing AML cells. There are several advantages to apply the nanobody switch to trigger CAR T cytotoxicity of cancer cells in vivo. First, as the nanobody switch is small, while it triggers potent AML killing in vivo, it has a relative short life to avoid CAR T mediated permanent damage of other normal cells. Second, the small size of nanobody switch makes it better to penetrate the tumor microenvironment and thus engage the CAR T cells in the tumor site to eliminate the target. Third, as the nanobody switch can reversibly trigger the CAR T

cells to kill AML cells, this likely reduces T cell exhaustion due to the chronic and persistent stimulation of the T cells as the CAR is constitutively activated in the tumor site, likely leading to CAR T cell exhaustion. While CD13 is expressed highly in AML blast cells, it is also expressed moderately in a few non-leukemia cells like colon epithelial cells, and kidney tubular epithelial cells. In this respect, the nanobody switch will likely kill the cancer cells, yet render the damage to the relevant normal tissues as reversible and manageable “inflammation” or “autoimmune response”.

It is conceivable that the STAR system is also be valuable for quickly discovering tumor-specific and CAR-compatible antibodies, as well as novel tumor targets for other types of cancers. Moreover, the current approach can generate multiple and CAR-compatible Nb switches targeting various TAAs or other targets. With these Nb switches, it is possible to treat cancer with temporally controlled use of the multiple switches of the CAR T cells to target the cell surface TAAs, based on monitoring the evolution of the personalized cancer surface TAAs, to improve cancer therapy. It is also possible to employ the dynamic and combinatory use of the multiple TAA-targeting switches to increase efficacy and reduce the toxicity. In summary, the present invention not only established a technical platform, the STAR system, to simultaneously isolate tumor-specific and CAR T-compatible Nbs, but also demonstrated that the isolated Nbs preferably bind AML and empower CAR T cells to eradicate AML, expediting the development of effective cancer CAR T therapy. This invention allows for the development of novel immunotherapies by generating STAR selected antibodies for CAR T cells or antibody conjugating drug (ADC) or tumor-diagnostic agents.

Example 9: Developing Nanobody-Directed Immunotherapy to Improve NET Treatment

Neuroendocrine tumors (NETs) pose serious threats to patients' well-being as malignant diseases, and most patients with metastatic NETs succumb to the disease. A remarkable feature common to most well differentiated NETs is the overexpression of somatostatin receptor 2 (SSTR2). Although Octreotide (a drug analog of somatostatin) and other similar derivatives have been used to target SSTR2 to suppress the secretion of hormones from the tumor cells and to inhibit tumor cell growth, the somatostatin analogs rarely kill tumor cells and reduce tumor mass. As such, it is sorely needed to develop novel therapies that selectively eradicate NET cells. Recently, adoptive T cell therapy involving engineered chimeric antigen receptors (CARs) targeting CD19-expressing leukemia or lymphoma has been shown to eradicate lymphocytic leukemia or lymphoma, and approved by FDA for therapy.

Moreover, antibody drug conjugates (ADCs), which are specific monoclonal antibodies (mAbs) linked to the cancer cell-killing drugs, have also been shown to effectively kill leukemia or solid tumor cells in patients. These advances raise an exciting possibility that NETs can be specifically targeted via CAR T cells or a NET-specific ADC to effectively eradicate NET cells. However, lack of specific cell surface tumor-associated antigens and corresponding antibodies constrain using CAR-T cell or ADC therapies to treat most of other types of cancers, such as NETs.

The camel family of animals including llama can produce heavy chain-only antibodies (which contain a single variable domain, a.k.a nanobody or VHH). These type of antibodies have a unique advantage as they can bind the small cavity of proteins, notably cell surface proteins such as G protein

coupled receptors (GPCRs), ion channels and transporters, which are usually hard to be targeted by regular antibodies. Moreover, VHHs also tend to have low immunogenicity in humans. Herein, NET cell surface antigen-specific VHH antibodies were generated via phage display and NET-specific VHH-CAR T cells developed. The effect of VHH-CAR T cells to kill NET cells in vitro and in vivo was demonstrated. Additionally, a NET-specific VHH based ADC drug was developed to treat NET cells in vitro and in vivo.

To achieve these goals, peripheral blood mononuclear cells (PBMCs), including B cells from 38 unimmunized naive llamas, were collected to make RNA and synthesize cDNAs. The variable regions of the heavy chain of the immunoglobulin heavy chain genes (VHH) were amplified with PCR, ligated to the phage display vector, and transformed into competent cells, yielding the VHH phage display library with $>10^{10}$ independent phage clones.

To screen for VHHs specifically binding to NET cell surfaces from the phage library, freshly prepared VHH phage, which express antigen binding VHH as a pIII fusion protein on their surface, was incubated with BON cells, followed by centrifuging to pellet BON cells to remove unbound phage. The pelleted cells were washed in flow-cytometry buffer, and the bound VHH phage was eluted from the cells. The eluted phage was amplified, and then counter-selected with human breast cancer BT474 cells, followed by four rounds of selection with BON cells (similar to illustration in FIG. 1A).

Next, to identify the protein antigen of the isolated VHH1 nanobody, ~3000 human cell membrane protein cDNAs were prepared and transfected into HEK293T cells individually, followed by flow cytometry analysis with the VHH1-expressing phage and FITC-labeled secondary antibody against phage M13 protein (FIG. 12A). After screening the ~3000 cell surface proteins, results showed that VHH1 only bound to human full length CDH17, also called cadherin 17 or liver-intestine cadherin (FIG. 12B). Further western blot results showed that BON cells have robust CDH17 protein expression compared with other tested cell lines (FIG. 12C). Human or mouse CDH17 cDNA was transiently transfected into HEK293T cells, followed by flow cytometry analysis with the VHH1-expressing phage and FITC-labeled secondary antibody against phage M13 protein. The results clearly showed that VHH1 bound both human and mouse CDH17 (FIG. 12D), which provides the opportunity to analyze the potential toxicity of VTH1-CAR T cells in mice.

In order to further confirm the antigen of VHH1 is CDH17, several CDH17 internal truncations were constructed. The truncations were transfected into HEK293T cells. Flow cytometry analysis was performed with the purified VHH1 nanobody and APC labeled anti-HA secondary antibody against HA on VHH1 nanobody, indicating that VHH1 binds to CDH17. Additionally, several CDH17 internal truncations were constructed (FIG. 22A) and transfected into HEK293T cells. Flow cytometry analysis was performed with the purified VHH1 nanobody and APC labeled anti-HA secondary antibody against HA on VHH1 nanobody. As shown in FIG. 22B, the EC1 domain deleted truncation, Del-1, completely lost the function to interact with VHH1 nanobody and the full length CDH17 and other truncations could be bound by VHH1 (FIGS. 22B-22C), indicating that VHH1 binds to the EC1 domain at the N-terminal part of CDH17. CDH17, a cell adhesion protein highly expressed in human NETs. CDH17 is a cell adhesion protein highly expressed in human NETs. All of the four fresh human pancreatic neuroendocrine tumor (PNET)

samples taken via biopsy from liver were robustly bound by VHH1 (FIG. 13C), indicating that CDH17 is highly expressed in PNETs. One nanobody, named VHH1 was identified that specifically binds to CDH17 protein. VHH1-CAR T cells specifically and robustly killed CDH17 expressing NET BON cells. The VHH1 sequence was cloned into the CAR vector (FIG. 13C), and the resulting CAR T cells showed marked killing of BON cells using LDH release assay, while the control UTD cells failed to kill the cells (FIG. 13D). VHH1 was sequenced and confirmed to contain the conserved VHH sequence (FIG. 14).

To determine if VHH1 directs CAR T cells to kill BON cells with certain length of the linker between VHH1 and the transmembrane domain (TM) of the CAR construct, an acute T cell leukemia cell line, JRT3 cells, was first used to perform a killing assay. It is well known that the distance between T cells and target cells is crucial for the formation of synapse of T cells and for T cells killing on target cells. Therefore, VHH1-CARs with several different hinge lengths were constructed in order to optimize the most effective CAR structure (FIG. 15A). The VHH1-CARs include different hinge lengths, a CD8 transmembrane domain, 4-1BB and CD3zeta domains and IRES-GFP for detection of positive CAR T cells (FIG. 15A, and the linker lengths summarized in FIG. 15B). Lentivirus was packaged and transduced into JRT3 cells, and all the resulting CARs were expressed in the T cells as illustrated by flow cytometry analysis for GFP expression (FIG. 15C). The resulting JRT3 cells expressing VHH1-CAR with CD8, IgG3s and IgG4m hinges specifically killed BON cells which can be bound by VHH1 (FIGS. 16A-16B), but not control BT474 cells which cannot be bound by VHH1 (FIG. 16C-16D). Results were from a lactate dehydrogenase (LDH) release assay. Consistently, western blot analysis confirmed different VHH1-CAR proteins expressed in JRT3 cells (FIG. 16E). In summary, JRT3 cells expressing VHH1-CAR with short hinges, CD8, IgG3s and IgG4m, killed BON cells, but the VHH1-CAR JRT3 cells with long hinges, IgG3+CD8 and IgG3, failed to kill BON cells (FIG. 16F).

Although, VHH1-CAR JRT3 cells showed potent killing capability on BON cells, JRT3 cells are cancer cells and can not be used to treat patients. Therefore, human primary T cells were chosen to be used. To confirm the above results, the same VHH1-CARs with different hinge lengths were also transduced into human primary T cells and the CAR expression on the T cell surface was detected by flow for GFP expression (FIG. 17A). The resulting human primary T cells expressing VHH1-CAR with CD8 and IgG4m hinge specifically killed BON cells (FIG. 17B), but not BT474 cells (FIG. 17C), measured using an LDH release assay. The VHH1-CAR T cells with an IgG4m hinge killed BON cells most effectively (FIG. 17B), but not BT474 breast cancer cells (FIG. 17C). Consistently, the VHH1-CAR T cells with an IgG4m hinge bound the BON cells under microscopic observation (FIG. 17B), but not BT474 cells (FIG. 17C). Western blot analysis confirmed different VHH1-CAR proteins were expressed in human primary T cells (FIG. 17D). In summary, the results showed that human primary T cells expressing VHH1-CAR with CD8 and IgG4m hinges specifically killed BON cells, but the VHH1-CAR T cells with IgG3+CD8 and IgG3 or IgG3s hinges failed to kill BON cells (FIG. 17E). The VHH1-CAR T cells with an IgG4m hinge killed BON cells most effectively (FIG. 17B), so the VHH1-CAR with IgG4m hinge was chosen for further studies (FIG. 18A).

Next, the killing of VHH1-CAR (aka CDH17CAR) T cells on BON cells in vivo was analyzed. To evaluate the

efficacy of VHH1-CAR T cells on NET xenograft, BON cells were transplanted into the NSG mice. Fourteen days later, VHH1 CAR T cells or untransduced (UTD) T cells as control were transfused into the NSG mice bearing the NETs via tail vein. The results showed that VHH1 CAR T cells inhibited NET tumor growth significantly, but did not eradicate the tumor.

To confirm whether the CDH17CAR T cells are capable of killing cancer cells in CDH17-specific manner in vivo, CDH17 was ectopically expressed into a CDH17 negative human acute leukemia cell line, NB4 cells and sorted the CDH17 positive cells. Flow cytometry data showed that most of the cells have CDH17 expression (FIG. 18A). CDH17-NB4 cells were transplanted into the flank of NSG mice, followed by injection of VHH1 second generation CAR T cells with 4-1BB and CD3z via tail vein (FIG. 18B). The results showed that VHH1-BBz CAR T cells completely eradicated the CDH17-NB4 tumors in vivo (FIG. 18C). Consistently, CAR T cells amplified in peripheral blood as compared to UTD T cells (FIG. 18D). Next, CDH17 was ectopically expressed into a CDH17 negative solid human ovary cancer cell line, SKOV3 cells, and the CDH17 positive cells were sorted and passaged (FIG. 19A). To evaluate the efficacy of VHH1-CAR T cells on SKOV3 xenograft, WT or CDH17-SKOV3 cells were transplanted into the NSG mice. 24 days later, second generation VHH1-BBz CAR, third generation VHH1-28BBz CAR with CD28 and 4-1BB co-stimulatory domains or UTD T cells were transfused into the NSG mice bearing the WT or CDH17-SKOV3 xenografts intravenously. The results showed that VHH1-28BBz CAR T cells completely eradicated the CDH17-SKOV3 tumors, but not the CDH17 negative WT SKOV3 tumors (FIG. 19C). However, VHH1-BBz CAR T cells only repressed tumor growth, but did not eradicate the tumors (FIG. 19C). Consistently, NSG mice with CDH17-SKOV3 tumors treated with VHH1-28BBz CAR T cells had the highest T cell number in peripheral blood as compared to other groups (FIG. 19D). Together, the results indicate that VHH1-CAR T cells indeed can eradicate the tumors in vivo.

Recently, a new well-differentiated human pancreatic neuroendocrine tumor cell line, NT-3, which has expression of typical pancreatic NETs like SSSTR2, chromogranin A and insulin, was established. Flow cytometry data showed that NT-3 cells were robustly bound by VHH1 (FIG. 20A), indicating that NT-3 cells have high CDH17 expression. Co-culturing VHH1-CAR T cells with NT-3 cells showed that both VHH1-BBz and VHH1-28BBz CAR T cells significantly killed NT-3 cells in vitro (FIG. 20B), using LDH release assay. Furthermore, VHH1-28BBz CAR T cells completely eradicated the NT-3 tumors in NSG mice without affecting the body weight of the mice (FIGS. 20C, 20D, and 20F). VHH1-28BBz CAR T cell number in peripheral blood increased at day 14 after T cell injection and then reduced at day 21 after tumor eradication (FIG. 20E). This series of studies indicates that the CDH17 CARs are capable of and potent in killing and eradicating NETs in vivo and CDH17 serves as valuable target for NETs and other CDH17-expressing tumors. Further, as the CDH17 CAR Ts did not cause obvious toxicities in mice, without loss of body weight, these findings indicate that CDH17 serves as an effective and safe target for immunotherapy against NETs and other types of cancers expressing CDH17.

The effect of VHH1-ADC on suppressing NET tumors was evaluated. To test whether VHH1 ADC specifically kills NET cells, Mertansine (DM1), which is toxic to cancer cells by blocking microtubule assembly and DM1-conjugated anti-HER2 antibody has been approved for treating breast

cancer, was conjugated to VHH1 homodimer connected by a linker (VHH1-linker-VHH1, aka VLV) (FIGS. 21A-21B). As a control, FITC labeled-VLV, when incubated with CDH17-expressing BON cells, was markedly internalized into BON cells (FIG. 21C, left), but not the CDH17-negative QGP1 cells (FIG. 21C, right). Flow data showed that both unconjugated VLV and VLV-DM1 bound to BON cells, albeit with slightly reduced affinity for VLV-DM1 (FIG. 21D), but not QGP1 cells (FIG. 21E). Furthermore, cytotoxicity assay showed that VLV-DM1 significantly killed BON cells (FIG. 21F), likely via internalized DM1, but not CDH17-negative QGP1 cells (FIG. 21G). The morphology data also clearly showed that the VLV-DM1 lysed the BON cells at 4 nM, but not unconjugated DM1 or VLV (FIG. 21H). These results demonstrated the impact of VHH1-ADC on killing NET cell line-derived xenograft and NET PDX in immunodeficient mice.

Other Embodiments

The recitation of a listing of elements in any definition of a variable herein includes definitions of that variable as any single element or combination (or subcombination) of listed elements. The recitation of an embodiment herein includes that embodiment as any single embodiment or in combination with any other embodiments or portions thereof.

The disclosures of each and every patent, patent application, and publication cited herein are hereby incorporated herein by reference in their entirety. While this invention has been disclosed with reference to specific embodiments, it is apparent that other embodiments and variations of this invention may be devised by others skilled in the art without departing from the true spirit and scope of the invention. The appended claims are intended to be construed to include all such embodiments and equivalent variations.

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Gly Arg Phe Thr Val Ser Arg Gly Asn Thr Gln Asn Ser Val Asn Leu	
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 Val Arg Gln Ala Pro Gly Lys Gly Pro Glu Trp Val Ser Gly Ile Tyr
 65 70 75 80
 Pro Ser Asp Gly Lys Thr Arg Tyr Ala Asp Phe Val Lys Gly Arg Phe
 85 90 95
 Ser Ile Ser Arg Asp Asn Ala Lys Asn Met Leu Tyr Leu Gln Met Asn
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 Asn Leu Glu Pro Glu Asp Thr Ala Leu Tyr Tyr Cys Ala Arg Gly Ile
 115 120 125
 Thr Gly Leu Gly Pro Arg Gly Gln Gly Thr Gln Val Thr Val Ser Ser
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 Ala Ala Ala Thr Ser Gly Gln Ala Gly Gln Ser Gly Glu Ser Lys Tyr
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 Gly Pro Pro Cys Pro Pro Cys Pro Ala Ser Tyr Ile Trp Ala Pro Leu
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 Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr
 180 185 190
 Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe
 195 200 205
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 210 215 220
 Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser
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Tyr Ala Asp Phe Val Lys Gly Arg Phe Ser Ile Ser Arg Asp Asn Ala
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Lys Asn Met Leu Tyr Leu Gln Met Asn Asn Leu Glu Pro Glu Asp Thr
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<223> OTHER INFORMATION: 4-1BB and CD3z

<400> SEQUENCE: 22

Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met
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 Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe
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 Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg
 35 40 45
 Ser Ala Asp Ala Pro Ala Tyr Lys Gln Gly Gln Asn Gln Leu Tyr Asn
 50 55 60
 Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg
 65 70 75 80
 Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro
 85 90 95
 Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala
 100 105 110
 Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His
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 Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp
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 Ala Leu His Met Gln Ala Leu Pro Pro Arg
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<210> SEQ ID NO 23

<211> LENGTH: 358

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: VHH163 CAR

<400> SEQUENCE: 23

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 His Ala Ala Arg Pro Gly Ser Ala Ala Gln Ala Ala Gln Val Gln Leu
 20 25 30

[illegible]

<400> SEQUENCE: 24

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Val Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Val Pro Ser Gly Phe
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Thr Phe Asp Gly Tyr Leu Ile Gly Trp Phe Arg Gln Ala Pro Gly Ser
 35 40 45

Glu Arg Lys Ala Val Ser Cys Ile Ser Val Asn Gly Asp Arg Thr Asn
 50 55 60

Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala
 65 70 75 80

Lys Asn Thr Val Tyr Leu Gln Met Asn Ser Leu Arg Pro Asn Asp Thr
 85 90 95

Ala Ile Tyr Tyr Cys Ala Thr Arg Arg Gly Asn Arg Leu Tyr Asn Asn
 100 105 110

Asn Cys Pro Tyr Phe Glu Tyr Gly Thr Trp Gly Gln Gly Thr Gln Val
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Thr Val Ser Ser Ala Ala
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<210> SEQ ID NO 25
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH157 CAR CDR1

<400> SEQUENCE: 25

Ser Tyr Ser Met Ala
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<210> SEQ ID NO 26
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH157 CAR CDR2

<400> SEQUENCE: 26

Gly Ile Tyr Pro Ser Asp Gly Lys Thr Arg Tyr Ala Asp Phe Val Lys
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Gly Arg

<210> SEQ ID NO 27
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH157 CAR CDR3

<400> SEQUENCE: 27

Ala Arg Gly Ile Thr Gly Leu Gly Pro
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<210> SEQ ID NO 28
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH163 CAR CDR1

<400> SEQUENCE: 28

Gly Tyr Leu Ile Gly
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<210> SEQ ID NO 29
 <211> LENGTH: 18

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<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH163 CAR CDR2

<400> SEQUENCE: 29

Cys Ile Ser Val Asn Gly Asp Arg Thr Asn Tyr Ala Asp Ser Val Lys
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Gly Arg

<210> SEQ ID NO 30
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH163 CAR CDR3

<400> SEQUENCE: 30

Ala Thr Arg Arg Gly Asn Arg Leu Tyr Asn Asn Asn Cys Pro Tyr Phe
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Glu Tyr Gly Thr
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<210> SEQ ID NO 31
 <211> LENGTH: 8721
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH157-BBz pLenti-CAR vector

<400> SEQUENCE: 31

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<210> SEQ ID NO 32

<211> LENGTH: 8754

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: VHH163-BBz pLenti-CAR vector

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<210> SEQ ID NO 33
 <211> LENGTH: 1047
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH1-BBz CAR

<400> SEQUENCE: 33

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cagctggggg gctcgacaag actctcctgt gtatcgtccc gcacctttag ttattatgac    180
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aatacccaga attcgggtgaa tctgcaaatg aacaacctga aacctgagga cacgggcatc    360
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<210> SEQ ID NO 34
 <211> LENGTH: 349
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH1-BBz CAR

<400> SEQUENCE: 34

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Met Ala Leu Pro Val Thr Ala Leu Leu Pro Leu Ala Leu Leu Leu
1           5           10          15

His Ala Ala Arg Pro Gly Ser Ala Ala Gln Ala Ala Gln Val Gln Leu
20          25          30

Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Thr Arg Leu
35          40          45

Ser Cys Val Ser Ser Arg Thr Phe Ser Tyr Tyr Asp Met Gly Trp Phe
50          55          60

Arg Gln Ala Pro Gly Lys Glu Arg Glu Phe Val Ala Leu Leu Ser Trp
65          70          75          80

Asn Gly Glu Asn Ala Glu Tyr Ser Asp Ser Val Met Gly Arg Phe Thr
85          90          95

Val Ser Arg Gly Asn Thr Gln Asn Ser Val Asn Leu Gln Met Asn Asn
100         105         110

Leu Lys Pro Glu Asp Thr Gly Ile Tyr Tyr Cys Ala Val Thr His Gly
115         120         125
  
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Gly Ala Arg Ser Val Arg Ser Trp Gly Gln Gly Thr Gln Val Thr Val
 130 135 140
 Ser Ser Ala Ala Ala Thr Ser Gly Gln Ala Gly Gln Ser Gly Glu Ser
 145 150 155 160
 Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro Ala Ser Tyr Ile Trp Ala
 165 170 175
 Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr
 180 185 190
 Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln
 195 200 205
 Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser
 210 215 220
 Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys
 225 230 235 240
 Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Lys Gln Gly Gln Asn Gln
 245 250 255
 Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu
 260 265 270
 Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg
 275 280 285
 Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met
 290 295 300
 Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly
 305 310 315 320
 Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp
 325 330 335
 Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 340 345

<210> SEQ ID NO 35
 <211> LENGTH: 1173
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VHH1-28BBz CAR

<400> SEQUENCE: 35

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gaccctgaga tggggggaaa gccgagaagg aagaaccctc aggaaggcct gtacaatgaa     1020
ctgcagaaag ataagatggc ggaggcctac agtgagattg ggatgaaagg cgagcgccgg     1080
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<210> SEQ ID NO 36
<211> LENGTH: 391
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: VHH1-28BBz CAR

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<400> SEQUENCE: 36

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
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His Ala Ala Arg Pro Gly Ser Ala Ala Gln Ala Ala Gln Val Gln Leu
 20          25          30

Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Thr Arg Leu
 35          40          45

Ser Cys Val Ser Ser Arg Thr Phe Ser Tyr Tyr Asp Met Gly Trp Phe
 50          55          60

Arg Gln Ala Pro Gly Lys Glu Arg Glu Phe Val Ala Leu Leu Ser Trp
 65          70          75          80

Asn Gly Glu Asn Ala Glu Tyr Ser Asp Ser Val Met Gly Arg Phe Thr
 85          90          95

Val Ser Arg Gly Asn Thr Gln Asn Ser Val Asn Leu Gln Met Asn Asn
100          105          110

Leu Lys Pro Glu Asp Thr Gly Ile Tyr Tyr Cys Ala Val Thr His Gly
115          120          125

Gly Ala Arg Ser Val Arg Ser Trp Gly Gln Gly Thr Gln Val Thr Val
130          135          140

Ser Ser Ala Ala Ala Thr Ser Ser Gly Glu Ser Lys Tyr Gly Pro Pro
145          150          155          160

Cys Pro Pro Cys Pro Asp Ile Phe Trp Val Leu Val Val Val Gly Gly
165          170          175

Val Leu Ala Cys Tyr Ser Leu Leu Val Thr Val Ala Phe Ile Ile Phe
180          185          190

Trp Val Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn
195          200          205

Met Thr Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr
210          215          220

Ala Pro Pro Arg Asp Phe Ala Ala Tyr Arg Ser Ala Ser Lys Arg Gly
225          230          235          240

Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val
245          250          255

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Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu
260 265 270

Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp
275 280 285

Ala Pro Ala Tyr Lys Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn
290 295 300

Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg
305 310 315 320

Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly
325 330 335

Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu
340 345 350

Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu
355 360 365

Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His
370 375 380

Met Gln Ala Leu Pro Pro Arg
385 390

<210> SEQ ID NO 37
 <211> LENGTH: 68
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CD28 transmembrane and CD28 intracellular
 domain

<400> SEQUENCE: 37

Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu
1 5 10 15

Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg Ser
20 25 30

Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro Gly
35 40 45

Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro Pro Arg Asp Phe Ala
50 55 60

Ala Tyr Arg Ser
65

<210> SEQ ID NO 38
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: modified IgG4h

<400> SEQUENCE: 38

Ala Thr Ser Ser Gly Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys
1 5 10 15

Pro

117

What is claimed is:

1. A chimeric antigen receptor (CAR) comprising a nanobody, a transmembrane domain, and an intracellular signaling domain, wherein the nanobody specifically binds to the first domain of CDH17, wherein the nanobody comprises a CDR1 region comprising the amino acid sequence of SEQ ID NO: 3, a CDR2 region comprising the amino acid sequence of SEQ ID NO: 4, and a CDR3 region comprising the amino acid sequence of SEQ ID NO: 5.
2. The CAR of claim 1, wherein the nanobody comprises the amino acid sequence of SEQ ID NO: 2.
3. The CAR of claim 1, wherein the nanobody is encoded by the nucleotide sequence of SEQ ID NO: 1.
4. The CAR of claim 1, wherein the CAR further comprises a hinge domain selected from the group consisting of a CD8 hinge, an IgG3s hinge, and an IgG4m hinge.
5. The CAR of claim 4, wherein the hinge domain comprises the amino acid sequence of any one of SEQ ID NOs: 20 or 38.
6. The CAR of claim 1, wherein the transmembrane domain is selected from the group consisting of CD8, CD28, and ICOS.
7. The CAR of claim 1, wherein the transmembrane domain comprises SEQ ID NO: 21.
8. The CAR of claim 1, wherein the intracellular signaling domain comprises 4-1BB and CD3 zeta.
9. The CAR of claim 1, wherein the intracellular signaling domain comprises SEQ ID NO: 22.

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10. The CAR of claim 1, wherein the CAR comprises the amino acid sequence of any one of SEQ ID NOs: 34 or 36.
11. The CAR of claim 1, wherein the CAR is encoded by the nucleotide sequence of any one of SEQ ID NOs: 33 or 35.
12. A modified T cell or precursor thereof, comprising the CAR of claim 1.
13. A method for treating cancer in a subject in need thereof, the method comprising administering to the subject a modified T cell or precursor thereof comprising the CAR of claim 1.
14. The method of claim 13, wherein the cancer is acute myeloid leukemia (AML).
15. The method of claim 13, wherein the cancer is a neuroendocrine tumor (NET).
16. The method of claim 13, wherein the cancer is colorectal cancer.
17. A composition comprising a nucleic acid encoding a chimeric antigen receptor (CAR), the CAR comprising a nanobody, a transmembrane domain, and an intracellular signaling domain, wherein the nanobody specifically binds to the first domain of CDH17, wherein the nanobody comprises a CDR1 region comprising the amino acid sequence of SEQ ID NO: 3, a CDR2 region comprising the amino acid sequence of SEQ ID NO: 4, and a CDR3 region comprising the amino acid sequence of SEQ ID NO: 5.

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